EXCEL ENGINEERING COLLEGE (Autonomous)

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

LAB MANUAL

23CB504 - OBJECT ORIENTED ANALYSIS AND DESIGN LABORATORY

Regulation 2023

Year / Semester: III / V

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DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

LABORATORY RECORD NOTEBOOK

Certified that this bona fide record of work done by

Selvan / Selvi____________________________Reg. No___________________________

of the Vth semester B. Tech - CSBS branch during the Academic Year 2025-2026 in

the **23CB504 - OBJECT ORIENTED ANALYSIS AND DESIGN** LABORATORY

Signature of Lab-Incharge **Signature of Head of the Department**

Submitted for the Practical Examination held on______________________

Internal Examiner **External Examiner**

EXCEL ENGINEERING COLLEGE, KOMARAPALAYAM
(AUTONOMOUS)

Department of Computer Science and Business Systems

Vision

To be a center of excellence in Computer Science and Business Systems by imparting futuristic knowledge and skills, fostering innovation, and nurturing ethically sound professionals to meet global industry and societal demands.

Mission

To deliver a comprehensive education in Computer Science and Business Systems integrating technical expertise with contemporary business principles.

To enhance employability skills by promoting innovation, problem solving and adaptability to emerging technologies.

To inculcate ethical values and social responsibility in students ensuring their contribution to sustainable development.

To foster partnerships with industries, academia, and research organizations for lifelong learning and professional growth.

Program Educational Objectives (PEOs)

Professional Competence

Graduates will demonstrate expertise in Computer Science and Business Systems, applying technical and business knowledge to solve real-world problems effectively.

Innovation and Lifelong Learning

Graduates will engage in innovative practices and continuous learning to adapt to evolving technologies and dynamic business environments.

Ethical and Social Responsibility

Graduates will uphold ethical values, contribute to sustainable development, and address societal challenges with a sense of responsibility.

Leadership and Collaboration

Graduates will exhibit leadership qualities and excel in multidisciplinary teamwork to achieve organizational and personal goals in global settings.

PSOs

To display competence in Computer Science while appreciating the significance of the management sciences and human ethics

To formulate, identify, and apply relevant strategies, resources, and advanced engineering and business technologies, including predictive modeling and data analytics, to complex engineering and business activities

23CB504 - OBJECT ORIENTED ANALYSIS AND DESIGN LABORATORY

OBJECTIVES:

- To capture the requirements specification for an intended software system
- To draw the UML diagrams for the given specification
- To map the design properly to code
- To test the software system thoroughly for all scenarios
- To improve the design by applying appropriate design patterns.

Draw standard UML diagrams using an UML modeling tool for a given case study and map design to code and implement a 3 layered architecture. Test the developed code and validate whether the SRS is satisfied.

1. Identify a software system that needs to be developed.
2. Document the Software Requirements Specification (SRS) for the identified system.
3. Identify use cases and develop the Use Case model.
4. Identify the conceptual classes and develop a Domain Model and also derive a Class Diagram from that.
5. Using the identified scenarios, find the interaction between objects and represent them using UML Sequence and Collaboration Diagrams.
6. Draw relevant State Chart and Activity Diagrams for the same system.
7. Implement the system as per the detailed design.
8. Test the software system for all the scenarios identified as per the use case diagram.
9. Improve the reusability and maintainability of the software system by applying appropriate design patterns.
10. Implement the modified system and test it for various scenarios.

SUGGESTED DOMAINS FOR MINI-PROJECT:

1. Student Information System.
2. Passport Automation System.
3. Book Bank.
4. Exam Registration.
5. Stock Maintenance System.
6. Online Course Reservation System.
7. Airline/Railway Reservation System.
8. Software Personnel Management System.
9. Credit Card Processing.
10. E-Book Management System.
11. Library Management System.

TABLE OF CONTENTS

EX.NO: 1 **CASE TOOLS**

Date:

INTRODUCTION:

CASE tools known as Computer-aided software engineering tools is a kind of component-based development which allows its users to rapidly develop information systems. The main goal of case technology is the automation of the entire information systems development life cycle process using a set of integrated software tools, such as modeling, methodology and automatic code generation. Component based manufacturing has several advantages over custom development. The main advantages are the availability of high quality, defect free products at low cost and at a faster time. The prefabricated components are customized as per the requirements of the customers. The components used are pre-built, ready-tested and add value and differentiation by rapid customization to the targeted customers. How ever the products we get from case tools are only a skeleton of the final product required and a lot of programming must be done by hand to get a fully finished, good product.

CHARACTERISTICS OF CASE:

Some of the characteristics of case tools that make it better than customized development are;

- ❖ It is a graphic oriented tool.
- ❖ It supports decomposition of process.

Some typical CASE tools are:

- ❖ Unified Modeling Language
- ❖ Data modeling tools, and
- ❖ Source code generation tools

INTRODUCTION TO UML (UNIFIED MODELING LANGUAGE):

The UML is a language for specifying, constructing, visualizing, and documenting the software system and its components. The UML is a graphical language with sets of rules and semantics. The rules and semantics of a model are expressed in English in a form known as OCL (Object Constraint Language). OCL uses simple logic for specifying the properties of a system. The UML is not intended to be a visual programming language. How ever it has a much closer mapping to object-oriented programming languages, so that the best of both can be obtained. The UML is much simpler than other methods preceding it. UML is appropriate for modeling systems, ranging from enterprise information system to distributed web based application and even to real time embedded system. It is a very expensive language addressing all views needed to develop and then to display system even though understand to use. Learning to apply UML effectively starts forming a conceptual mode of languages which requires learning.

Three major language elements:

- UML basic building blocks
- Rules that dictate how this building blocks put together
- Some common mechanism that apply throughout the language

The primary goals in the design of UML are:

1. Provides users ready to use, expressive visual modeling language as well so they can develop and exchange meaningful models.
2. Provide extensibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development processes.
4. Provide formal basis for understanding the modeling language.
5. Encourage the growth of the OO tools market.
6. Support higher-level development concepts.
7. Integrate best practices and methodologies.

Every complex system is best approached through a small set of nearly independent views of a model. Every model can be expressed at different levels of fidelity. The best models are connected to reality.

UML defines nine graphical diagrams:

1. Class Diagram
2. Use-Case Diagram
3. Behavior Diagram

4. Interaction Diagram
5. Sequence Diagram
6. Collaboration Diagram
7. State Chart Diagram
8. Activity Diagram
9. Implementation diagram
10. component diagram
11. deployment diagram

UML Class Diagram:

The UML class diagram is also known as object modeling. It is a static analysis diagram. These diagrams show the static structure of the model. A class diagram is a connection of static model elements, such as classes and their relationships, connected as a graph to each other and to their contents.

Use-Case Diagram:

The functionality of a system can be described in a number of different use-cases, each of which represents a specific flow of events in a system. It is a graph of actors, a set of use-cases enclosed in a boundary, communication, associations between the actors and the use-cases, and generalization among the use-cases.

Behavior Diagram:

It is a dynamic model unlike all the others mentioned before. The objects of an object-oriented system are not static and are not easily understood by static diagrams. The behavior of the class's instance (an object) is represented in this diagram. Every use-case of the system has an associated behavior diagram that indicates the behavior of the object. In conjunction with the use-case diagram we may provide a script or interaction diagram to show a time line of events. It consists of sequence and collaboration diagrams.

Interaction Diagram:

It is the combination of sequence and collaboration diagram. It is used to depict the flow of events in the system over a timeline. The interaction diagram is a dynamic model which shows how the system behaves during dynamic execution.

State Chart Diagram:

It consists of state, events and activities. State diagrams are a familiar technique to describe the behavior of a system. They describe all of the possible states that a particular object can get into and how the object's state changes as a result of events that reach the object. In most OO techniques, state diagrams are drawn for a single class to show the lifetime behavior of a single object.

Activity Diagram:

It shows organization and their dependence among the set of components. These diagrams are particularly useful in connection with workflow and in describing behavior that has a lot of parallel processing. An activity is a state of doing something: either a real-world process, or the execution of a software routine.

Implementation Diagram:

It shows the implementation phase of the systems development, such as the source code structure and the run-time implementation structure. These are relatively simple high-level diagrams compared to the others seen so far. They are of two sub-diagrams, the component diagram and the deployment diagram.

Component Diagram:

These are organizational parts of a UML model. These are boxes to which a model can be decomposed. They show the structure of the code itself. They model the physical components such as source code, user interface in a design. It is similar to the concept of packages.

Deployment Diagram:

The deployment diagram shows the structure of the runtime system. It shows the configuration of runtime processing elements and the software components that live in them. They are usually used in conjunction with deployment diagrams to show how physical modules of code are distributed on the system.

NOTATION ELEMENTS:

These are explanatory parts of UML model. They are boxes which may apply to describe and remark about any element in the model. They provide the information for understanding the necessary details of the diagrams.

Relations in the UML:

These are four kinds of relationships used in an UML diagram, they are:

- Dependency
- Association
- Generalization
- Realization

Dependency:

It is a semantic relationship between two things in which a change one thing affects the semantics of other things. Graphically a dependency is represented by a non-continuous line.

Association:

It is a structural relationship that describes asset of links. A link is being connected among objects. Graphically association is represented as a solid line possibly including label.

Generalization:

It is a specialized relationship in which the specialized elements are substitutable for object of the generalized element. Graphically it is a solid line with hollow arrow head parent.

Realization:

It is a semantic relation between classifiers. Graphically it is represented as a cross between generalization and dependency relationship.

Where UML can be used:

UML is not limited to modeling software. In fact it is expressive to model non-software such as to show in structure and behavior of health case system and to design the hardware of the system.

Conceptual model be UML:

UML you need to form the conceptual model of UML. This requires three major elements:

UML basic building blocks.

Rules that dictate how this building blocks are put together.

Some common mechanism that apply throughout the language.

Once you have grasped these ideas, you may be able to read. UML create some basic ones. As you gain more experience in applying conceptual model using more advanced features of this language.

Building blocks of the UML:

The vocabulary of UML encompasses these kinds of building blocks.

Use CASE definition:

Description:

A use case is a set of scenarios tied together by a common user goal. A use case is a behavioral diagram that shows a set of use case actions and their relationships.

Purpose:

The purpose of use case is login and exchange messages between sender and receiver (Email client).

Main flow:

First, the sender gives his id and enters his login. Now, he enters the message to the receiver id.

Alternate flow:

If the username and id by the sender or receiver is not valid, the administrator will not allow entering and “Invalid password” message is displayed.

Pre-condition:

A person has to register himself to obtain a login ID.

Post-condition:

The user is not allowed to enter if the password or user name is not

valid.

Class diagram:

Description:

A class diagram describes the type of objects in system and various kinds of relationships that exists among them. Class Diagrams and collaboration diagrams are alternate representations of object models. During analysis, we use class diagram to show roles and responsibilities of entities that provide email client system behaviors design. We use to capture the structure of classes that form the email client system architecture.

A class diagram is represented as:

- <<Class name>>
- <<Attribute 1>>
- <<Attribute n>>
- <<Operation ()>>

Relationship used:

A change in one element affects the other

Generalization:

It is a kind of relationship

State chart:

Description:

The state chart diagram made the dynamic behavior of individual classes.

State chart shows the sequences of states that an object goes through events and state transitions.

A state chart contains one state 'start' and multiple 'end' states.

The important objectives are;

Decision:

It represents a specific location state chart diagram where the work flow may branch based upon guard conditions.

Synchronization:

It gives a simultaneous workflow in a state chart diagram. They visually define forks and joints representing parallel workflow.

Forks and joins:

A fork construct is used to model a single flow of control.

Every work must be followed by a corresponding join.

Joints have two or more flow that unit into a single flow.

State:

A state is a condition or situation during a life of an object in which it satisfies condition or waits for some events.

Transition:

It is a relationship between two activities and between states and activities.

Start state:

A start state shows the beginning of a workflow or beginning of a state machine on a state chart diagram.

End state:

It is a final or terminal state.

Activity diagram Description:

Activity diagram provides a way to model the workflow of a development process. We can also model this code specific information such as class operation using activity diagram. Activity diagrams can model different types of diagrams. There are various tools involved in the activity diagram.

Activity:

An activity represents the performance of a task on duty. It may also represent the execution of a statement in a procedure.

Decision:

A decision represents a condition on situation during the life of an object, which it satisfies some condition or waits for an event.

Start state:

It represents the condition explicitly the beginning of a workflow on an activity.

Object flow:

An object on an activity diagram represents the relationship between activity and object that creates or uses it.

Synchronization:

It enables us to see a simultaneous workflow in an activity.

End state:

An end state represents a final or terminal state on an activity diagram or state chart diagram.

Sequence diagram:

Description:

A sequence diagram is a graphical view of scenario that shows object interaction in a time based sequence what happens first what happens next. Sequence diagrams are closely related to collaboration diagram.

The main difference between sequence and collaboration diagram is that sequence diagram show time based interaction while collaboration diagram shows objects associated with each other.

The sequence diagram for the e-mail client system consists of the following objectives:

Object:

An object has state, behavior and identity. An object is not based is referred to as an instance.

The various objects in e-mail client system are:

- ❖ User
- ❖ Website
- ❖ Login
- ❖ Groups

Message icon:

A message icon represents the communication between objects indicating that an action will follow. The message icon is the horizontal solid arrow connecting lifelines together.

Collaboration diagram:

Description:

Collaboration diagram and sequence diagrams are alternate representations of an interaction. A collaboration diagram is an interaction diagram that shows the order of messages that implement an operation or a transaction. Collaboration diagram is an interaction diagram that shows the order of messages that implement an operation or a transaction.

Collaboration diagram shows object s, their links and their messages. They can also contain simple class instances and class utility instances.

During, analysis indicates the semantics of the primary and secondary interactions. Design, shows the semantics of mechanisms in the logical design of system.

Toggling between the sequence and collaboration diagrams.

When we work in either a sequence or collaboration diagram, it is possible to view the corresponding diagram by pressing F5 key.

EX.NO: 2

STUDENT INFORMATION SYSTEM

Date:

AIM

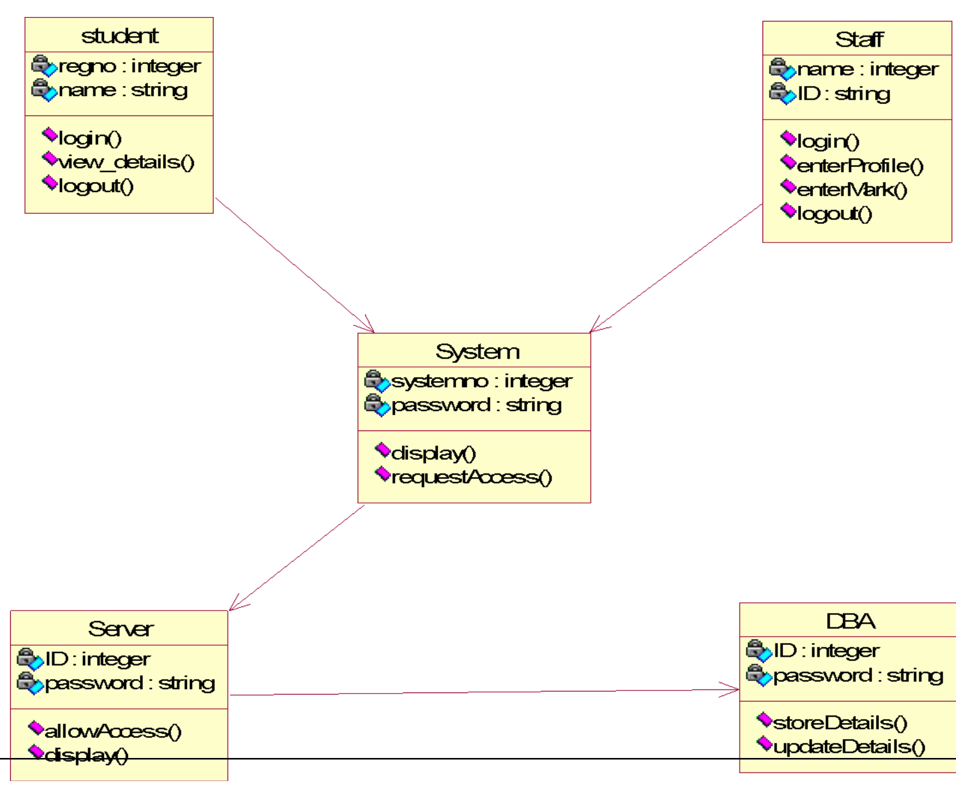
To design an object-oriented model for Student information system using Rational Rose software.

PROBLEM STATEMENT

The student must register by entering the name and password to login the form. The admin select the particular student to view the details about that student and maintaining the student details. This process of student information system is described sequentially through following steps. The student registers the system. The admin login to the student information system. He/she search for the list of students. Then select the particular student. Then view the details of that student. After displaying the student details then logout.

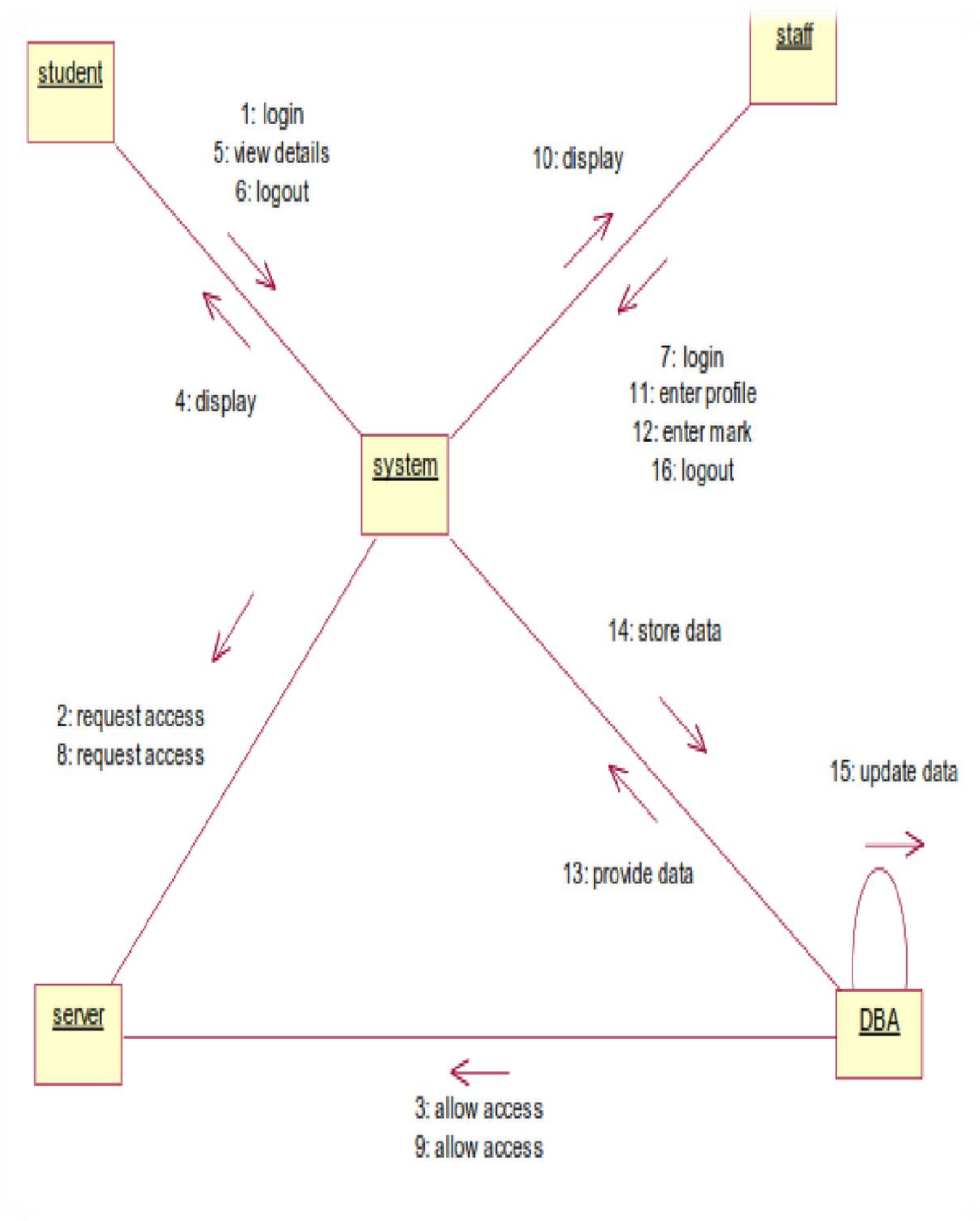
CLASS DIAGRAM

The class diagram is the graphical representation of all classes used in the system. The class diagram is drawn as rectangular box with three components or compartments like class name, attributes and operations. The student information system makes use of the following classes like student, staff, system, DBA and server.



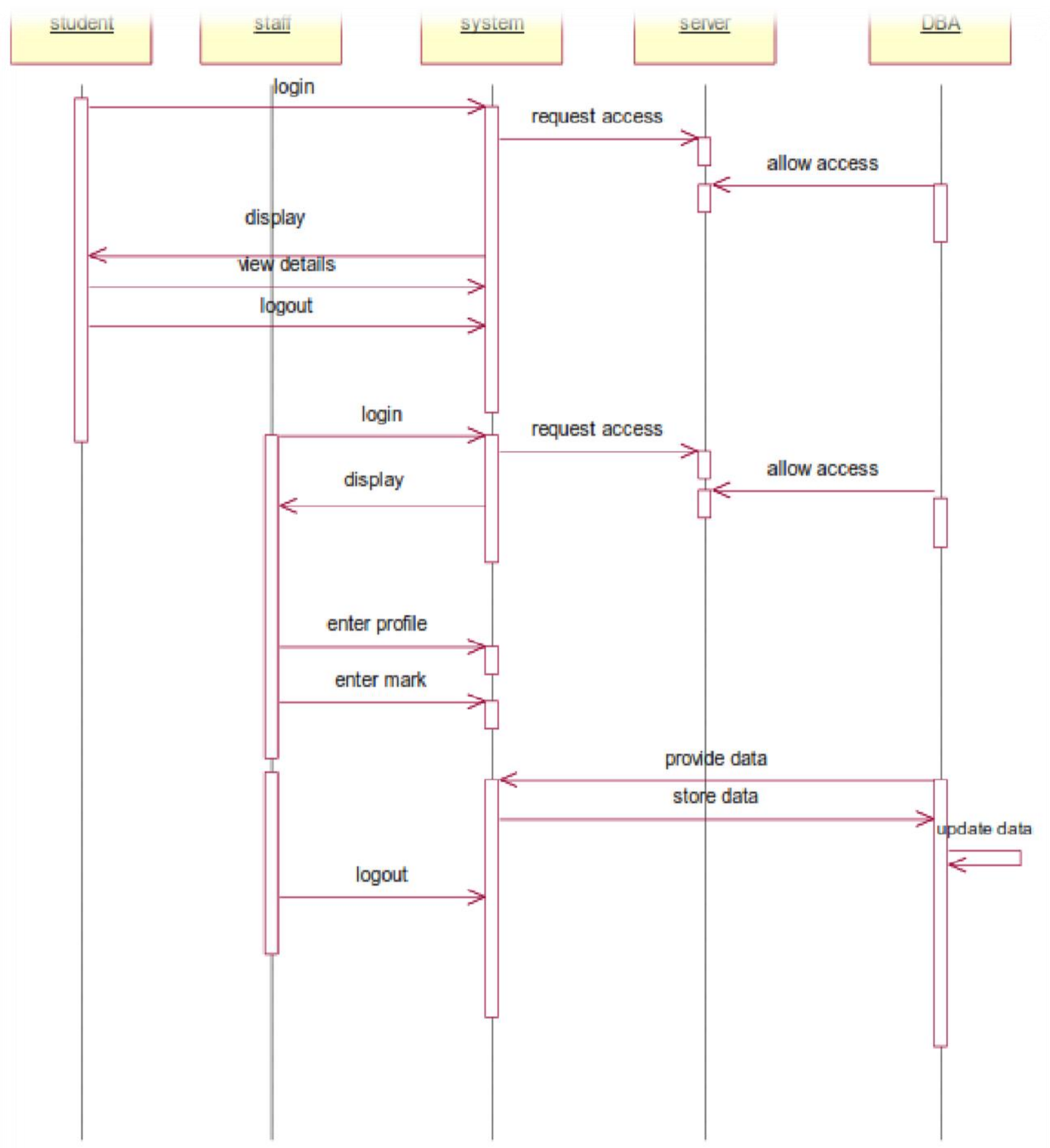
COLLABORATION DIAGRAM

A Collaboration diagram represents the collaboration in which is a set of objects related to achieve a desired outcome. In collaboration, the sequence is indicated by numbering the message several numbering schemes are available. Login, request access, allow access, display, view details, logout, login, request access, allow access, display, enter profile, enter mark, provide data, logout, store data, update data.



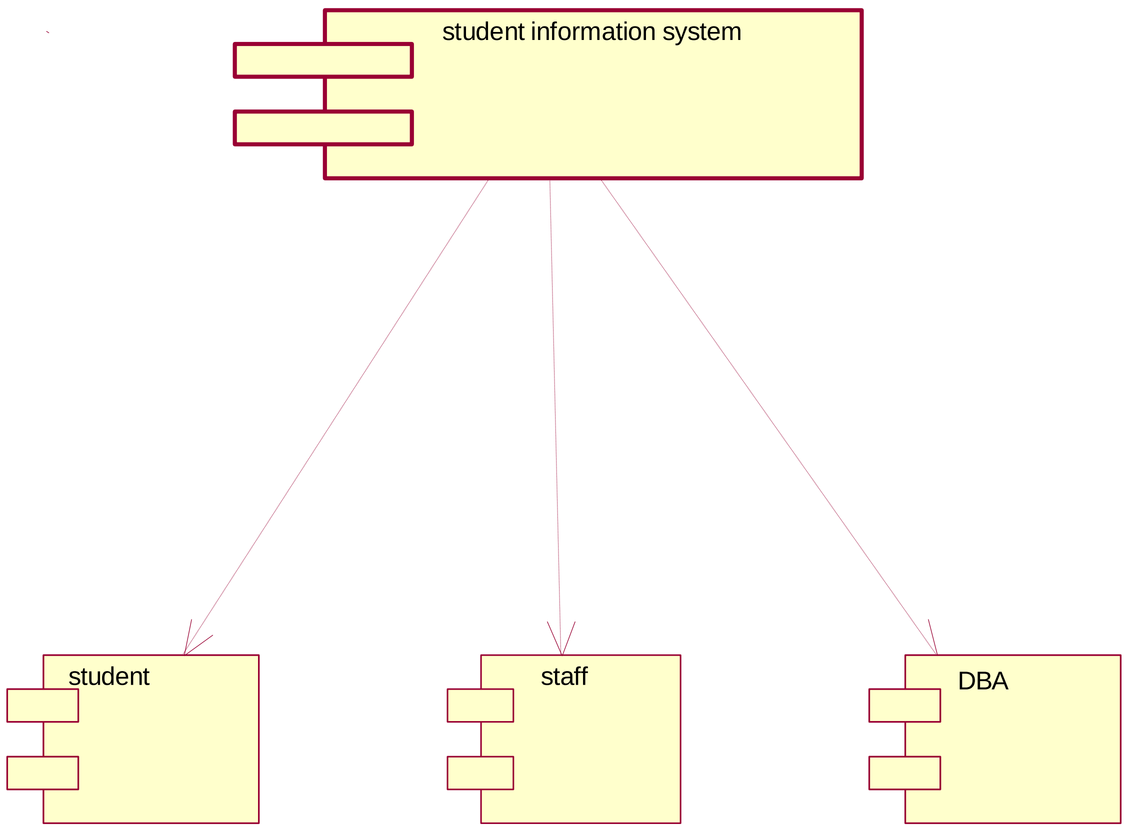
SEQUENCE DIAGRAM

A Sequence diagram represent the sequence and interaction of a given usecase or scenario. Sequence diagram capture most of the information about the system. Here the sequence starts between the student and the system. The second half of interaction takes place between staff and system then by police and followed by database. The student first login to the system and then view the details of the details. Staff login to the system enter mark and enter the details of the student. DBA store and update the details of the student



COMPONENT DIAGRAM

Component diagram carries the major living actors of the system. The component diagram main purpose is to show the structural relationship between components of the system.The main component of the system is student information system and the other components of the system are student, staff and DBA.

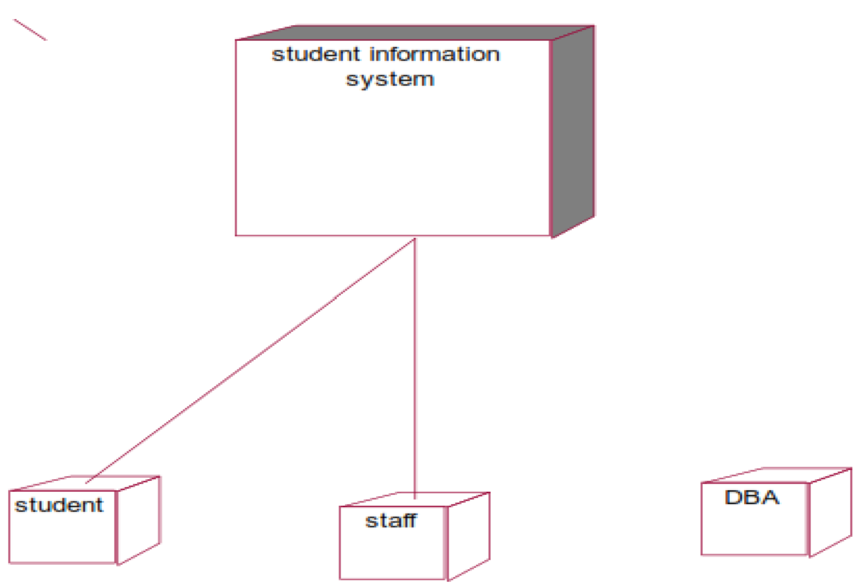


USE CASE DIAGRAM

Use case diagram is a graph of actors, a set of use cases, association between the actors and the use cases and generalization among the cases. Use case diagram is a list of actions or events. Use case diagram was drawn to represent the static design view of the system. Steps typically defined the interactions between a role and a system to achieve a goal. The use case diagram consists of various functionality performed by the actors like student, staff, system, DBA and server. The use case diagram consists of various functionality like login, display, enter profile, enter mark, view details, update details, allow access, request access, store details, logout.

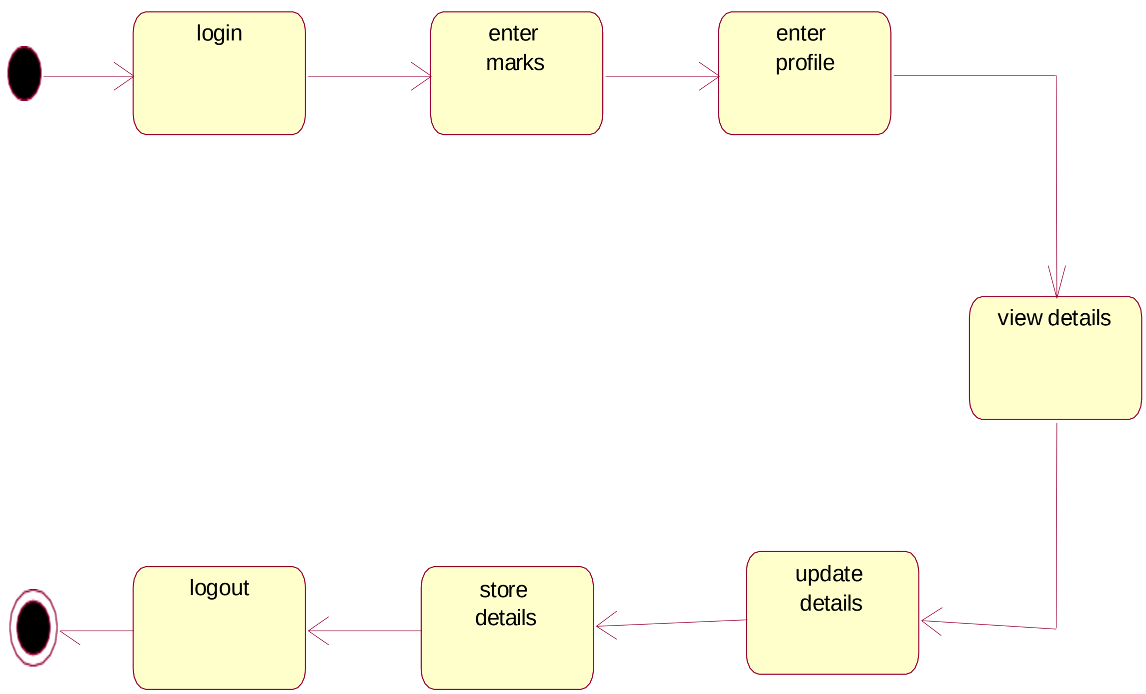
DEPLOYMENT DIAGRAM

Deployment diagram shows the configuration of runtime processing elements and the software components processes and objects that live in them. Component diagram are used in conjunction with deployment diagram to show how physical modules code are distributed on various hardware platform. The processor node in the system is student information system and the execution environment nodes or device nodes are student, staff and DBA.



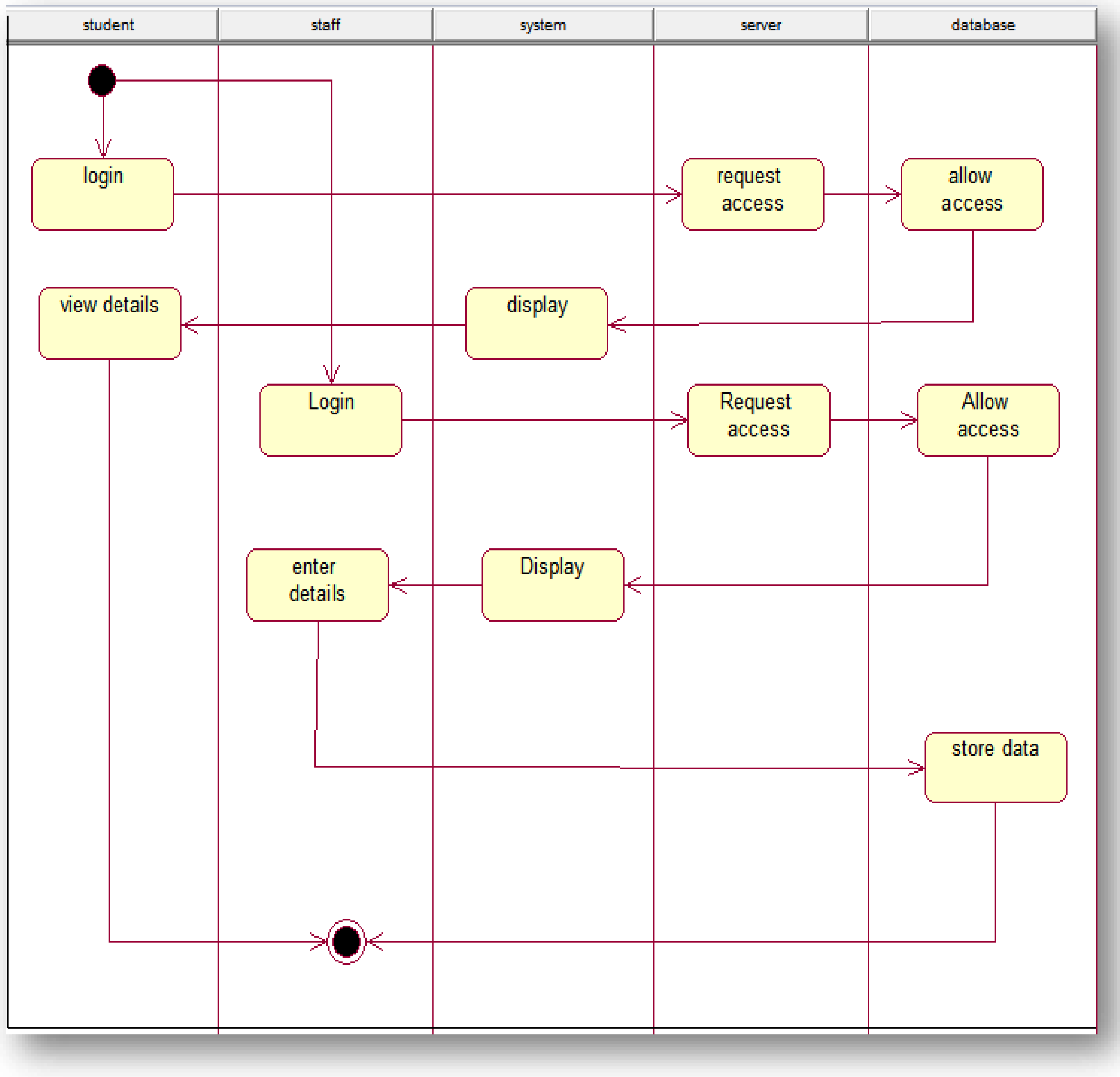
STATE CHART DIAGRAM

A State chart diagram is also called as state machine diagram. The state chart contains the states in the rectangular boxes and the states are indicated by the dot enclosed. The state chart diagram describes the behavior of the system. The state chart involves six stages such as login, enter mark, enter profile, view details, provide details, update details, store details and logout.



ACTIVITY DIAGRAM

Activity diagram are graphical representation of stepwise activities and actions with support for choice, interaction and concurrency. Here in the activity diagram the student login to the system and view the details of the student. The staff login to the system for entering the student details and update the details in the database. The final interaction is the DBA store the details of the student.



RESULT

Thus the various UML diagrams for student information system were drawn and code was generated successfully.

EX.NO:3

PASSPORT AUTOMATION SYSTEM

Date:

AIM:

To create an automated system to perform the Passport Process.

PROBLEM STATEMENT:

Passport Automation System is used in the effective dispatch of passport to all of the applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as name, address etc.,) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to the already existing information in the database.

SOFTWARE REQUIREMENT SPECIFICATION:

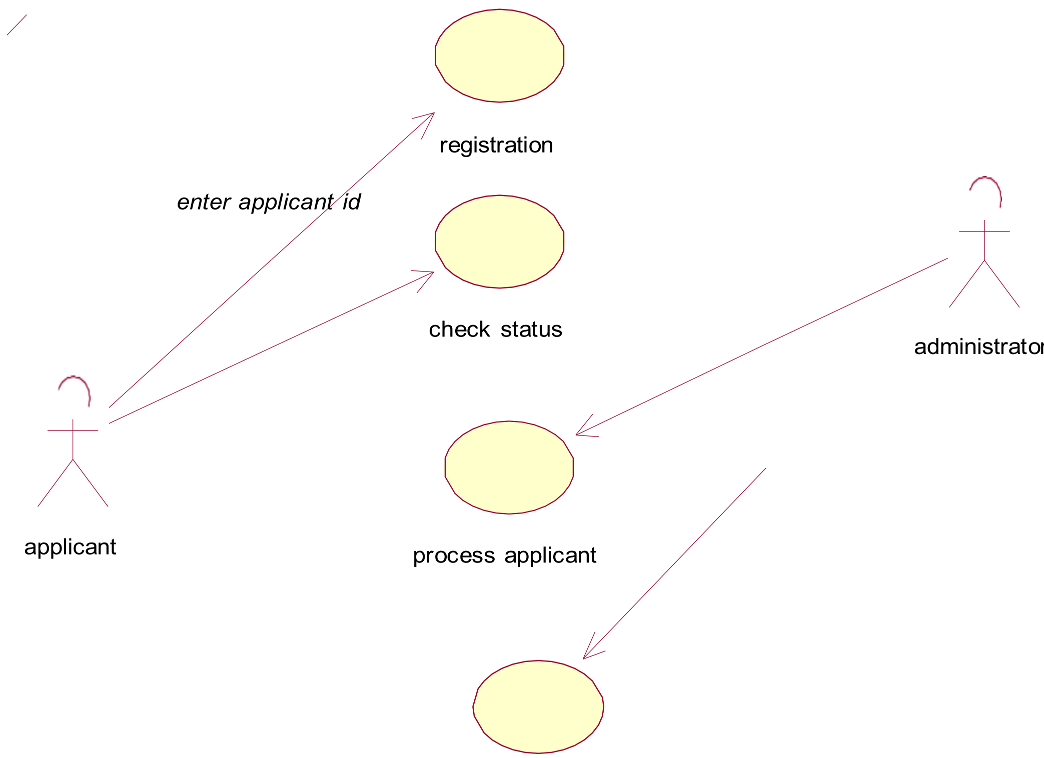
SOFTWARE INTERFACE

- **Front End Client -** The applicant and Administrator online interface is built using JSP and HTML. The Administrators 's local interface is built using Java.
- **Web Server -** Glassfish application server(Oracle Corporation).
- **Back End -** Oracle database.

HARDWARE INTERFACE

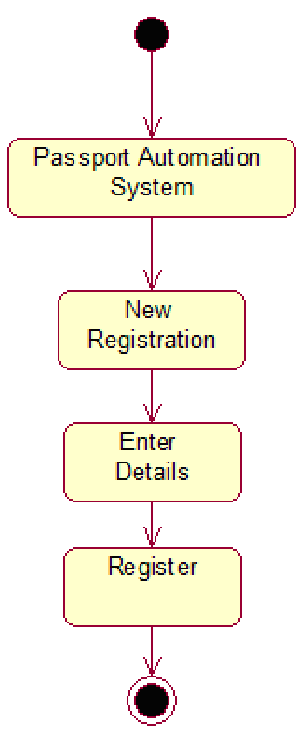
The server is directly connected to the client systems. The client systems have access to the database in the server.

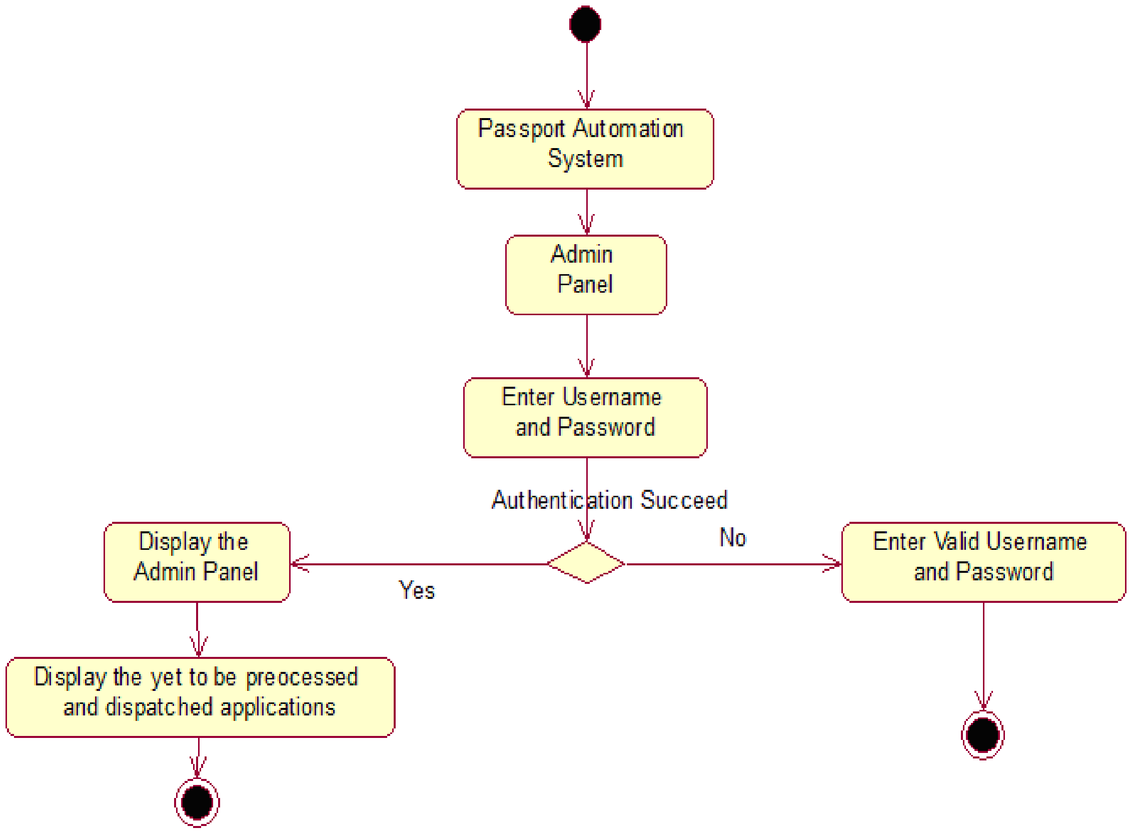
USE CASE DIAGRAM :

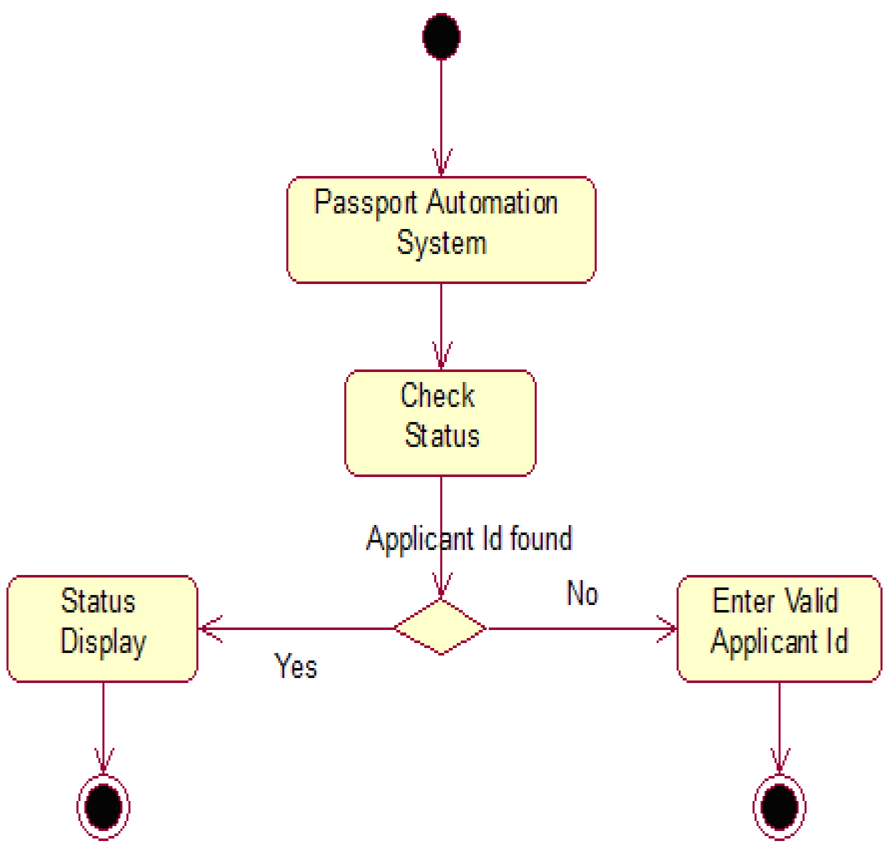


dispatch passport

ACTIVITY DIAGRAM:




Passport Automation System
Admin Panel
Enter Username and Password
Authentication Succeed
Yes
No
Display the Admin Panel
Enter Valid Username and Password
Display the yet to be preocessed and dispatched applications

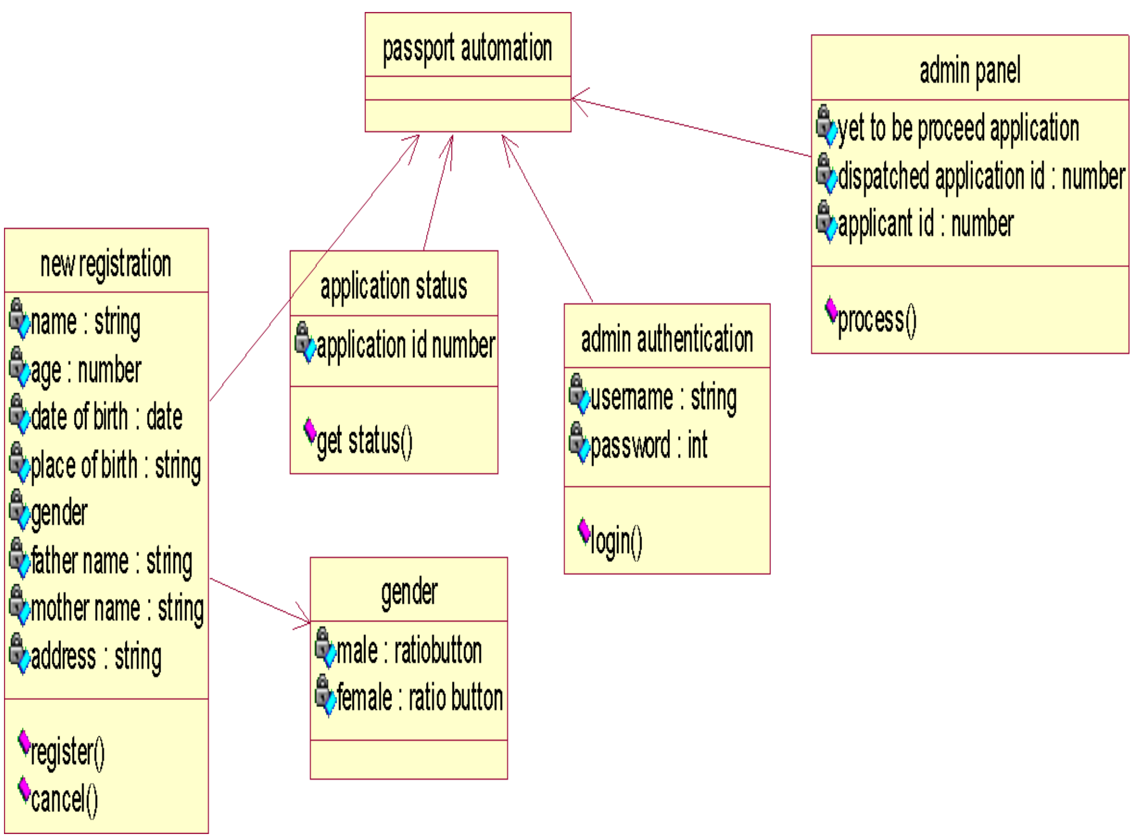

Passport Automation System
Check Status
Applicant Id found
Yes
No
Status Display
Enter Valid Applicant Id

CLASS DIAGRAM:

The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.

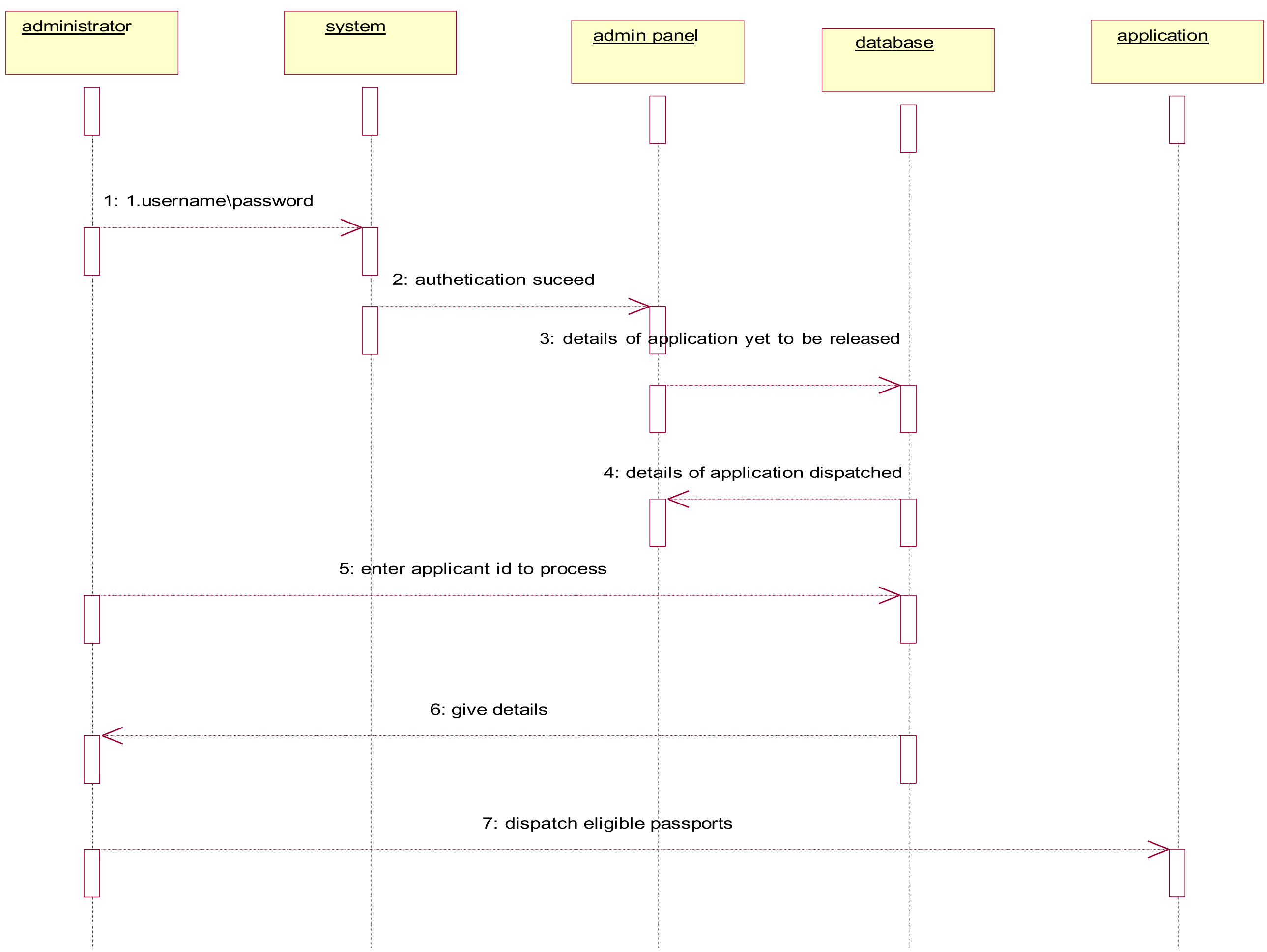
The Passport Automation system class diagram consists of four classes Passport Automation System

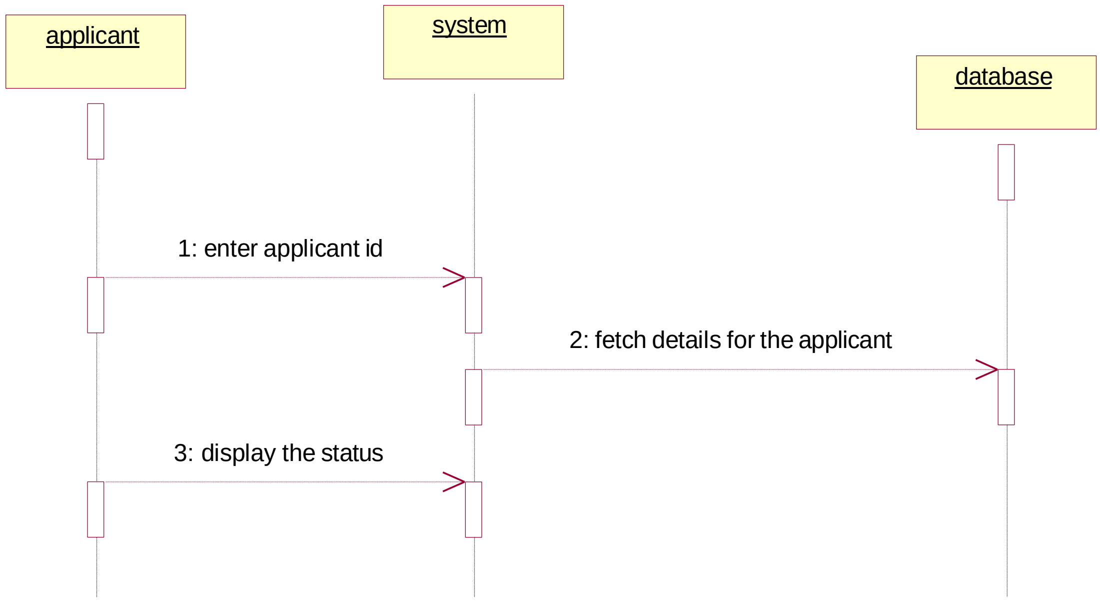
1. New registration
2. Gender
3. Application Status
4. Admin authentication
5. Admin Panel

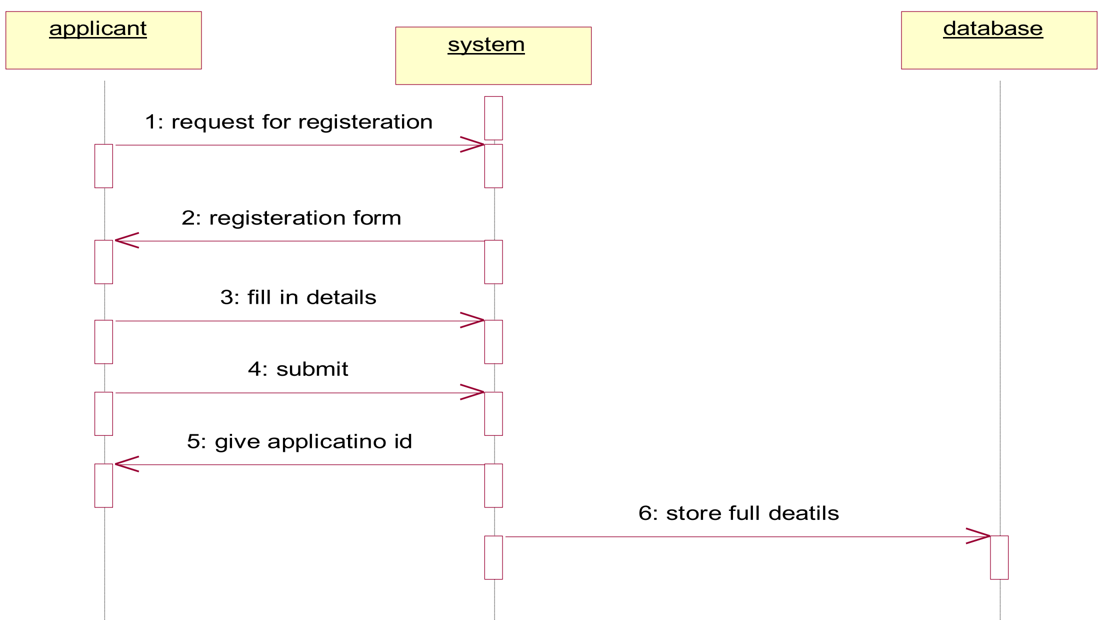


INTERACTION DIAGRAM AND SEQUENCE DIAGRAM:

A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another, in which the “form” object is requesting an operation be performed by the “to” object. The sequence diagram for each USE-CASE that exists when a user administrator, check status and new registration about passport automation system are given.

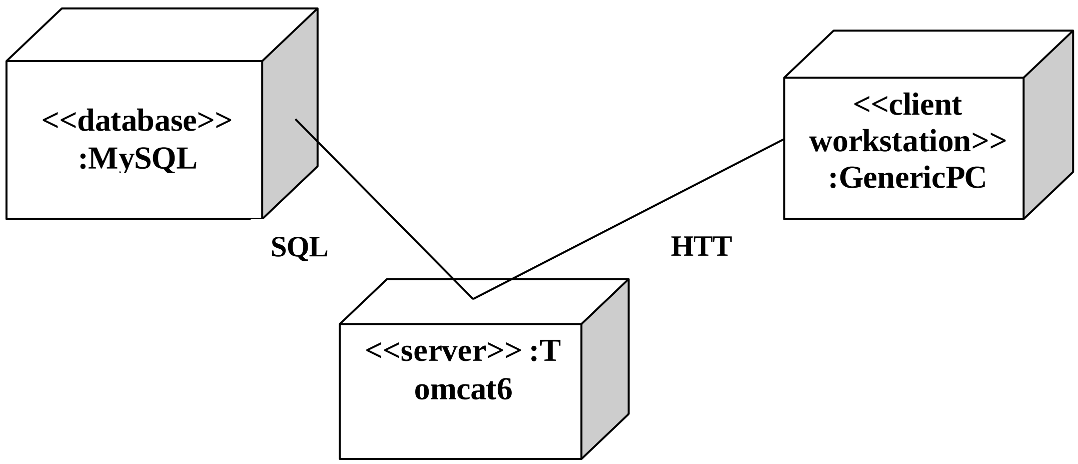


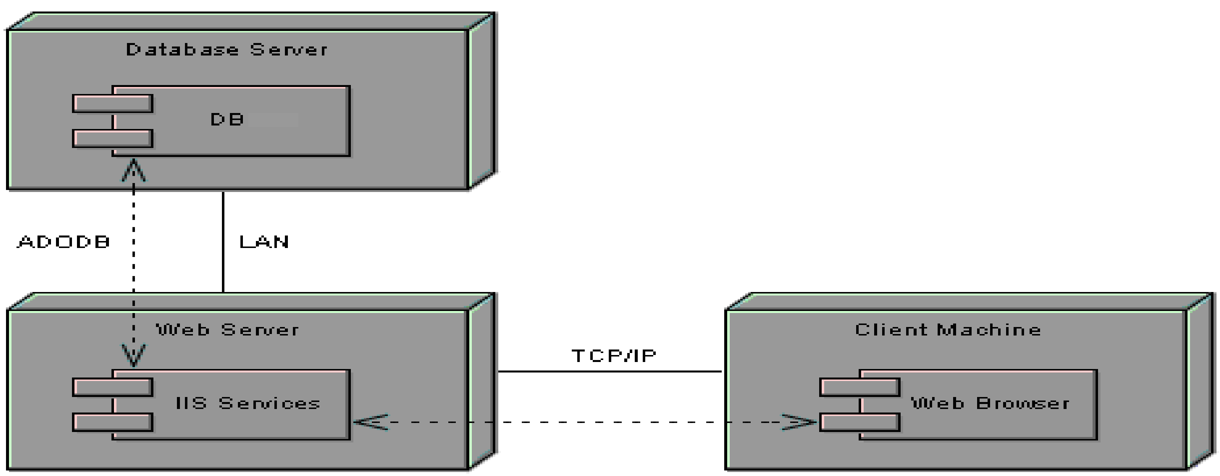

applicant
system
database
1: enter applicant id
2: fetch details for the applicant
3: display the status


applicant
system
database
1: request for registeration
2: registeration form
3: fill in details
4: submit
5: give applicatino id
6: store full deatils

DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM:

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed. Component diagrams are used to visualize the organization and relationship among components in system.





RESULT:

Thus the mini project for passport automation system has been successfully executed and codes are generated.

EX.NO: 4

BOOK BANK SYSTEM

Date:

AIM:

To create a system to perform book bank operation.

PROBLEM STATEMENT:

A Book Bank lends books and magazines to member, who is registered in the system. Also it handles the purchase of new titles for the Book Bank. Popular titles are brought into multiple copies. Old books and magazines are removed when they are out or date or poor in condition. A member can reserve a book or magazine that is not currently available in the book bank, so that when it is returned or purchased by the book bank, that person is notified. The book bank can easily create, replace and delete information about the tiles, members, loans and reservations from the system.

SOFTWARE REQUIREMENTS SPECIFICATION:

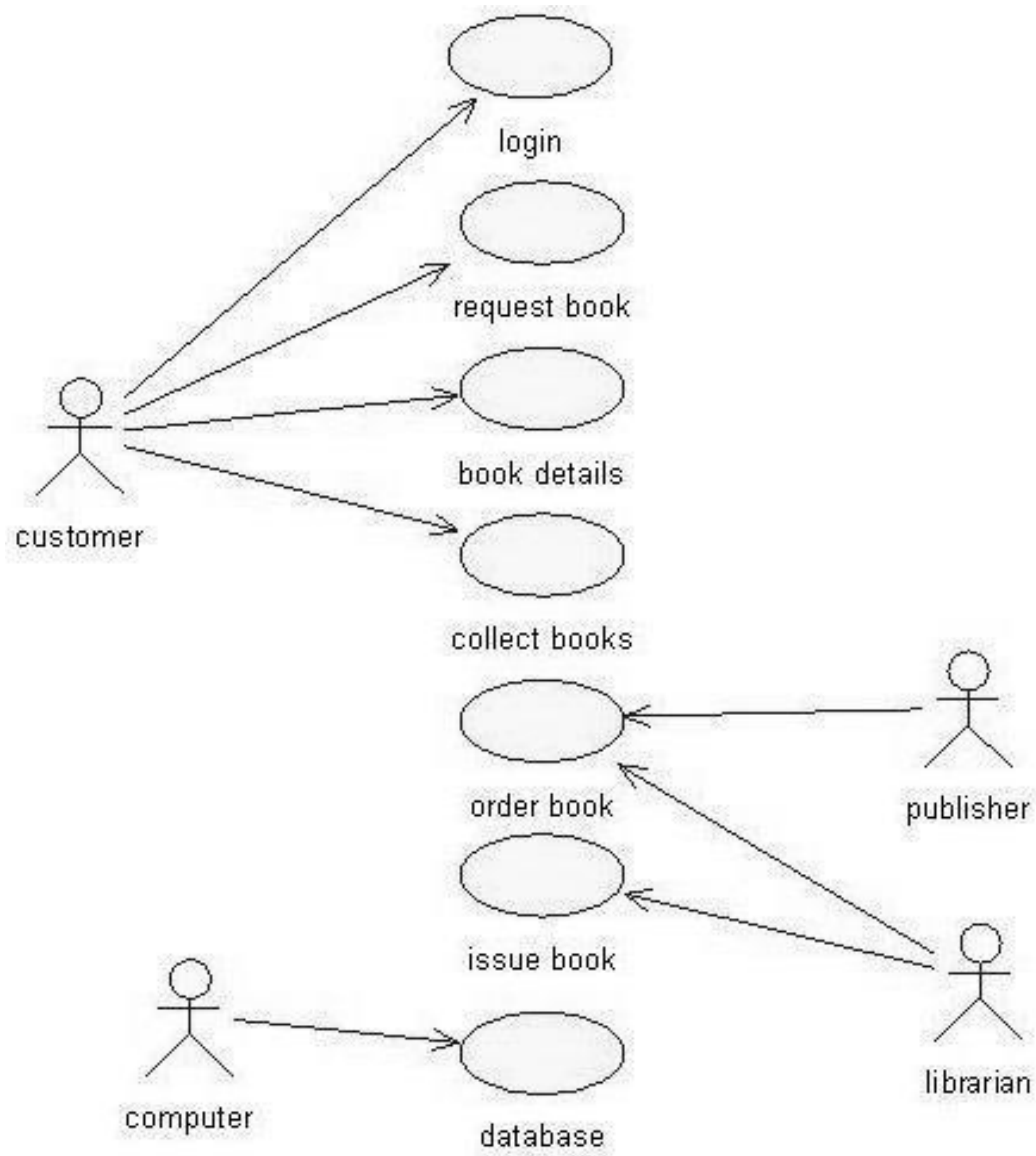
2.1 SOFTWARE INTERFACE

- **Front End Client -** The Student and Librarian online interface is built using JSP and HTML. The Librarians local interface is built using Java.
- **Web Server -** Glassfish application server (Oracle Corporation).
- **Back End -** Oracle database

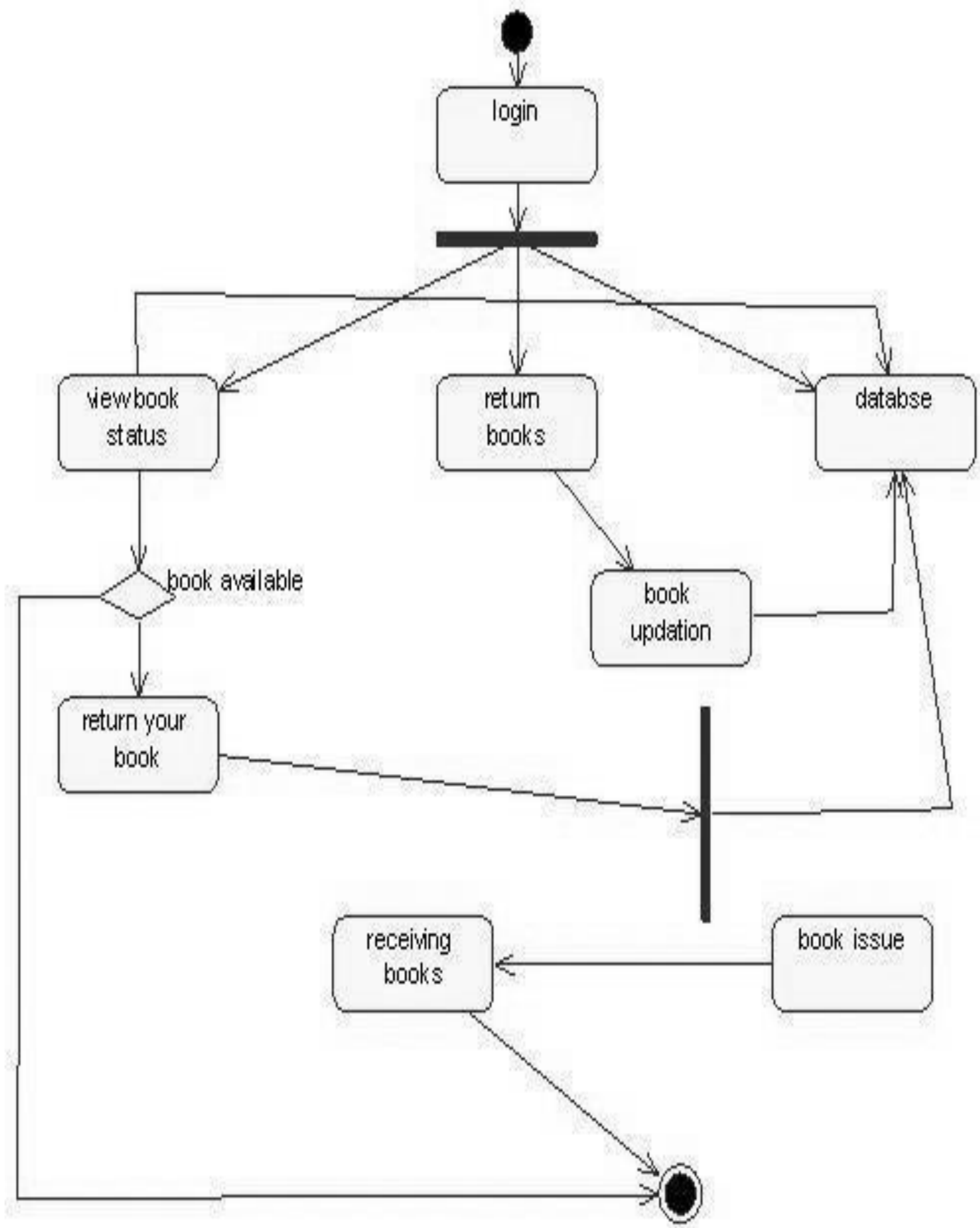
2.2 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

USE-CASE DIAGRAM:

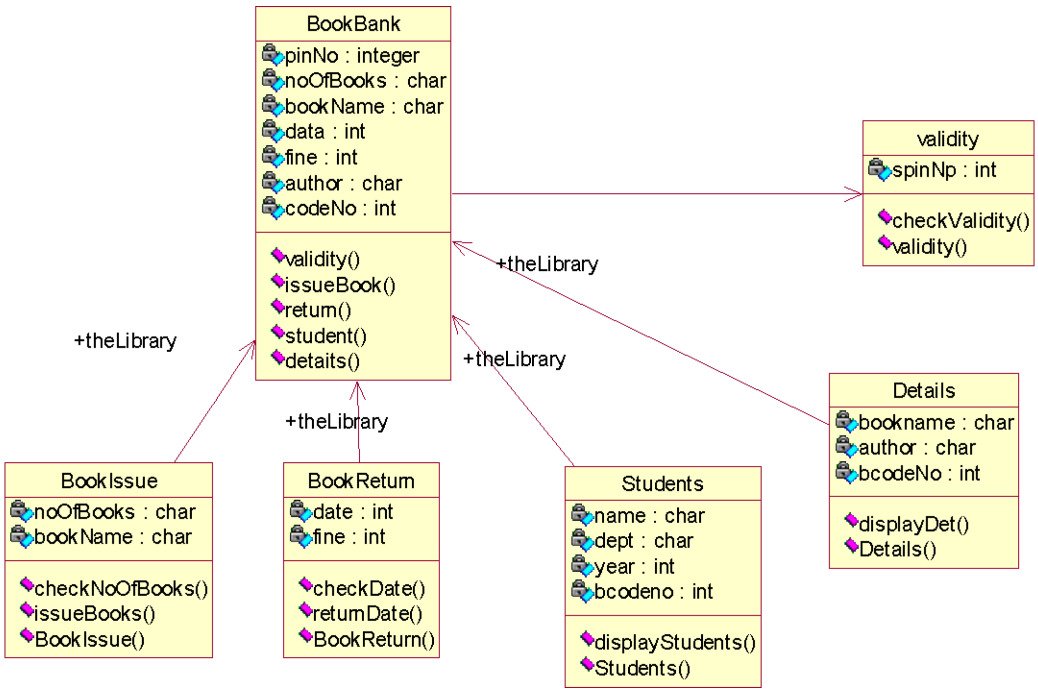


ACTIVITY DIAGRAM:



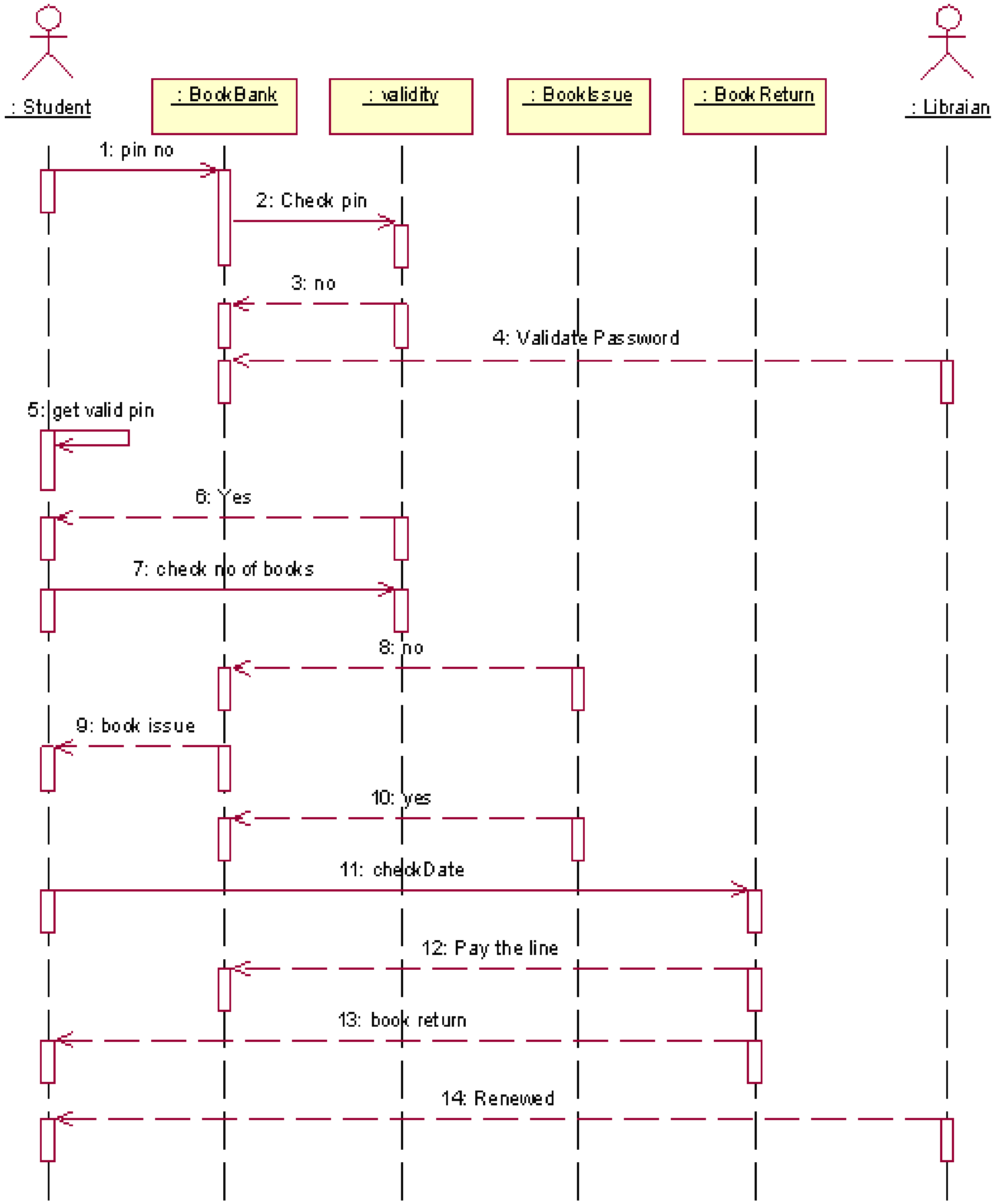
CLASS DIAGRAM:

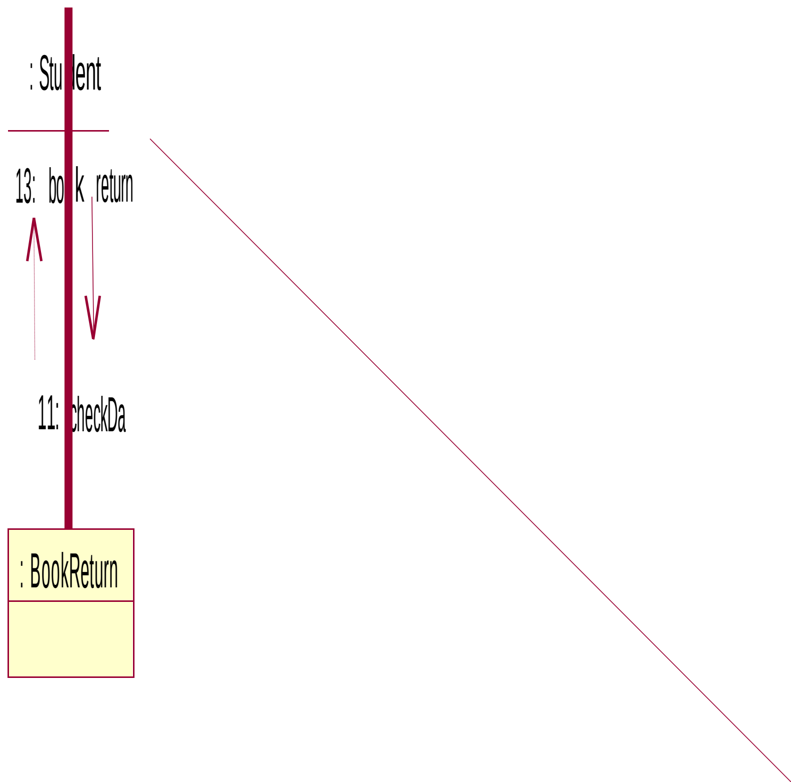
The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.



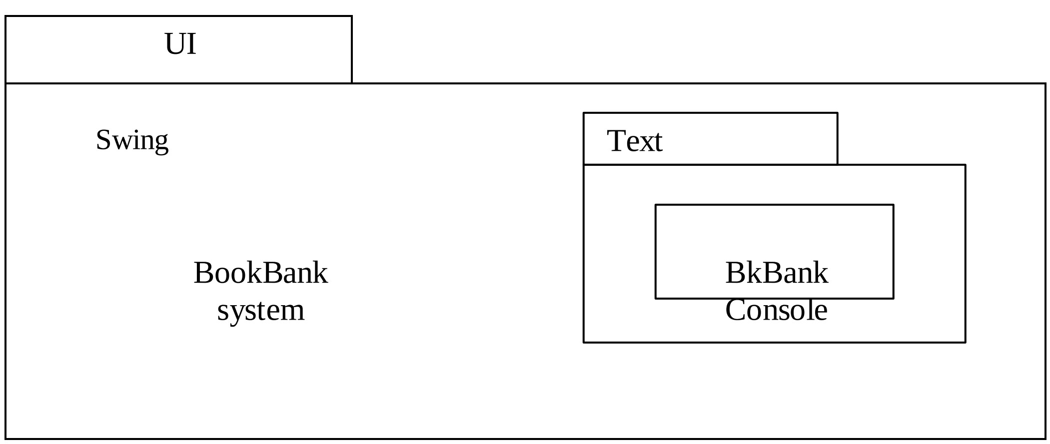
SEQUENCE DIAGRAM:

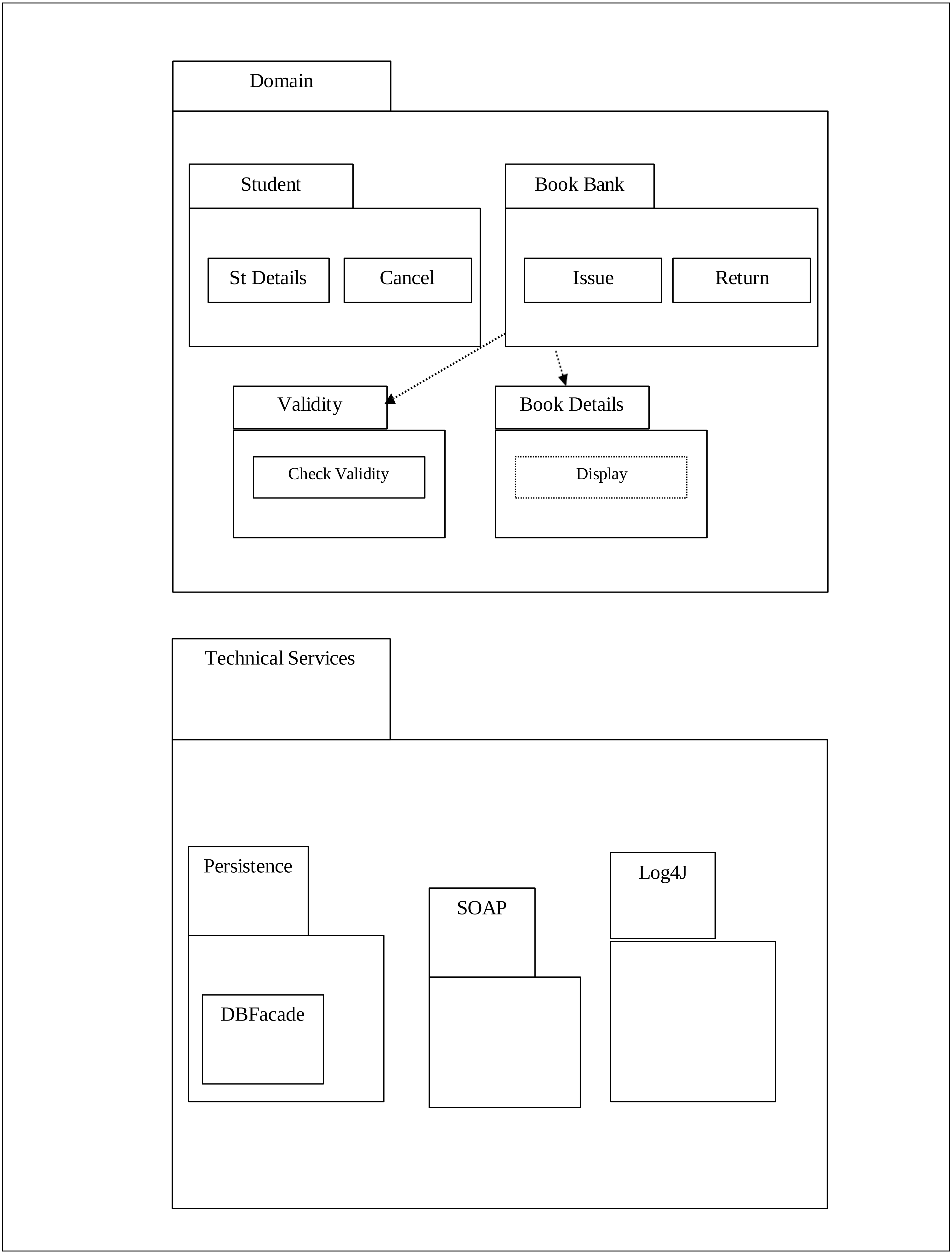
A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another, in which the “form” object is requesting an operation be performed by the “to” object. The “to” object performs the operation using a method that the class contains.It is also represented by the order in which things occur and how the objects in the system send message to one another. The diagrams show the pin no is entered and check the pin .Get no and validate password check the condition based on condition book issue and return are done. Pay the online and renewed.





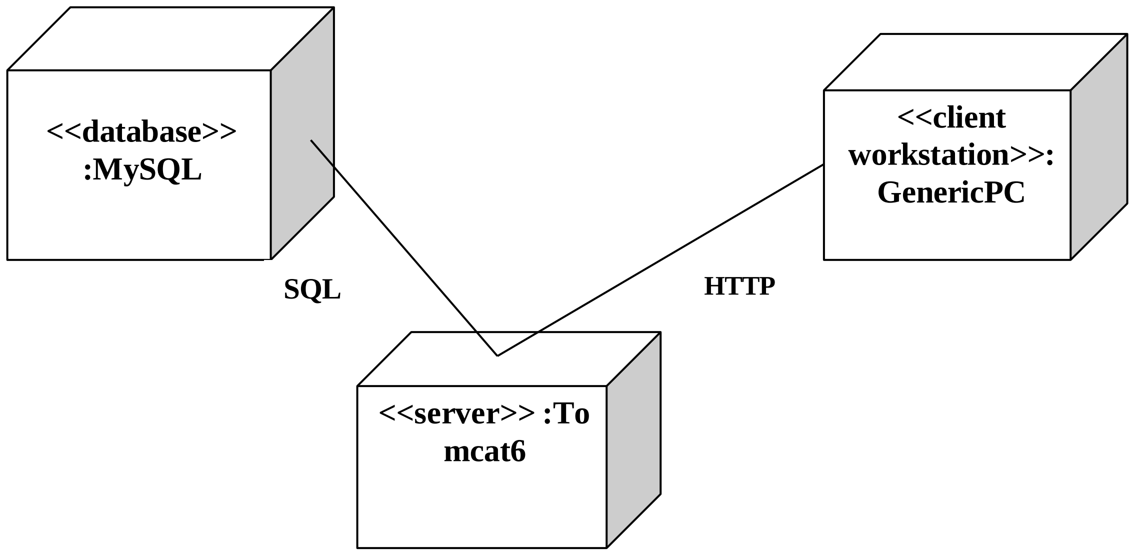
PARTIAL LAYERD LOGICAL ARCHITECTURE DIAGRAM:




Domain
Student
St Details
Cancel
Book Bank
Issue
Return
Validity
Check Validity
Book Details
Display
Technical Services
Persistence
DBFacade
SOAP
Log4J

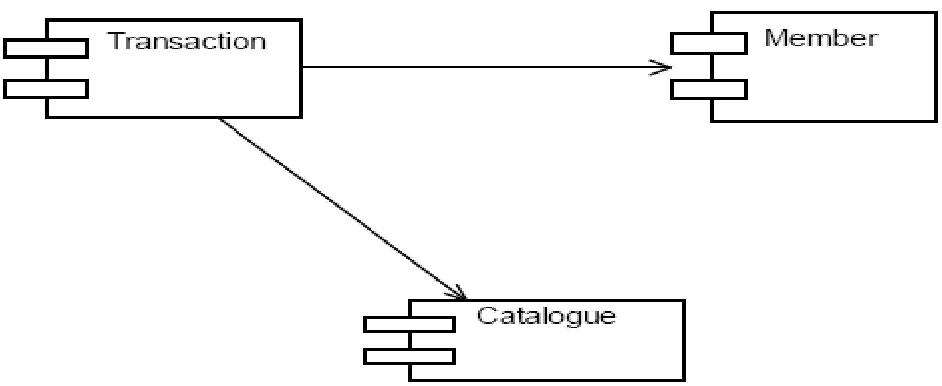
DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



COMPONENT DIAGRAM

Component diagrams are used to visualize the organization and relationships



RESULT:

Thus the mini project for Book Bank System has been successfully executed and codes are generated.

EX.NO:5

EXAM REGISTRATION SYSTEM

Date:

AIM:

To create a system to perform the Exam Registration system

PROBLEM STATEMENT:

Exam Registration system.is used in the effective dispatch of registration form to all of the students. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as name, reg.no etc.,) filled by the student whose testament is verified for its genuineness by the Exam Registration System with respect to the already existing information in the database.

SOFTWARE REQUIREMENT SPECIFICATION:

2.1 SOFTWARE INTERFACE

• **Front End Client -** The student and Controller online interface is built using
JSP and HTML. The Exam Controller's local interface is built using Java.

• **Web Server -** Glassfish application server(SQlCorporation).

• **Back End -** SQL database.

2.2 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have
access to the database in the server.

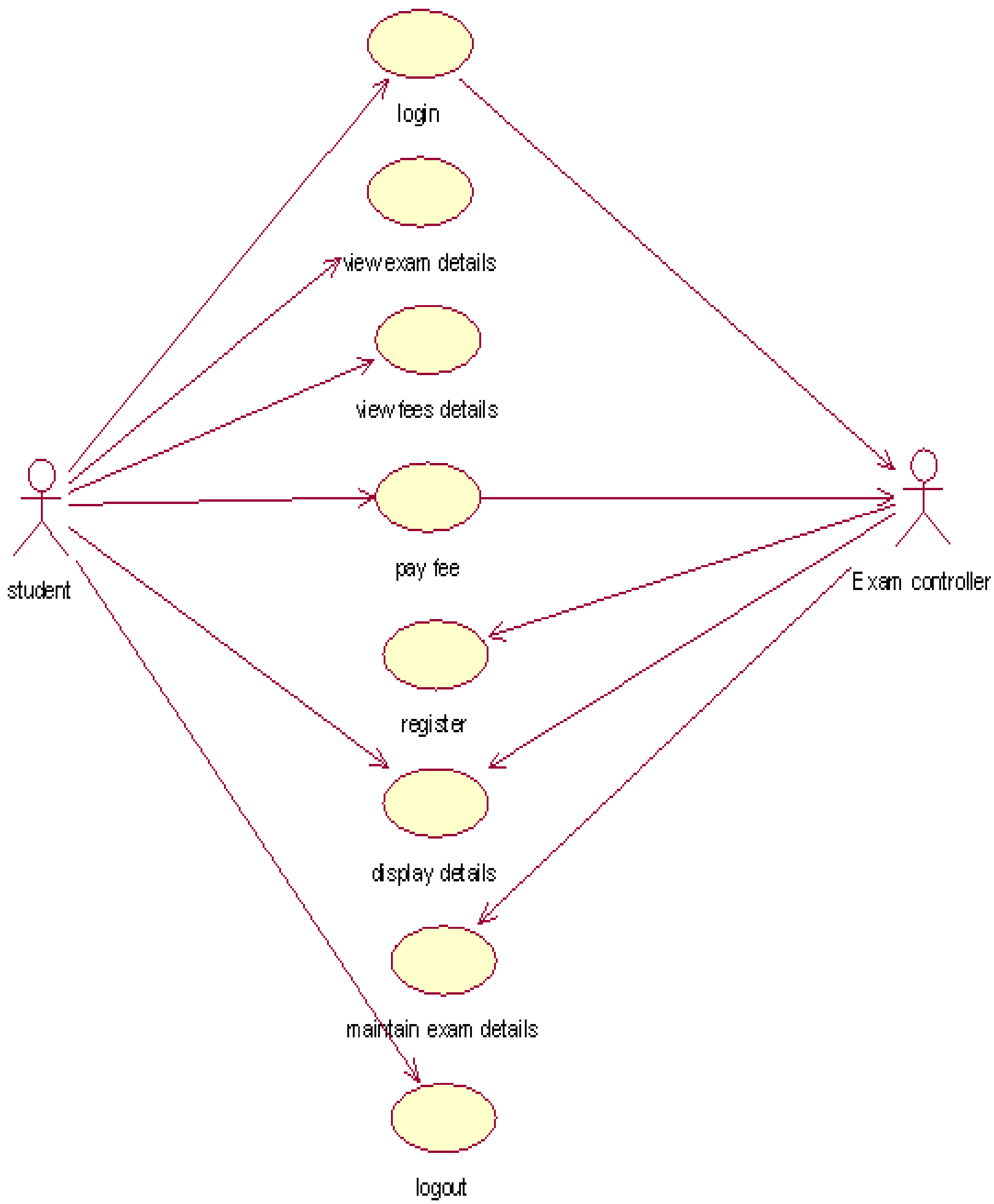
USECASE DIAGRAM:

The Exam Registration use cases in our system are:

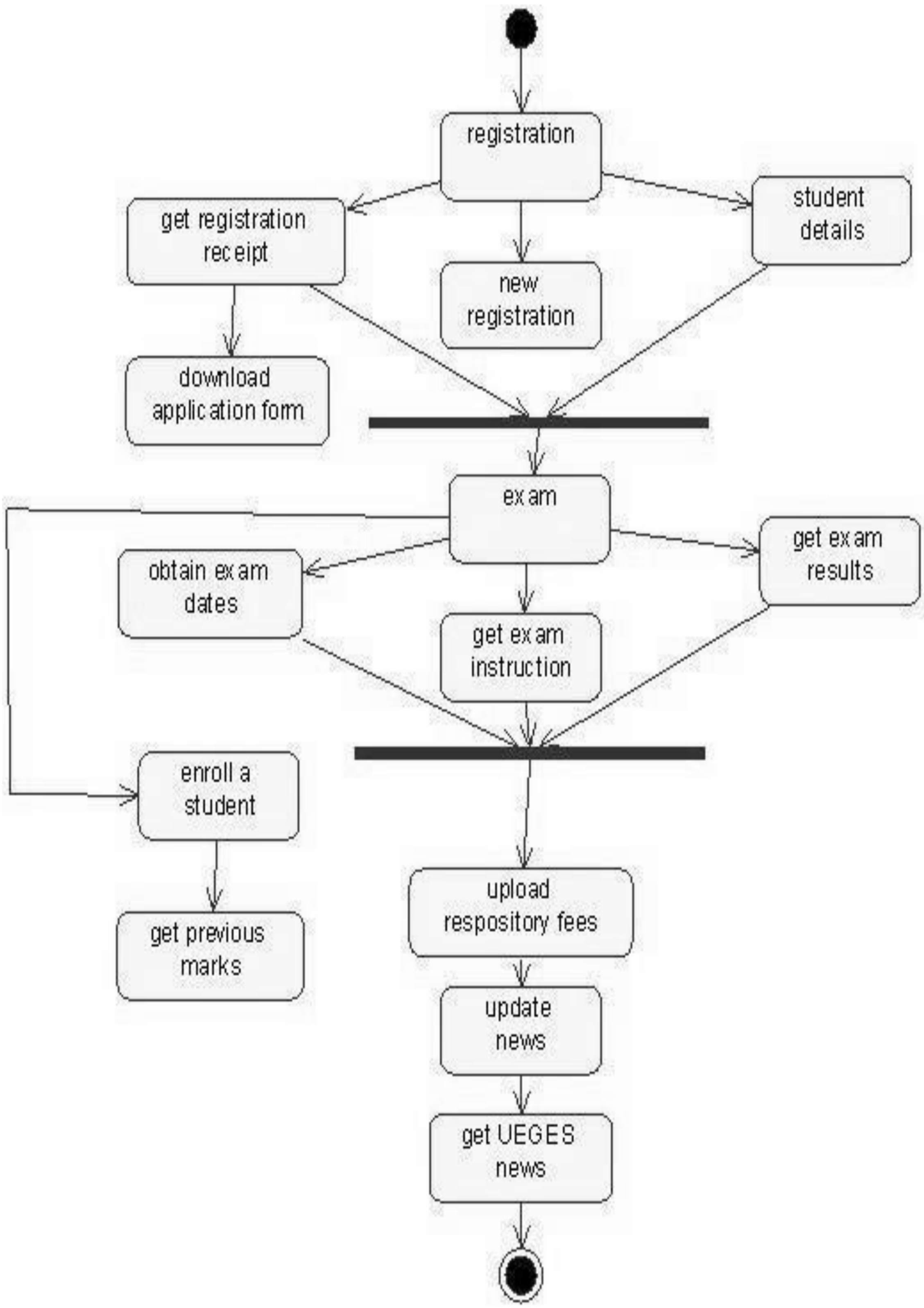
1. Login
2. View exam details
3. View fees details

4. Pay fee
5. Display details
6. Logout

USECASE DIAGRAM :

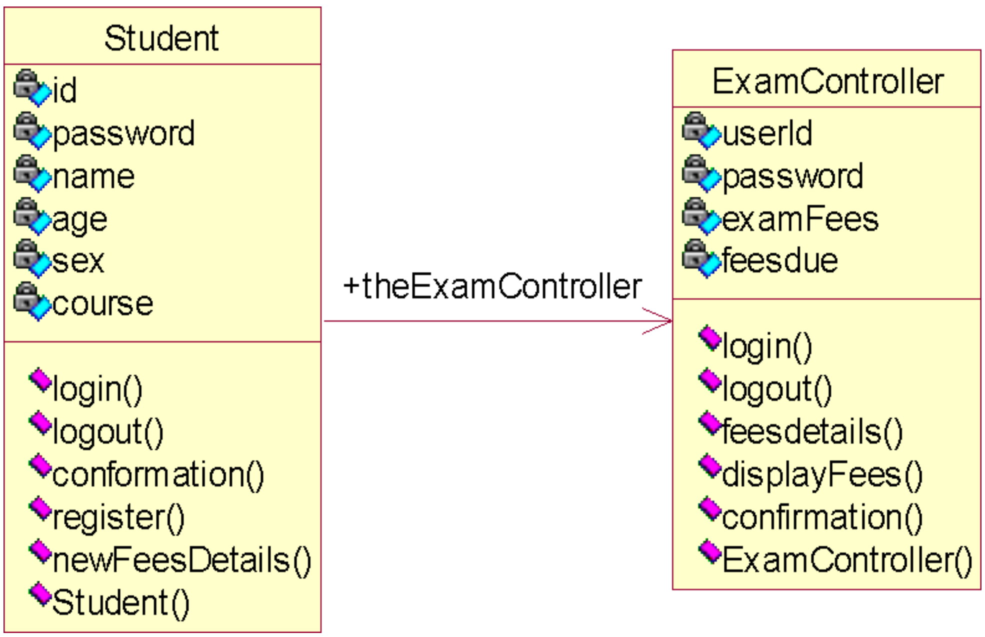


ACTIVITY DIAGRAM:



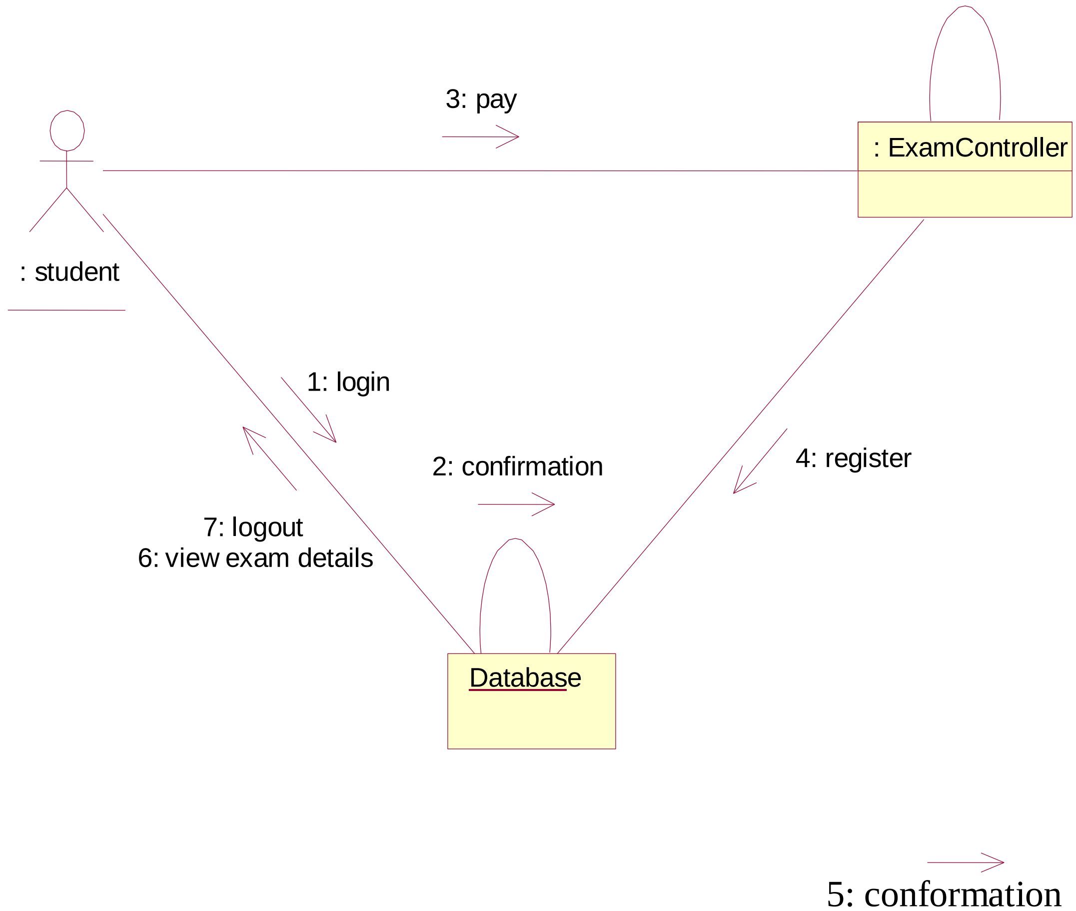
CLASS DIAGRAM:

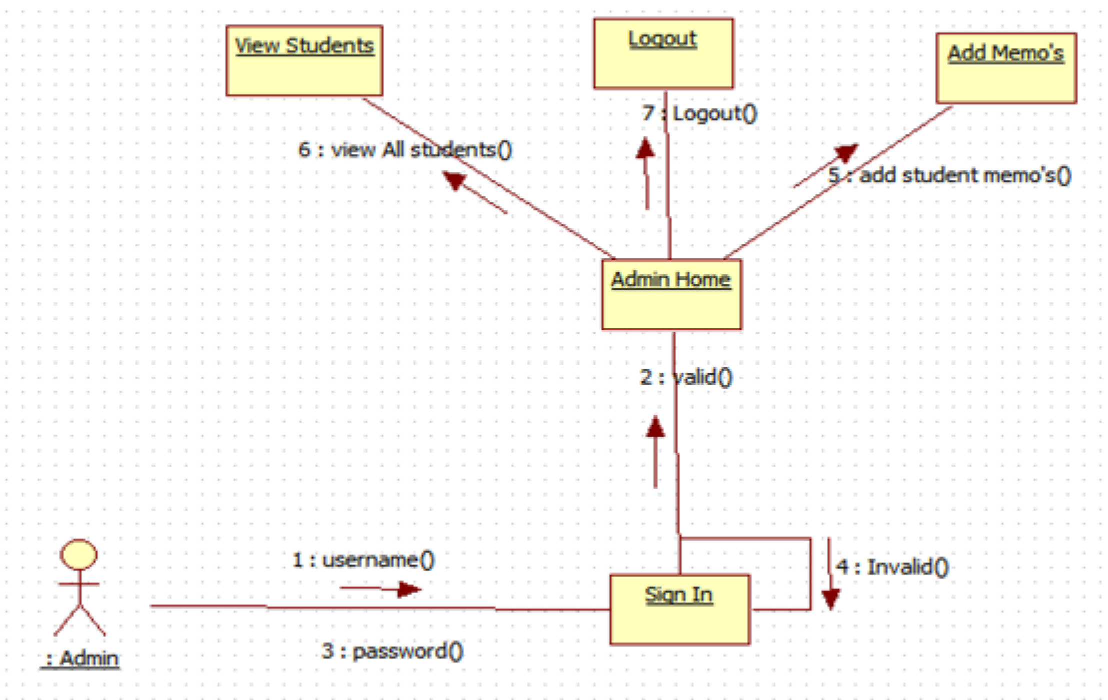
The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.



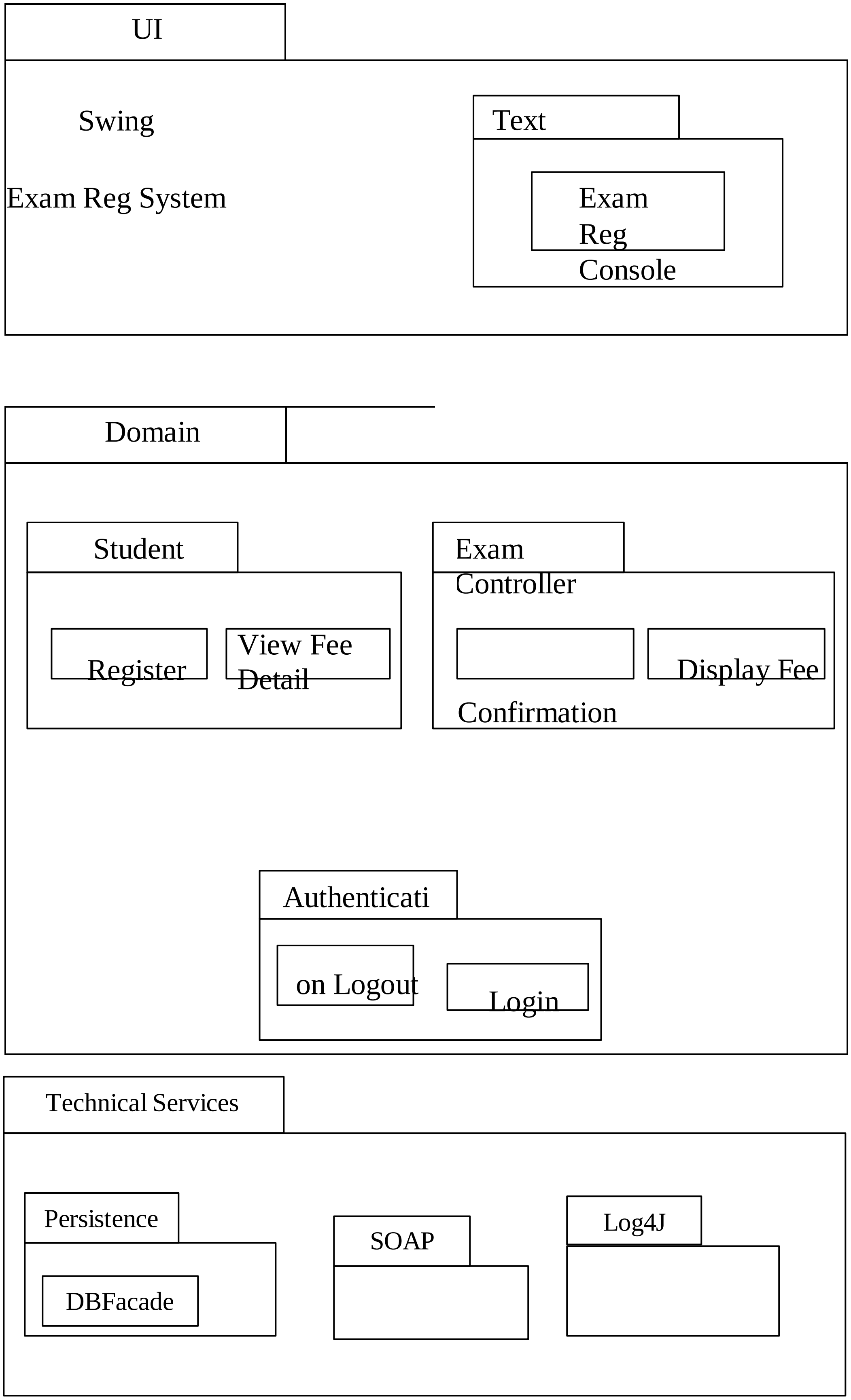
INTERACTION DIAGRAM:

A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another


3: pay
: ExamController
: student
1: login
2: confirmation
4: register
7: logout
6: view exam details
Database
5: conformation

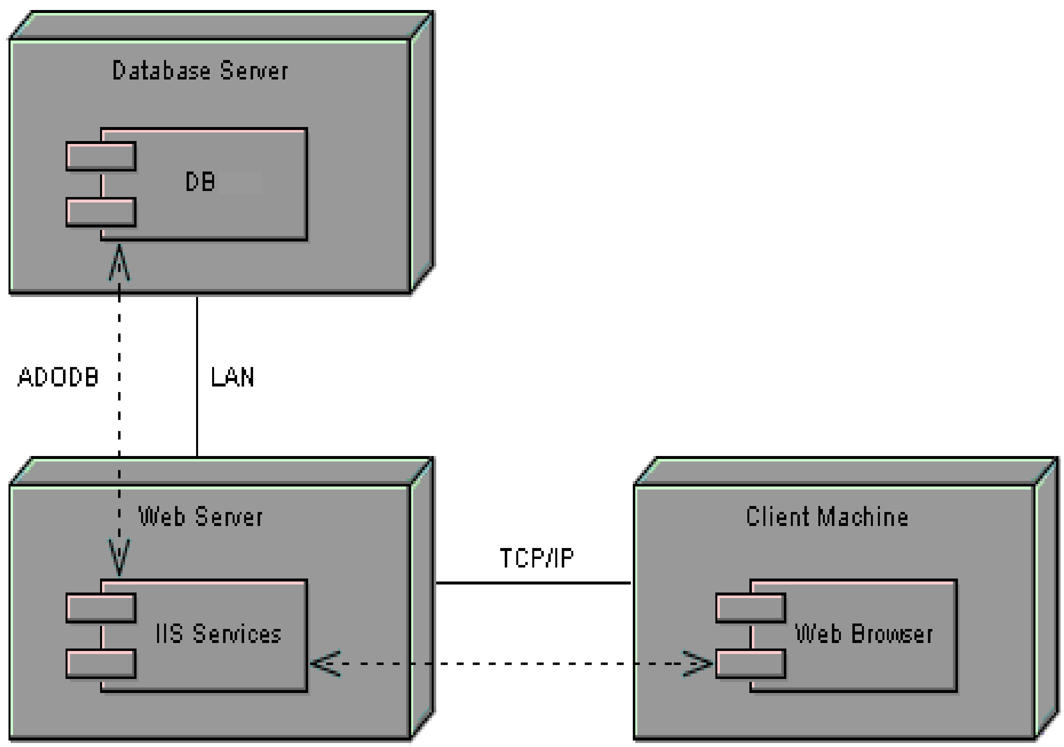

View Students
Logout
Add Memo's
7 : Logout()
6 : view All students()
5 : add student memo's()
Admin Home
2 : valid()
1 : username()
4 : Invalid()
Sign In
: Admin
3 : password()

PARTIAL LAYERD LOGICAL ARCHITECTURE DIAGRAM:



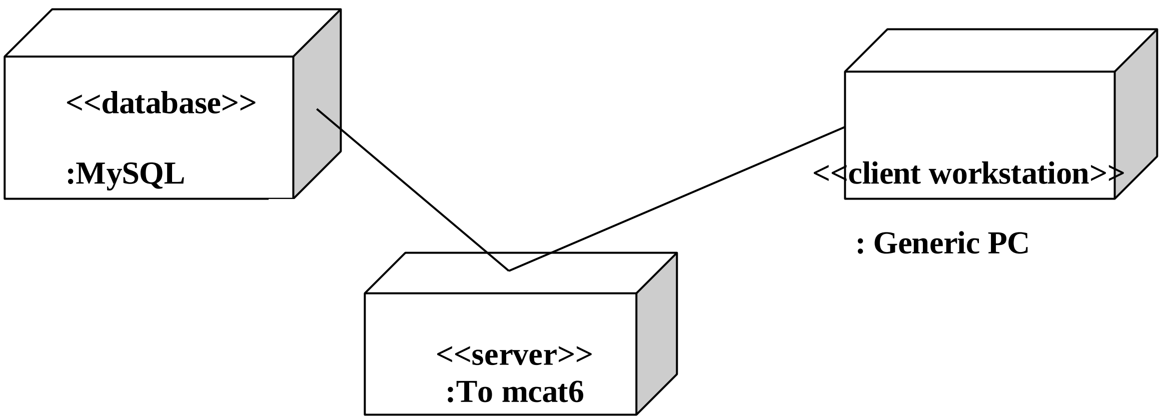
COMPONENT DIAGRAM

Component diagrams are used to visualize the organization and relationships among components in a system.



DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



RESULT:

Thus the mini project for Exam Registration system has been successfully executed and codes are generated.

EX.NO: 6

STOCK MAINTENANCE

Date:

AIM :

To create a system to perform the Stock maintenance

PROBLEMSTATEMENT

The stock maintenance system must take care of sales information of the company and must analyze the potential of the trade. It maintains the number of items that are added or removed.The sales person initiates this Use case. The sales person is allowed to update information and view the database.

SOFTWAREREQUIREMENTSPECIFICATION

PURPOSE

The entire process of Stock maintenance is done in a manual manner Considering the fact that the number of customers for purchase is increasing every year, a maintenance system is essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process.

SCOPE

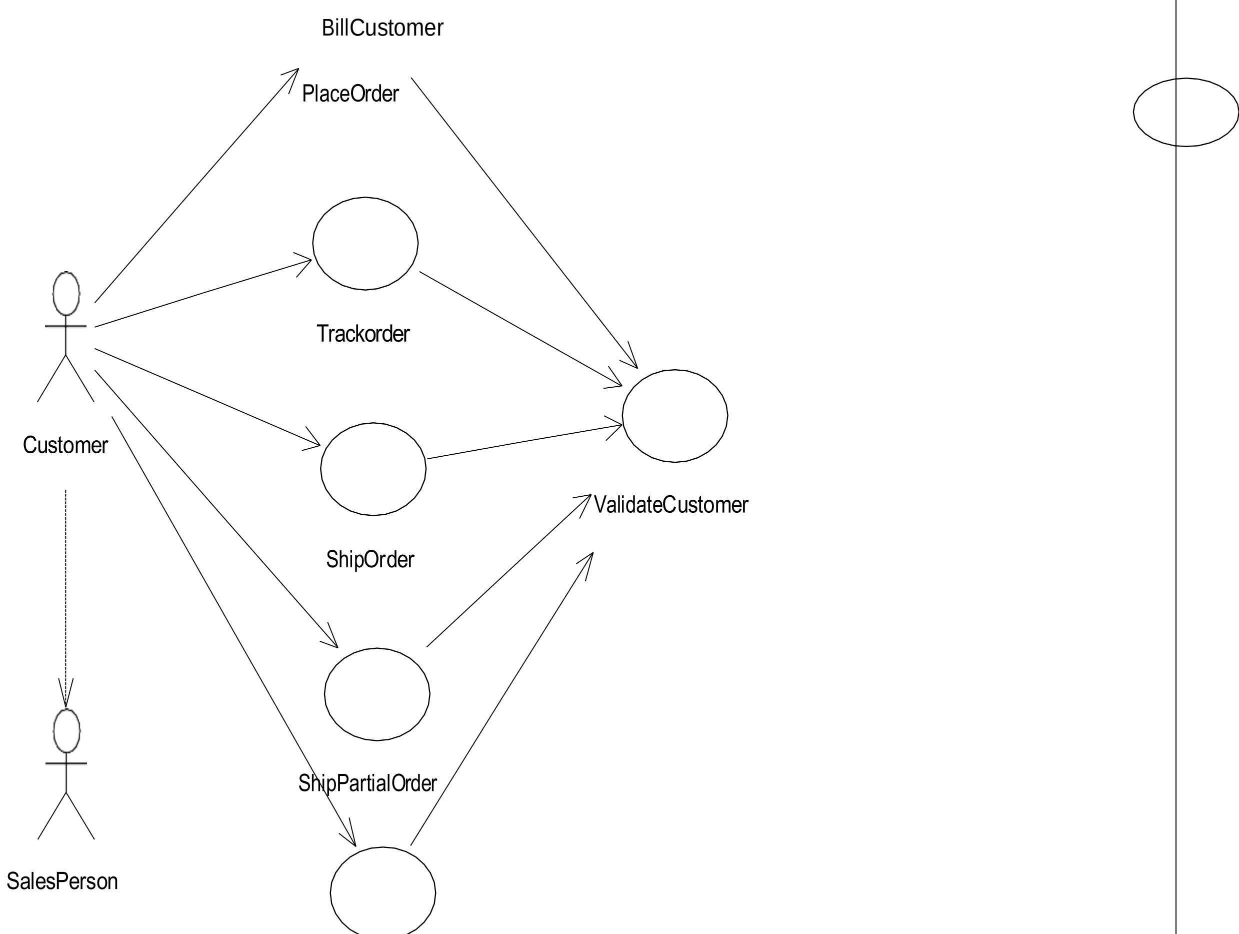
- The System provides an interface to the customer where they can fill in orders for the item needed.
- The sales person is concerned with the issue of items and can use this system.
- Provide a communication platform between the customer and the sales person.

TOOLS TO BE USED

- Eclipse IDE (Integrated Development Environment)
- Rational Rose tool(for developing UML Patterns)

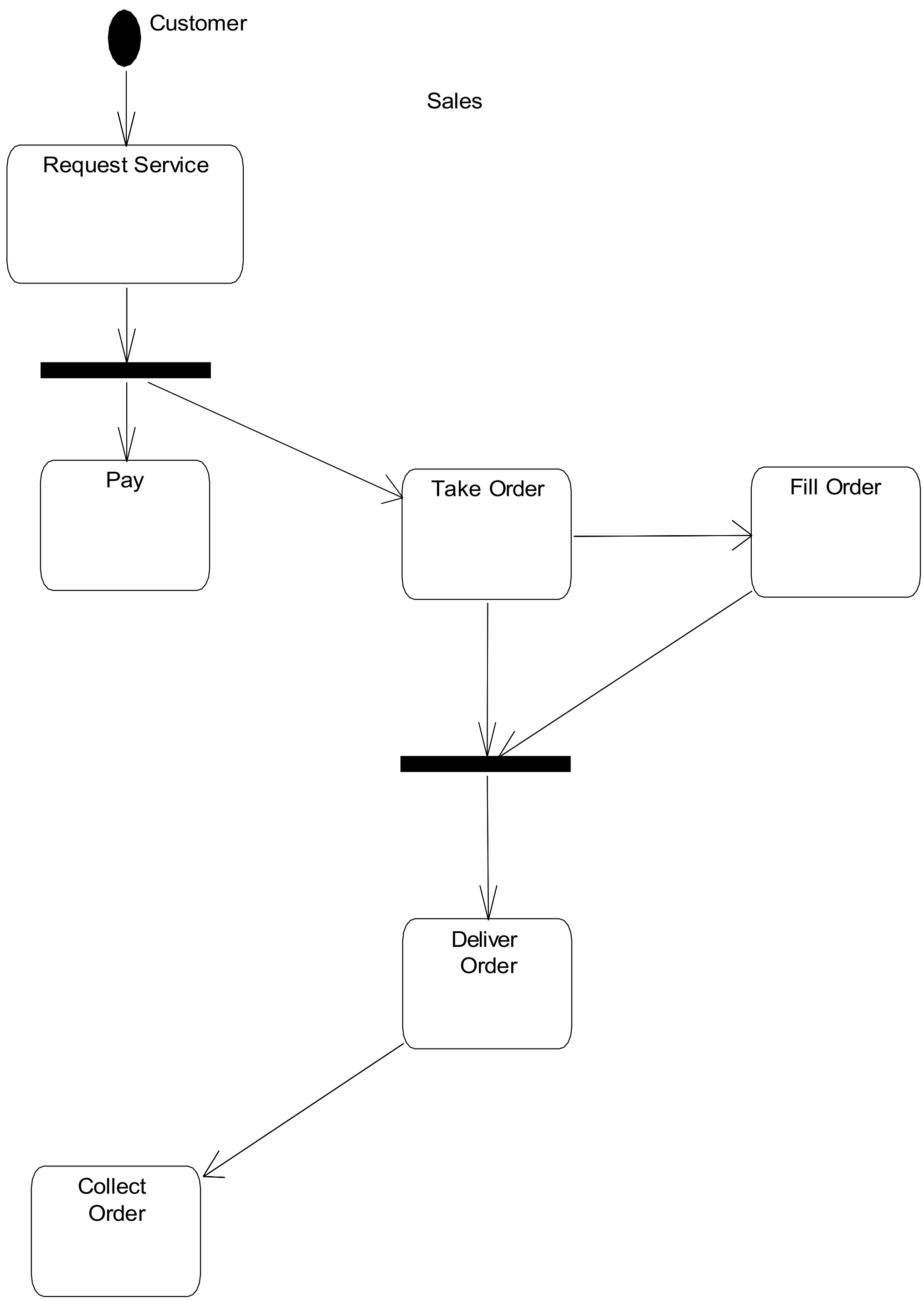
USE CASE DIAGRAM

The functionality of a system can be described in a number of different use-cases, each of which represents a specific flow of events in a system. It is a graph of actors, a set of use-cases enclosed in a boundary, communication, associations between the actors and the use-cases, and generalization among the use-cases



ACTIVITY DIAGRAM

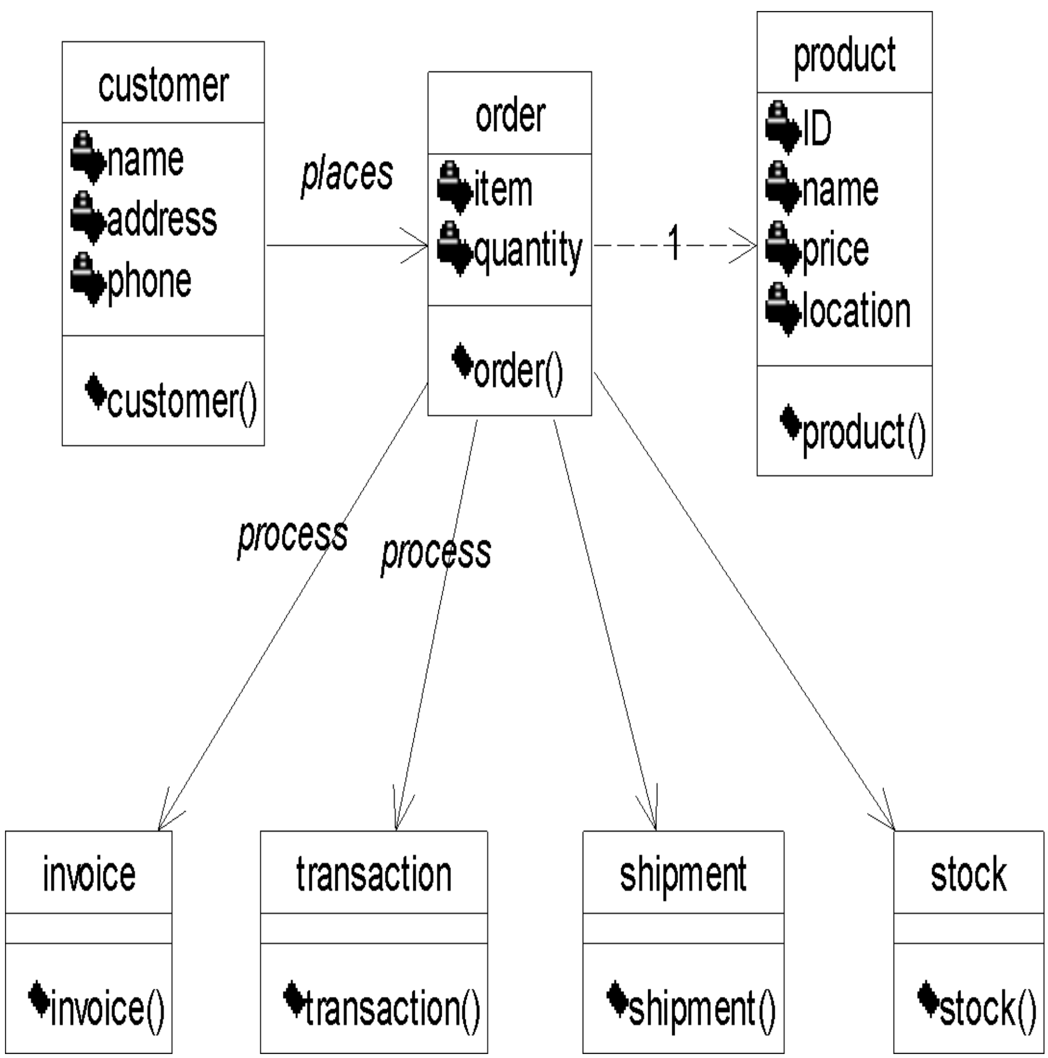
It shows organization and their dependence among the set of components. These diagrams are particularly useful in connection with workflow and in describing behavior at has a lot of parallel processing. An activity is a state of doing something: either a real-world process, or the execution of a software routine.



CLASS DIAGRAM

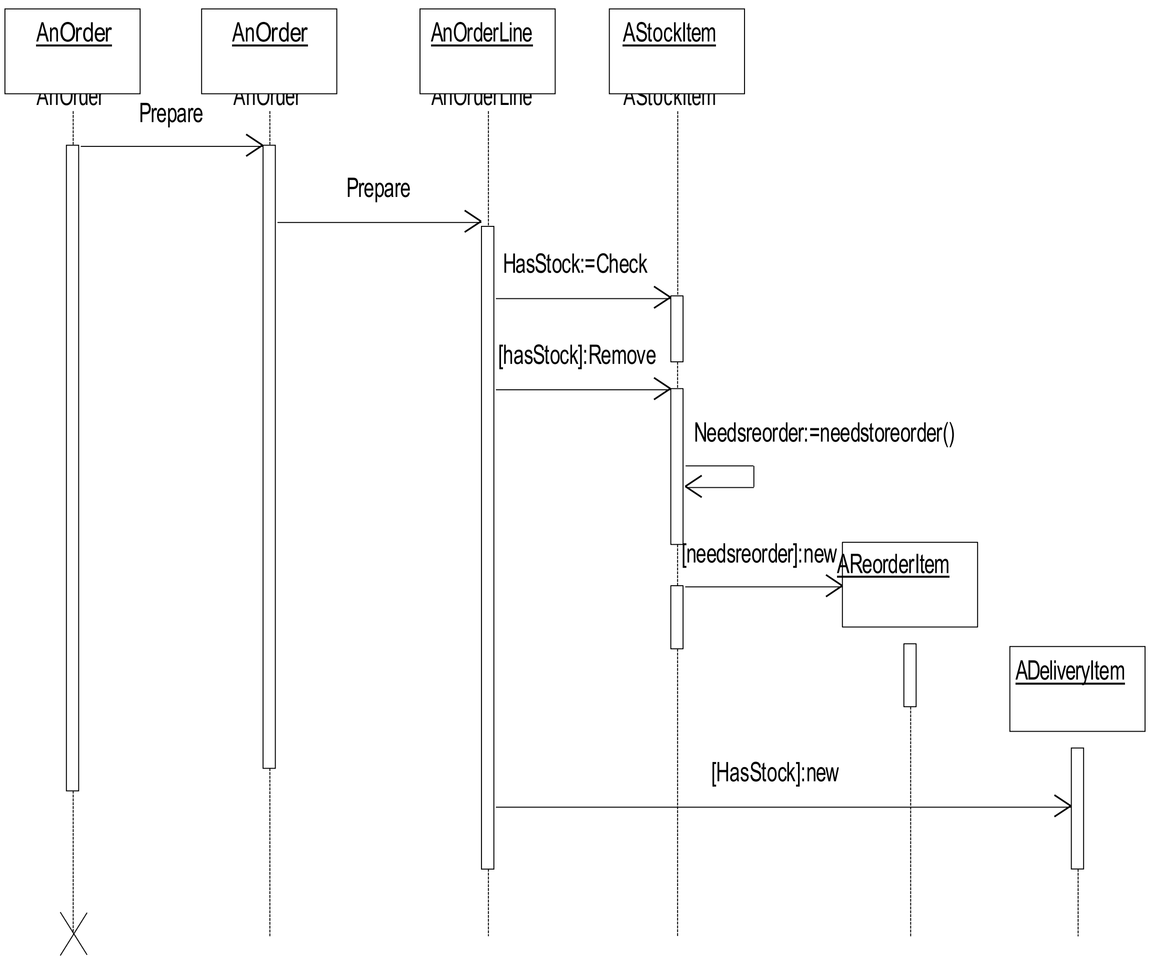
Description:

- A class diagram describes the type of objects in system and various kinds of relationships that exists among them.
- Class diagrams and collaboration diagrams are alternate representations of object models.



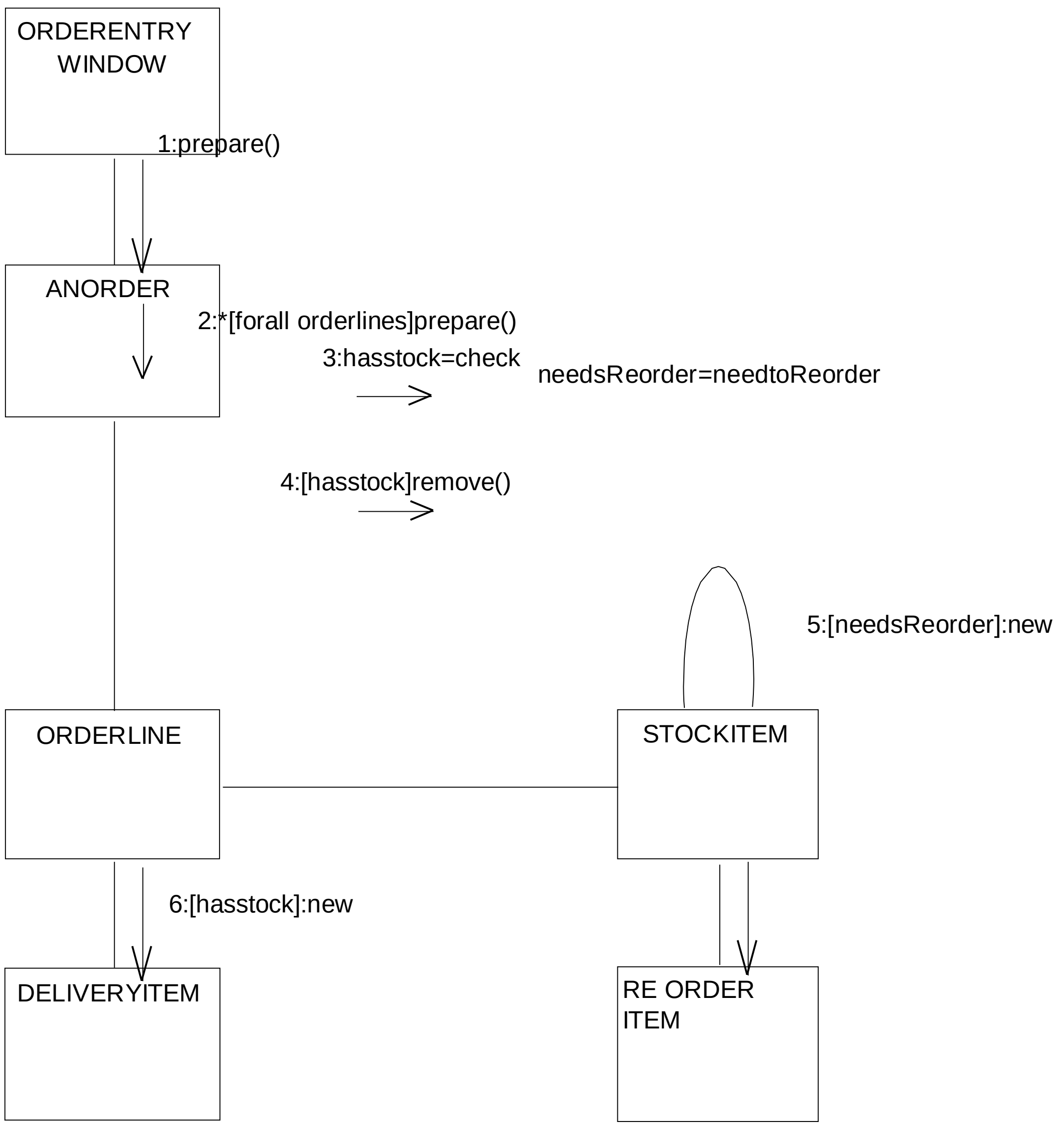
UML INTERACTION DIAGRAMS

It is the combination of sequence and collaboration diagram. It is used to depict the flow of events in the system over a timeline. The interaction diagram is a dynamic model which shows how the system behaves during dynamic execution.

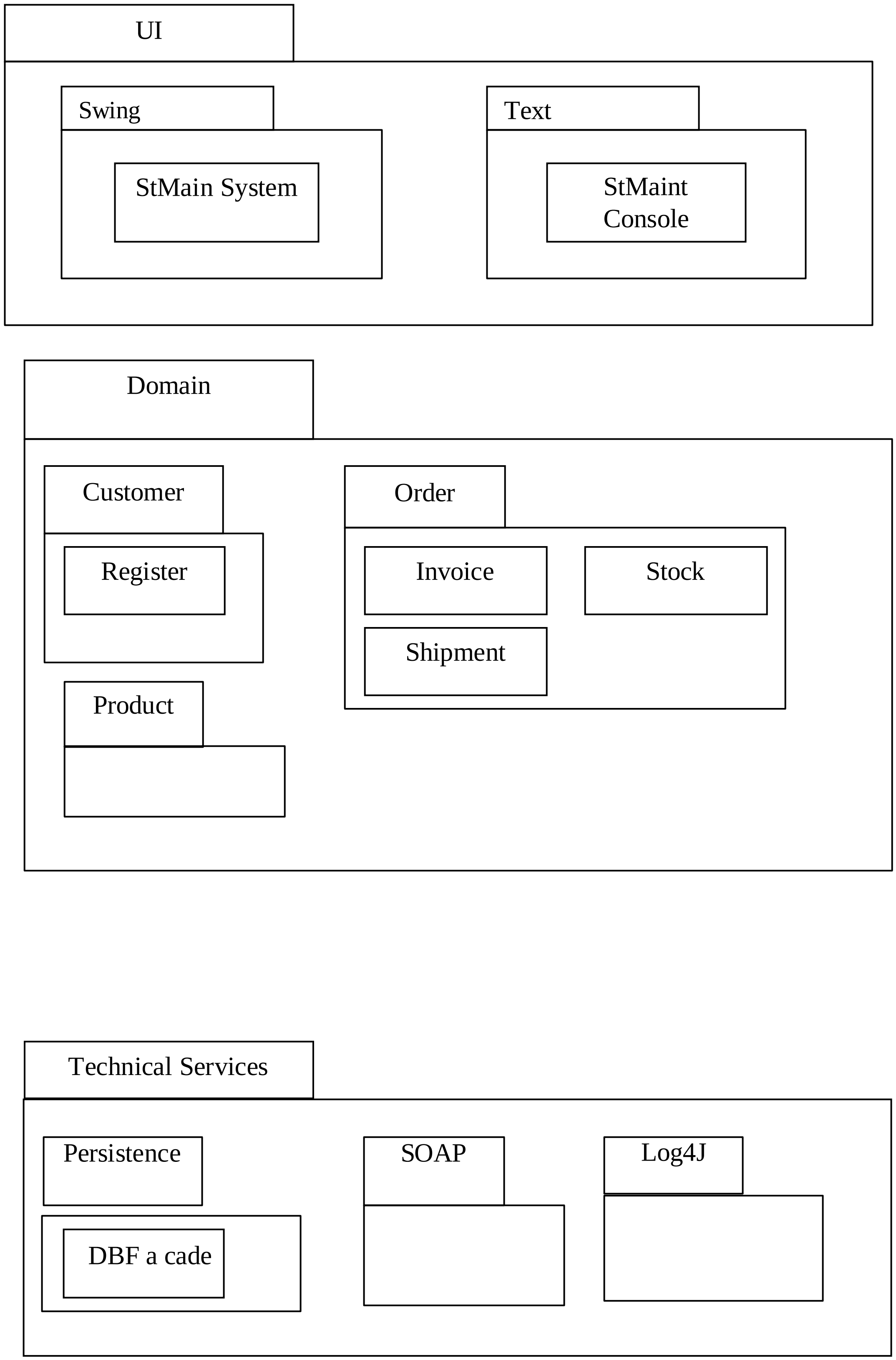


COLLABORATIONDIAGRAM

Collaboration diagram and sequence diagrams are alternate representations of an interaction. A collaboration diagram is an interaction diagram that shows the order of messages that implement an operation or a transaction.

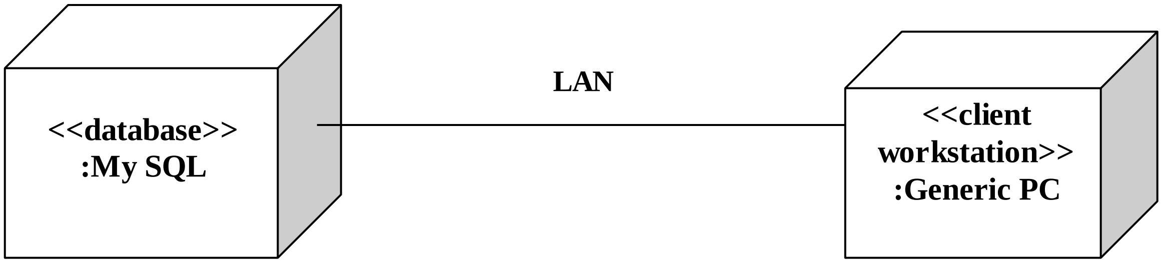


PARTIALLAYERD LOGICAL ARCHITECTURE DIAGRAM



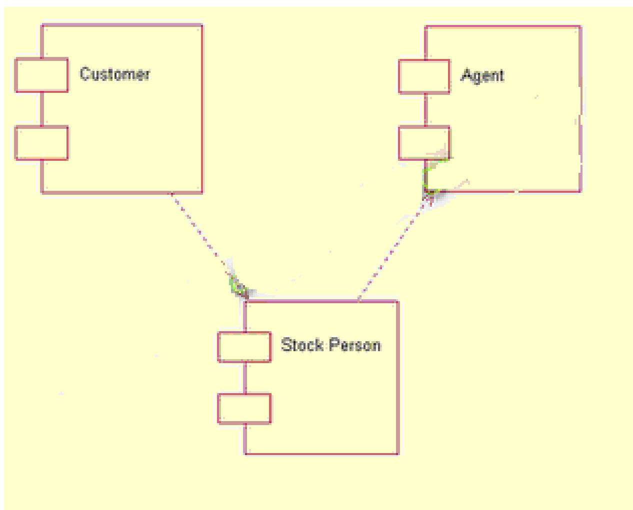
DEPLOYMENTDIAGRAMANDCOMPONENTDIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



Component Diagram

Component diagrams are used to visualize the organization and relationships among components in a system.



RESULT:

Thus the mini project for stock maintenance system has been successfully executed and codes are generated.

EX.NO: 7

ONLINE COURSE RESERVATION SYSTEM

Date:

AIM

To design an object oriented model for course reservation system.

PROBLEM STATEMENT

a) When ever the student comes to join the course he/she should be provided with the list of course available in the college.

b) The system should maintain a list of professor who is teaching the course. At the end of the course the student must be provided with the certificate for the completion of the course.

SYSTEM REQUIEMENT SPECIFICATION

OBJECTIVES

a) The main purpose of creating the document about the software is to know about the list of the requirement in the software project part of the project to be developed.
b) It specifies the requirement to develop a processing software part that completes the set of requirement.

SCOPE

a) In this specification, we define about the system requirements that are about from the functionality of the system.

b) It tells the users about the reliability defined in use case specification

FUNCTIONALITY

Many members of the process line to check for its occurrences and transaction, we are have to carry over at sometimes

USABILITY

The user interface to make the transaction should be effectively

PERFORMANCE

It is the capability about which it can performed function for many user at sometimes efficiently (ie) without any ever occurrences

RELIABILITY

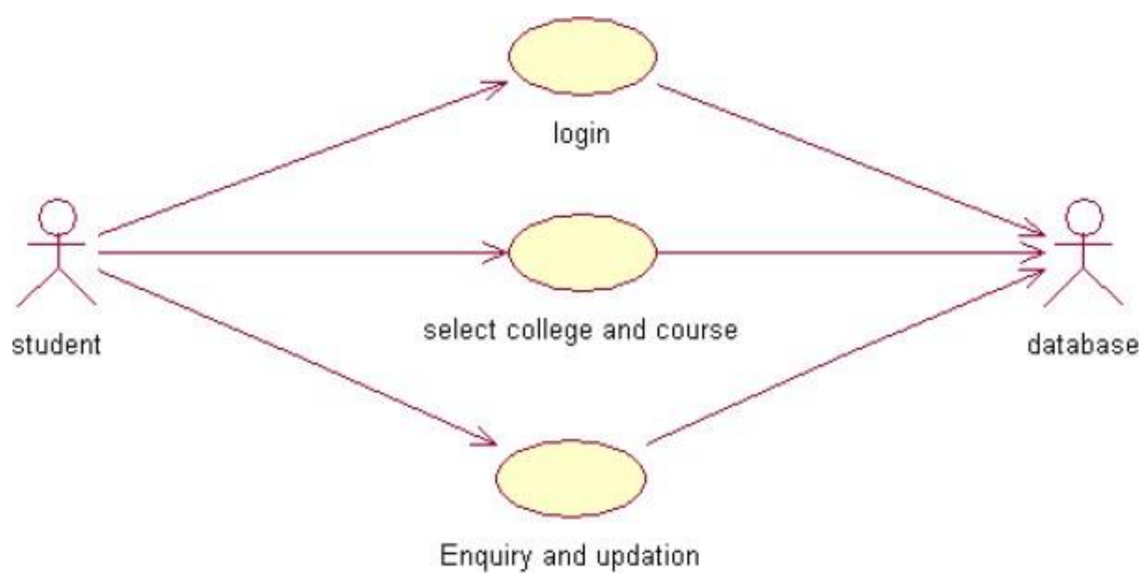
The system should be able to the user through the day today transaction

USER CASE DIAGRAM

a) Use case is a sequence of transaction in a system who set ask is to yield result of measurable value to individual author of the system

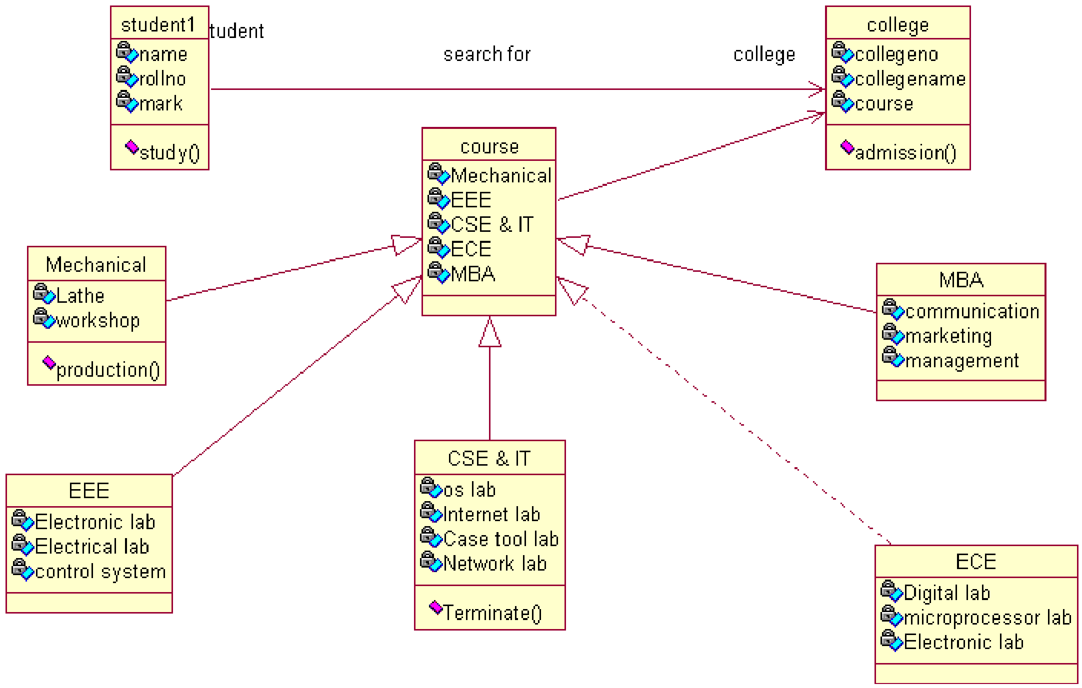
b) Use case is a set of scenarios to gether by a common user goal

c) A scenario is a sequence of step describing as interaction between a user and a system



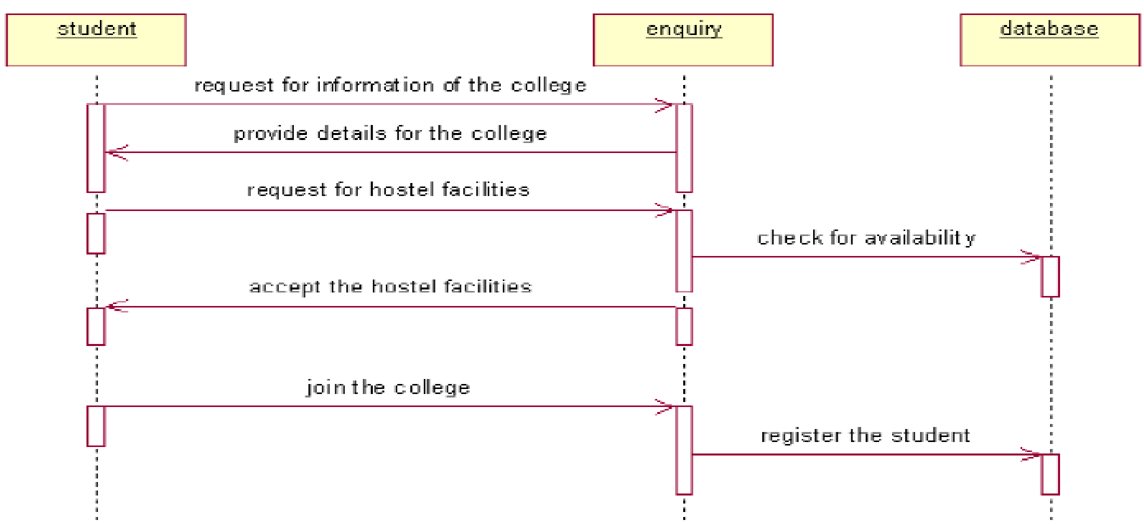
CLASSDIAGRAM:

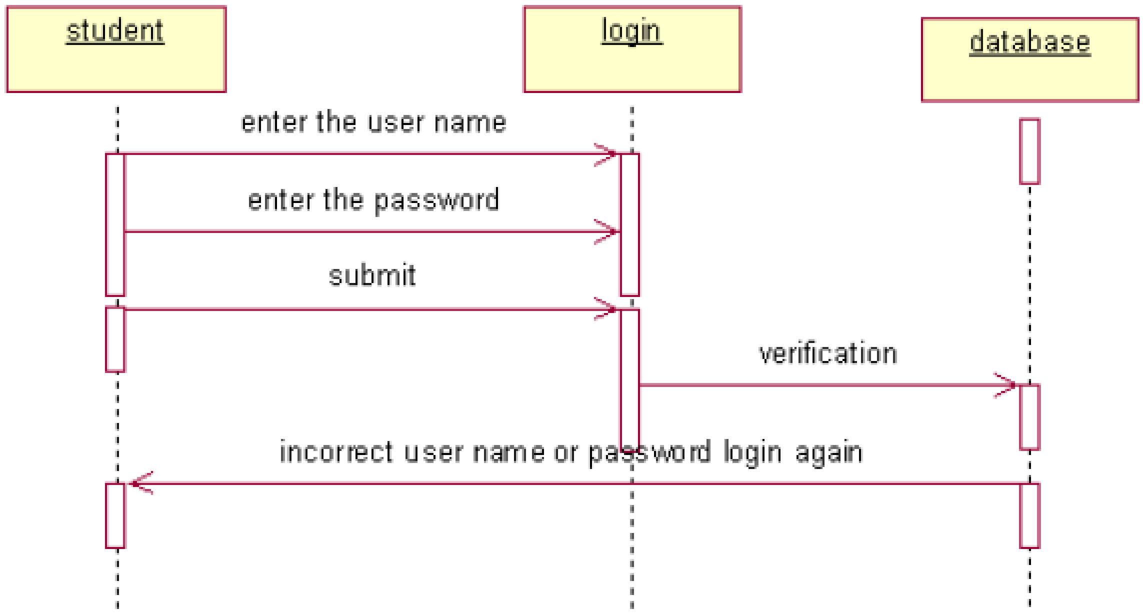
A class diagram describes the type of objectors in the system the various kinds of static relationship that exist among them.

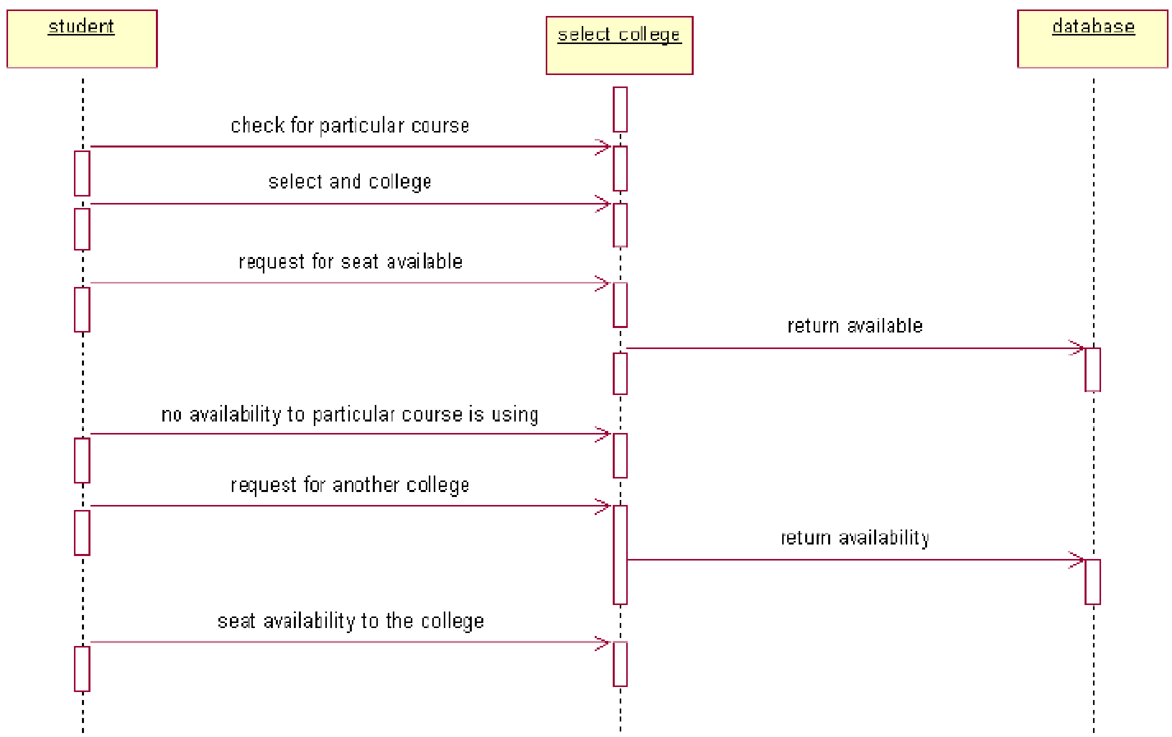


SEQUENCEDIAGRAM

A sequence diagram is one that includes the object of the projects and tells the lifetimes and also various action performed between objects.

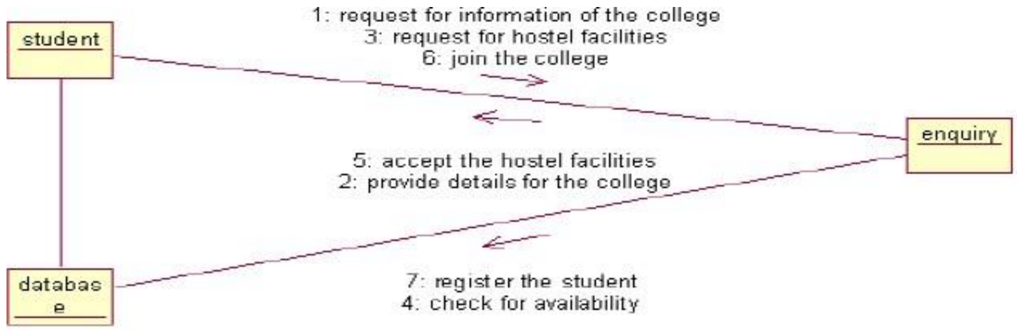


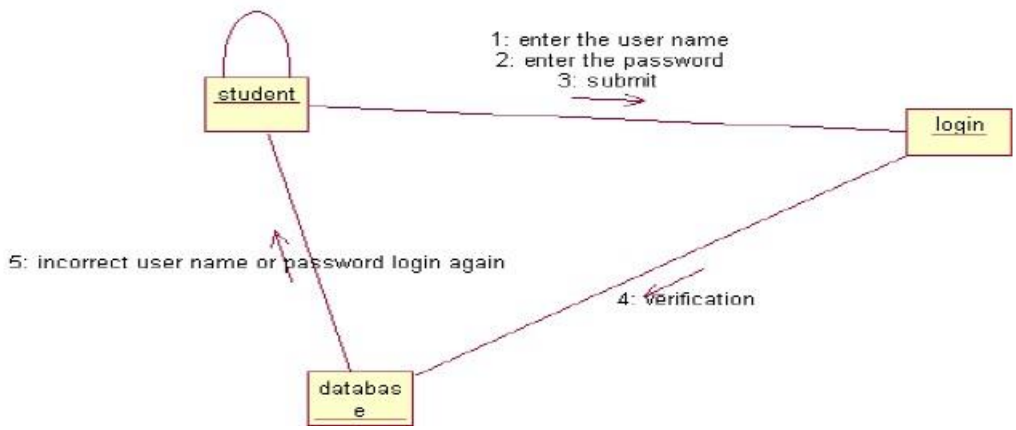


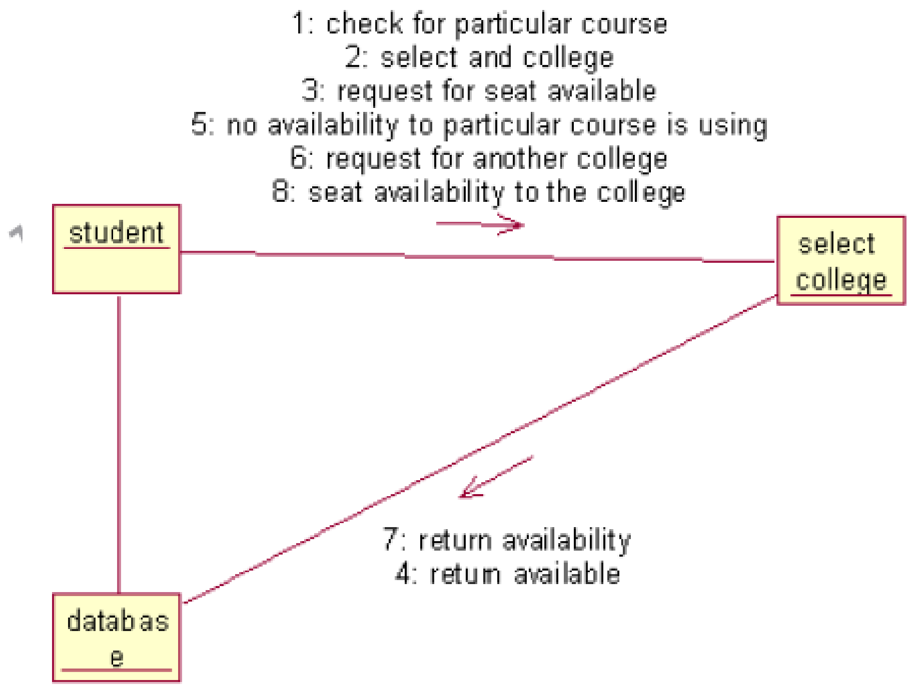


COLLOBORATIIONDIAGRAM

It is same as the sequence diagram that involved the project with the only difference that we give the project with the only difference that we give sequence number to each process.

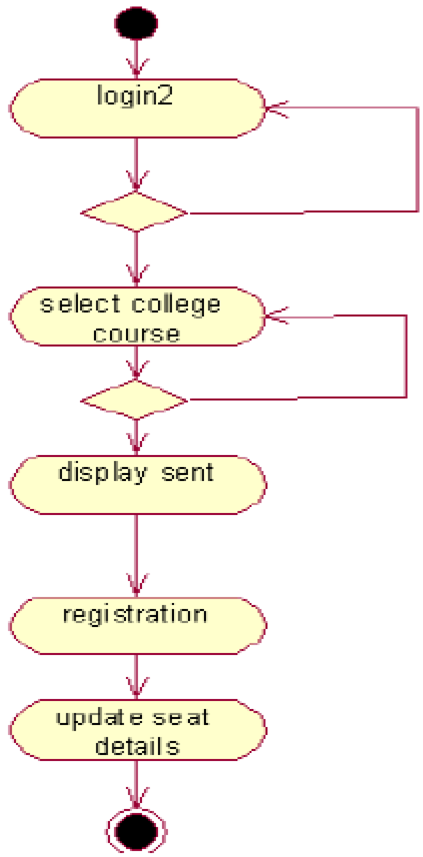

student
1: request for information of the college
3: request for hostel facilities
6: join the college
enquiry
5: accept the hostel facilities
2: provide details for the college
7: register the student
4: check for availability
databas
e


student
1: enter the user name
2: enter the password
3: submit
login
5: incorrect user name or password login again
4: verification
databas
e


1: check for particular course
2: select and college
3: request for seat available
5: no availability to particular course is using
6: request for another college
8: seat availability to the college
student
select
college
7: return availability
4: return available
databas
e

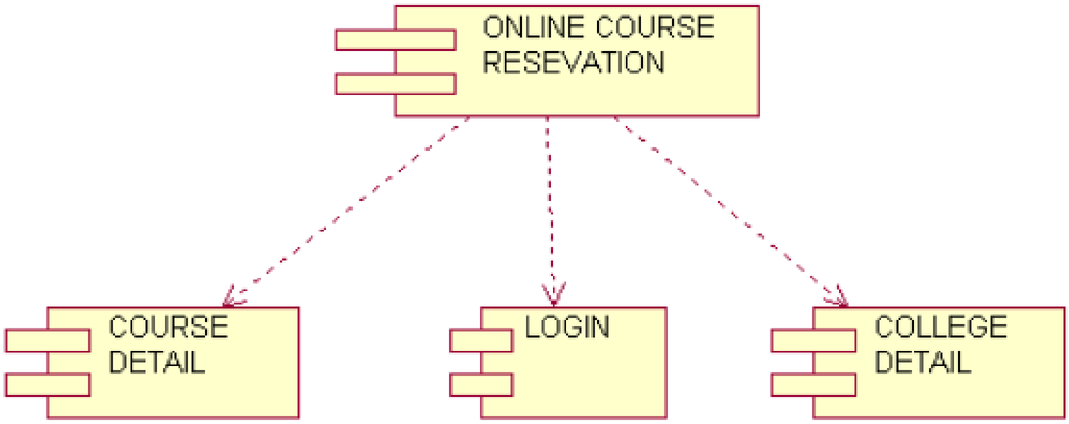
ACTIVIYDIAGRAM

It includes all the activities of particular project and various steps using join and forks



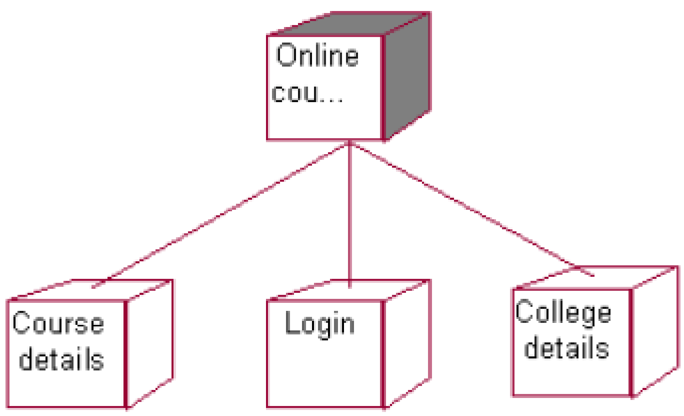
COMPONENTDIAGRAM

The component diagram is represented by figure dependency and it is a graph of design of figure dependency. The component diagram's main purpose is to show the structural relationships between the components of a systems. It is represented by boxed figure. Dependencies are represented by communication association



DEPLOYMENTDIAGRAM

It is a graph of nodes connected by communication association. It is represented by a three dimensional box. A deployment diagram in the unified modeling language serves to model the physical deployment of artifacts on deployment targets. Deployment diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3-dimentional box. Dependencies are represented by communication association. The basic element of a deployment diagram is a node of two types



PACKAGE DIAGRAM

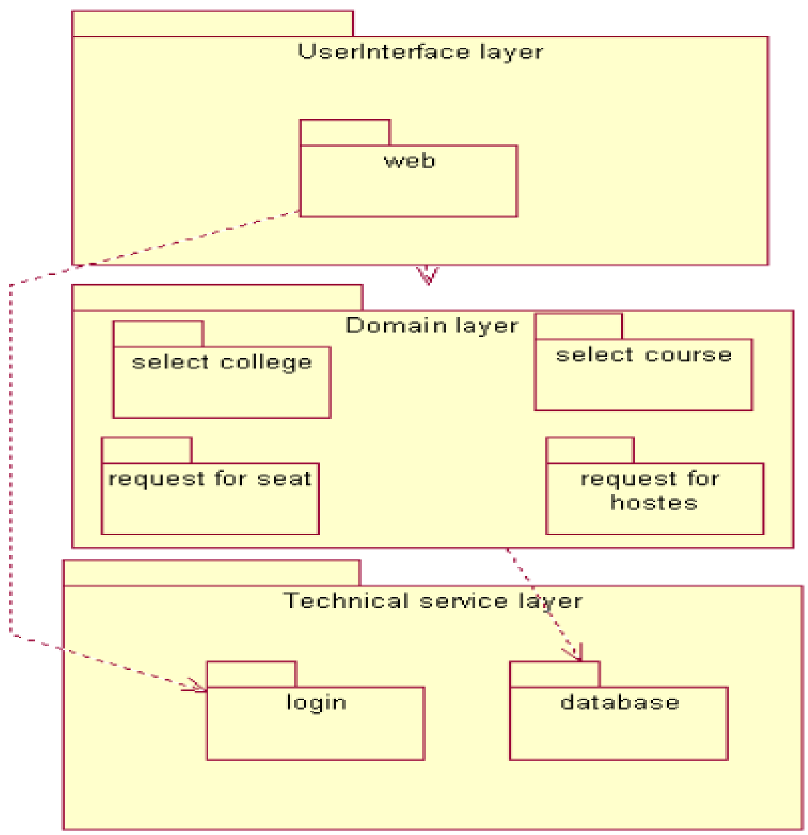
A package diagram is represented as a folder shown as a large rectangle with a top attached to its upper left corner. A package may contain both sub ordinate package and ordinary model elements. All uml models and diagrams are organized into package. A package diagram in unified modeling language that depicts the dependencies between the packages that makeup a model. A Package Diagram (PD)shows a grouping of elements in the OO model, and is a Cradle extension to UML. PDs can be used to show groups of classes in Class Diagrams (CDs), groups of components or processes in Component Diagrams (CPDs), or groups of processors in Deployment Diagrams (DPDs).

There are three types of layer. They are

a. User interface layer

b. Domain layer

c. Technical services layer



RESULT

Thus the mini project for online course reservation system has been successfully executed and codes are generated.

EX.NO: 8

AIRLINE/RAILWAYRESERVATIONSYSTEM

Date:

AIM

To develop the Airline / Railway reservation System using Rational Rose Software.

PROBLEMANALYSISANDPROJECTPLANNING

In the Airline/Railway reservation System the main process is a applicant have to login the database then the database verifies that particular user name and password then the user must fill the details about their personal details then selecting the flight and the database books the ticket then send it to the applicant then searching the flight or else cancelling the process.

OVERALL DESCRIPTION

Functionality

The data base should be act as an main role of thee-ticketing system it can be booking the ticket in easy way.

Usability

TheUserinterfacemakestheCreditCardProcessingSystemtobeefficient.

Performance

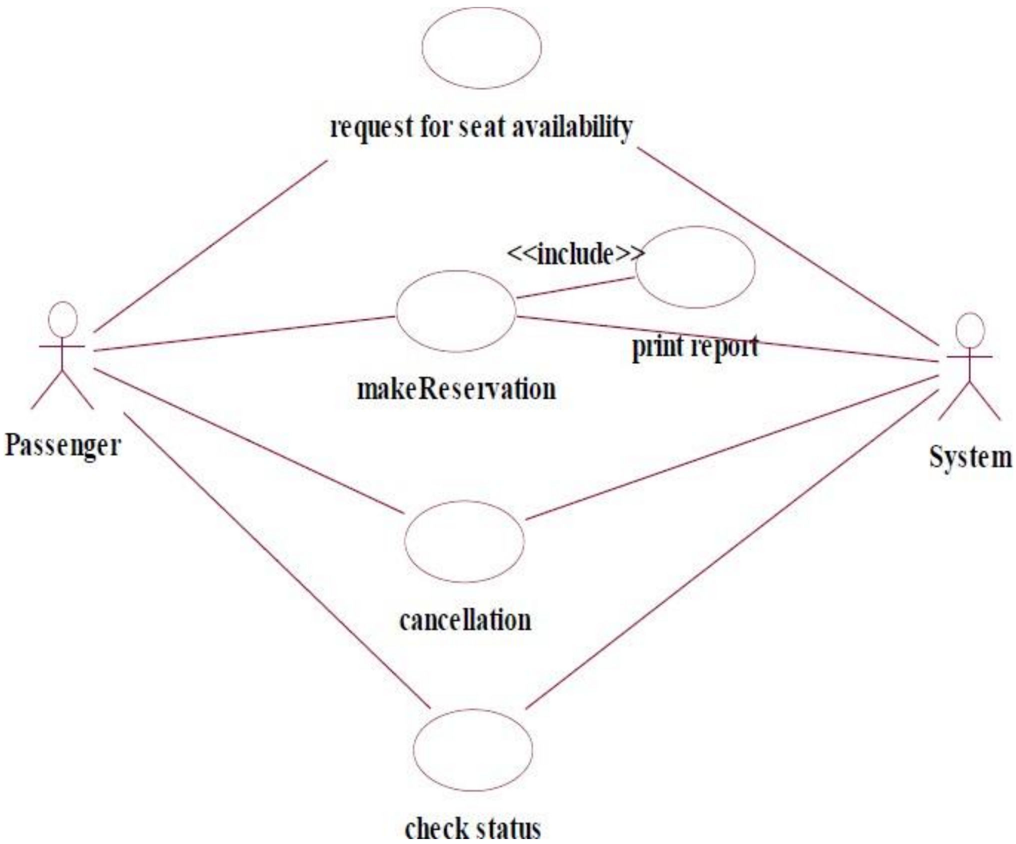
It is of the capacities about which it can perform function for many users at the same times efficiently that are without any error occurrence.

Reliability

The system should be able to process the user for their corresponding request.

USECASEDIAGRAM

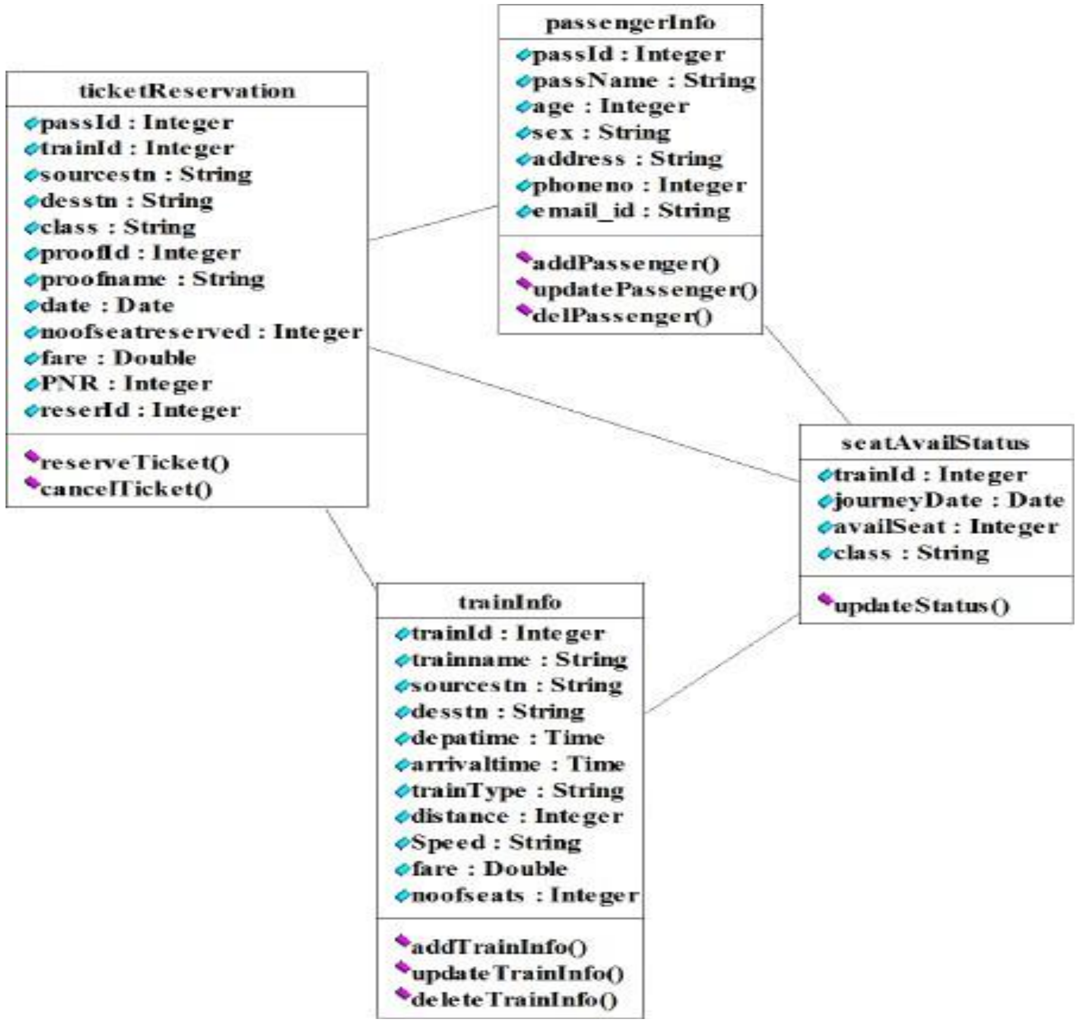
The passenger can view the status of there served tickets. So the passenger can confirm his/her travel.



CLASSDIAGRAM

The online ticket reservation system makes use of the following classes:

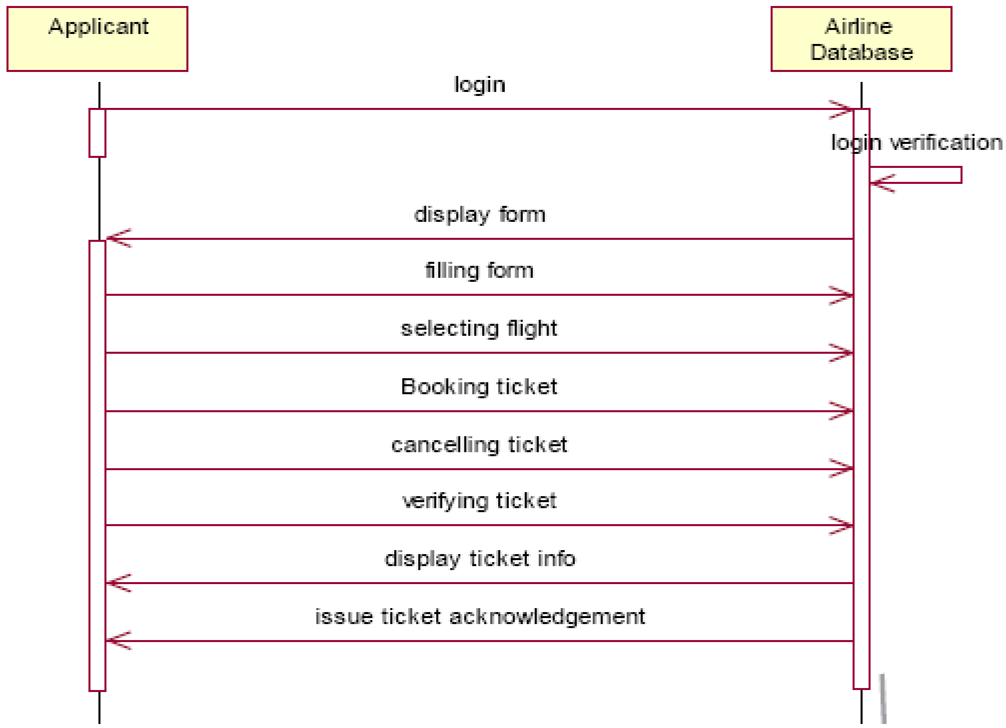
1. Ticket Reservation
2. Train Info
3. Passenger Info
4. Seat Avail Status



SEQUENCEDIAGRAM

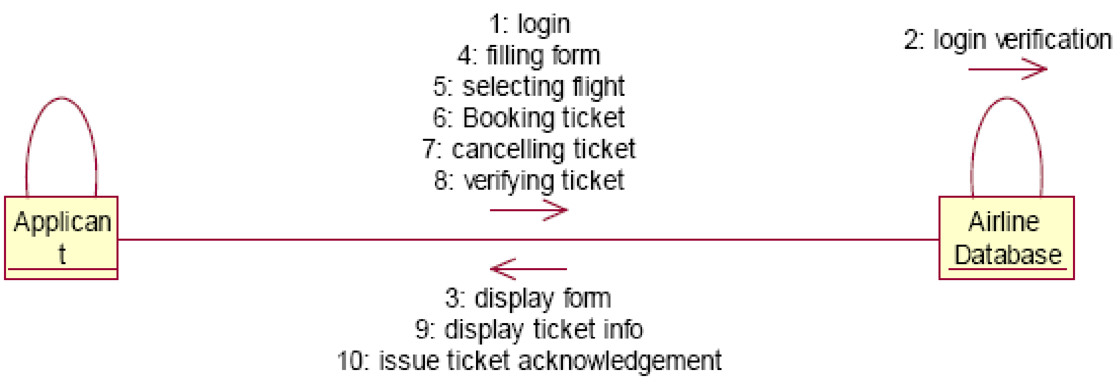
A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. There are two dimensions.

1. Vertical dimension-represent time.
2. Horizontal dimension-represent different objects.



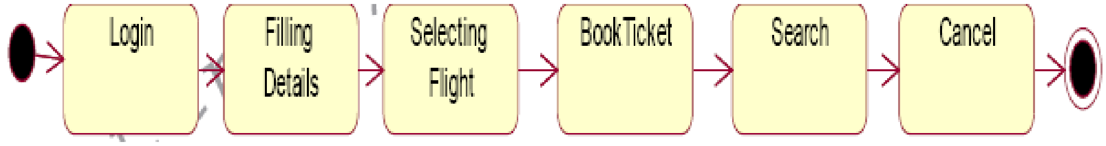
COLLABRATIONDIAGRAM

A collaboration diagram, also called a communication diagram or interaction diagram,. A sophisticated modeling tool can easily convert a collaboration diagram into a sequence diagram and the vice versa. A collaboration diagram resembles a flowchart that portrays the roles, functionality and behavior of individual objects as well as the overall operation of the system in real time.



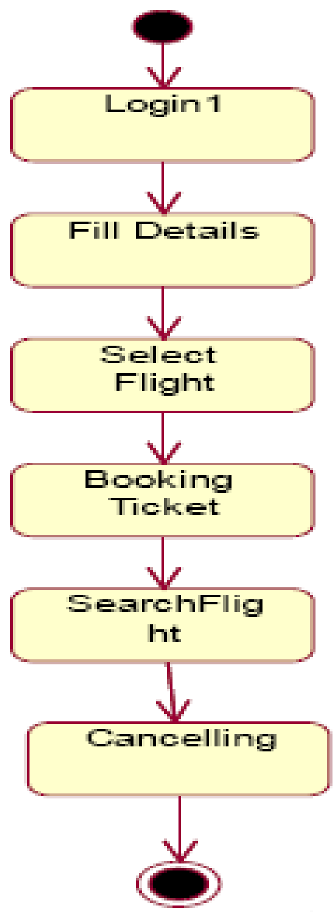
STATECHARTDIAGRAM

The purpose of state chart diagram is to understand the algorithm involved in performing a method. It is also called as state diagram. A state is represented as a round box, which may contain one or more compartments. An initial state is represented as small dot. A final state is represented as circle surrounding a small dot.



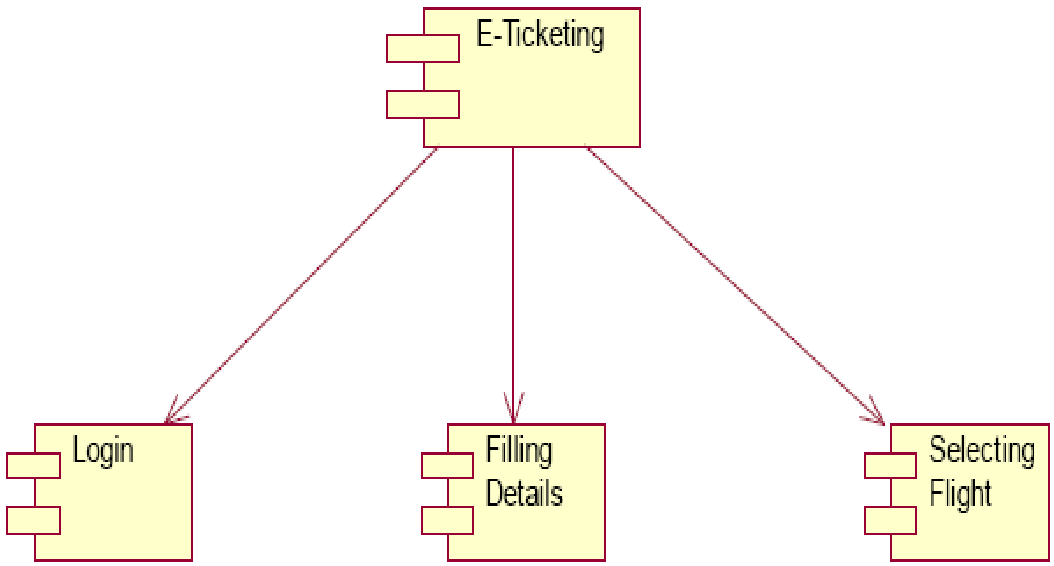
ACTIVITYDIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. An activity is shown as an rounded box containing the name of the operation.



COMPONENTDIAGRAM

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association.



RESULT

Thus the mini project for Airline / Railway reservation System has been successfully executed and codes are generated.

EX.NO: 9

SOFTWARE PERSONNEL MANAGEMENT SYSTEM

Date:

AIM:

To implement a software for software personnel management system.

PROBLEMSTATEMENT:

Human Resource management system project involves new and/or system upgrades of software of send to capture information relating to the hiring termination payment and management of employee. He uses system to plan and analyze all components and performance of metrics driven human resource functions, including recruitment, attendance, compensation, benefits and education. Human resources management systems should align for maximum operating efficiency with financial accounting operations customer relationship management, security and business lines as organization.

SOFTWARE REQUIREMENT SPECIFICATION:

SOFTWARE INTERFACE

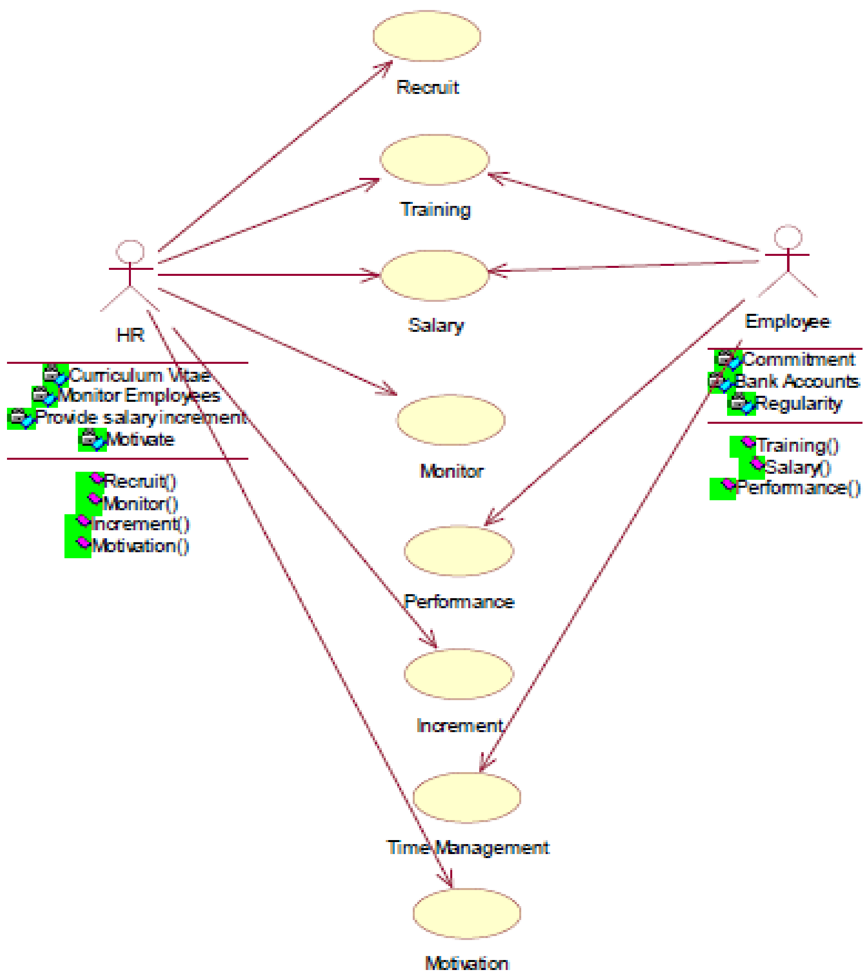
- **Front End Client-**The applicant and Administrator online interface is built using JSP and HTML. The HR's local interface is built using Java.
    - **Server-**Glass fish application server (SQL Corporation).
    - **Back End-** SQL database.

HARDWAREINTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

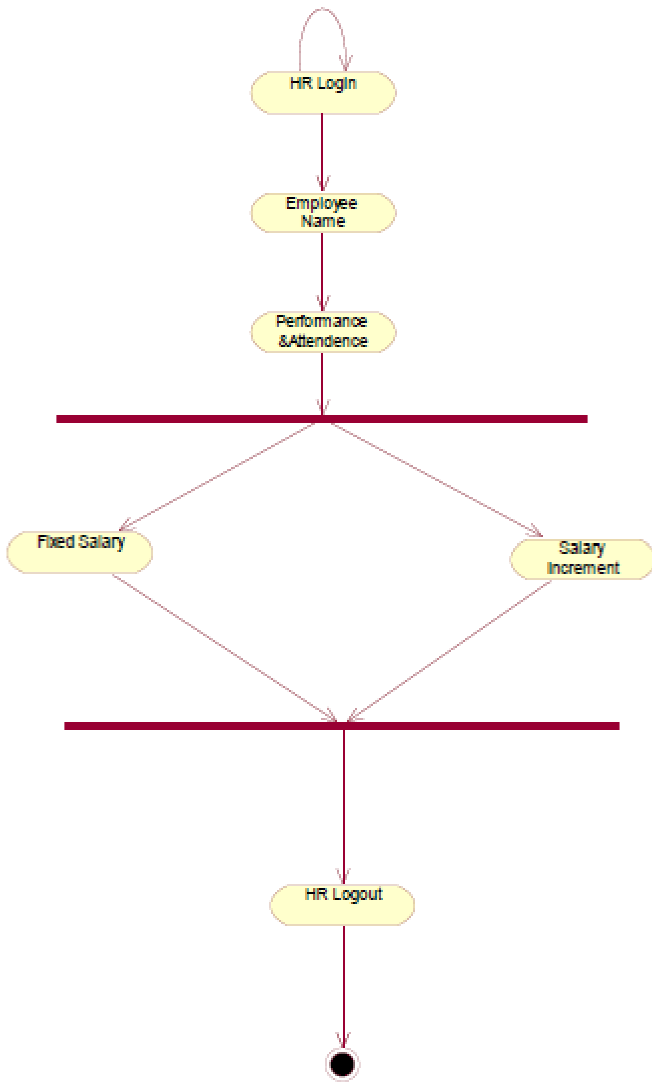
USECASE DIAGRAM:

The HR of an organization involves recruitment training, monitoring and motivation of an employee. The HR also involves gives salary as observed in the payroll sheet. The employee undergoes training, receives the salary , gives the expected performance and manages time in order to complete a given task within the required period.



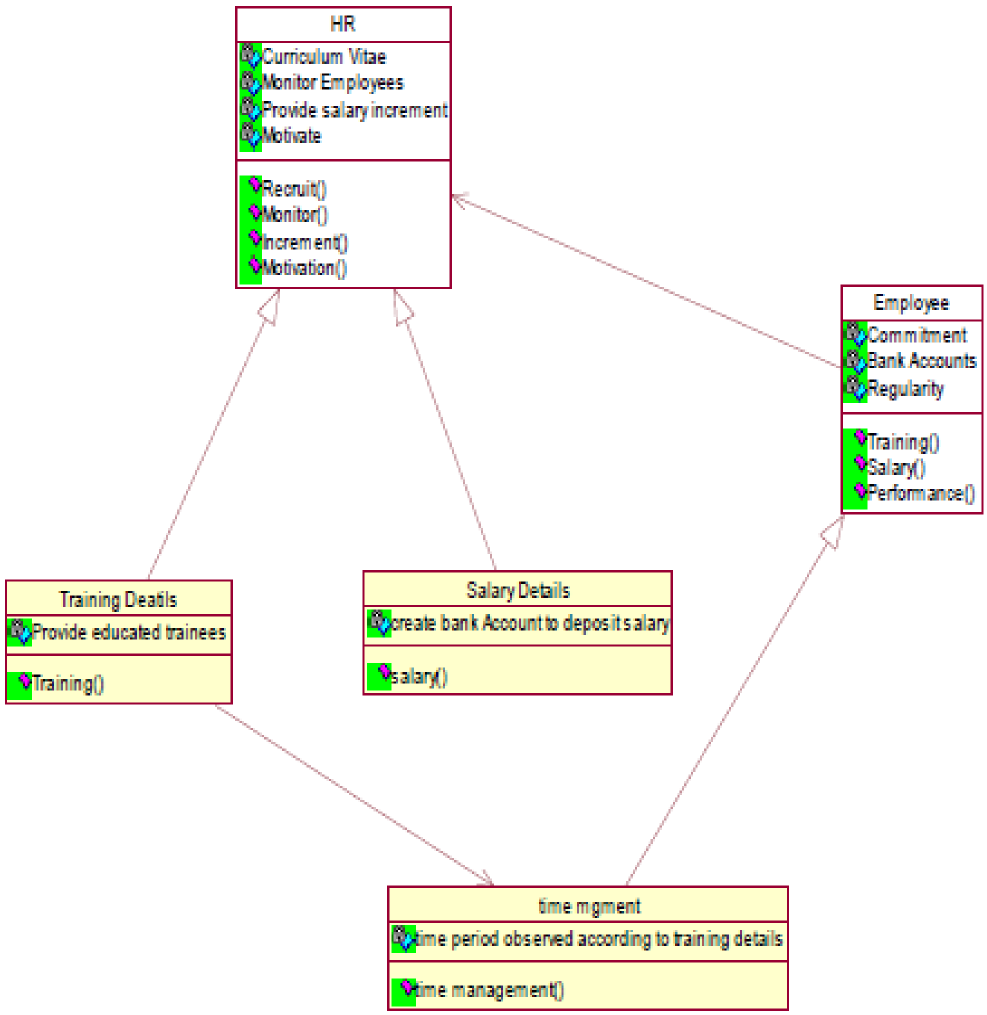
ACTIVITYDIAGRAM:

The activity diagram notation is an action, partition, fork join and object node. Most of the notation is self explanatory, two points. Once an action finished, there is an automatic outgoing transaction. The diagram can show both control flow and data flow.



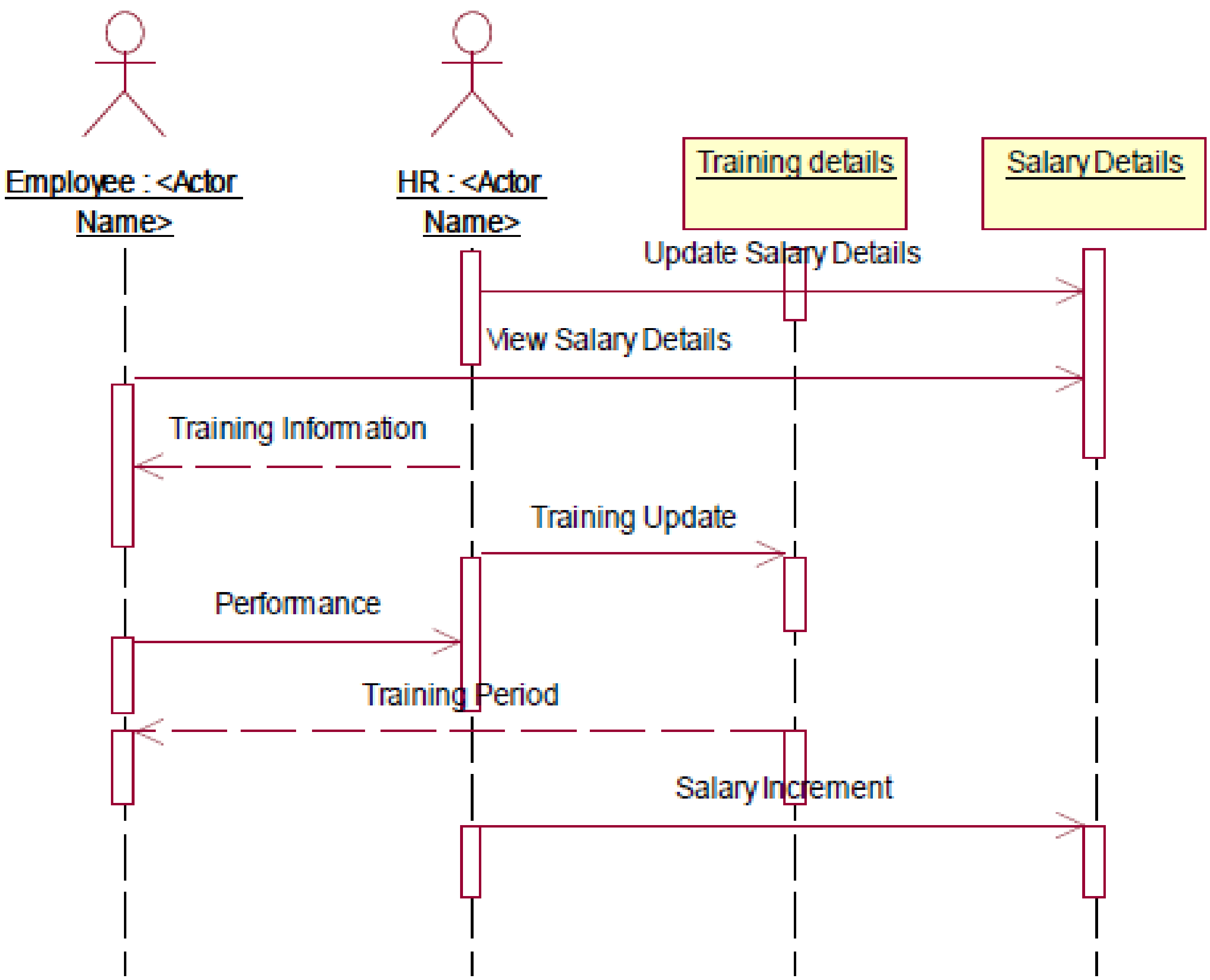
CLASS DIAGRAM:

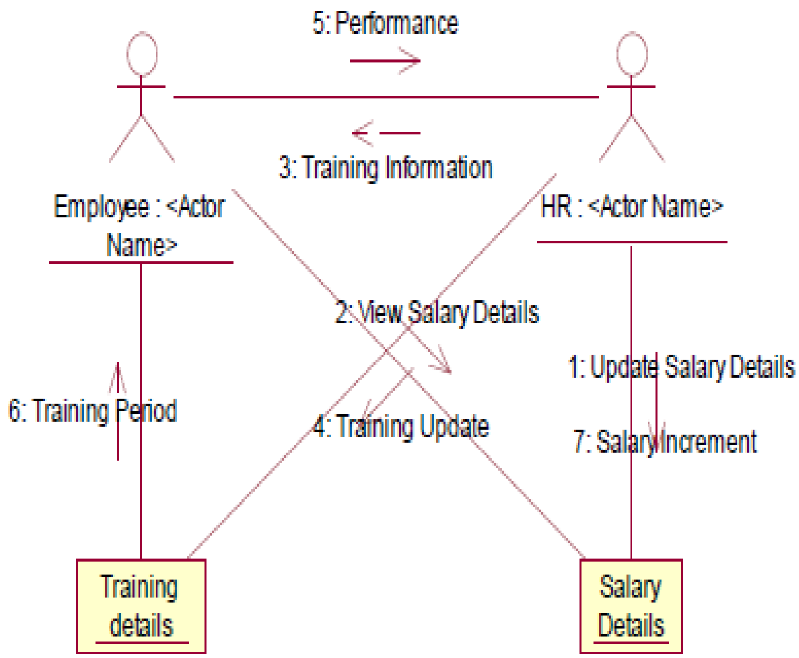
The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.



INTERACTIONDIAGRAM:

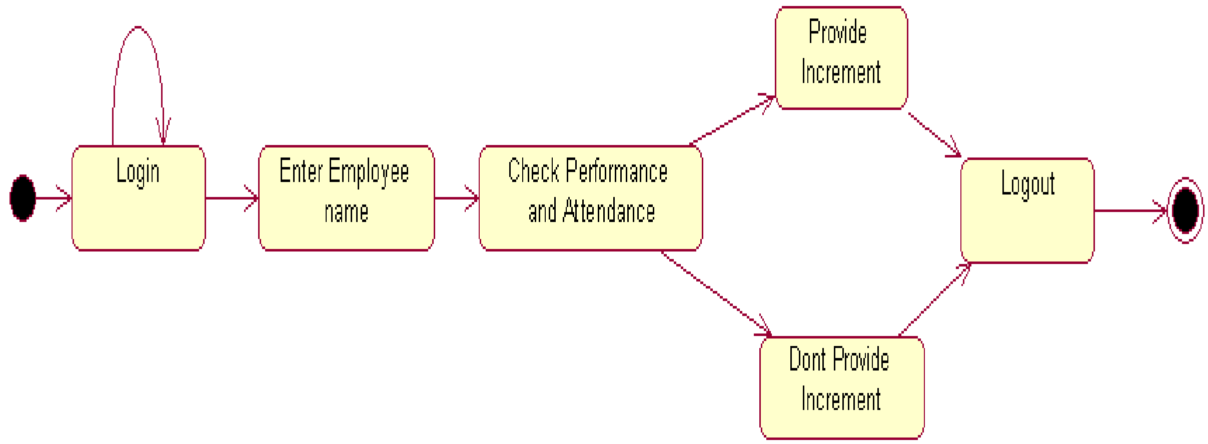
A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.





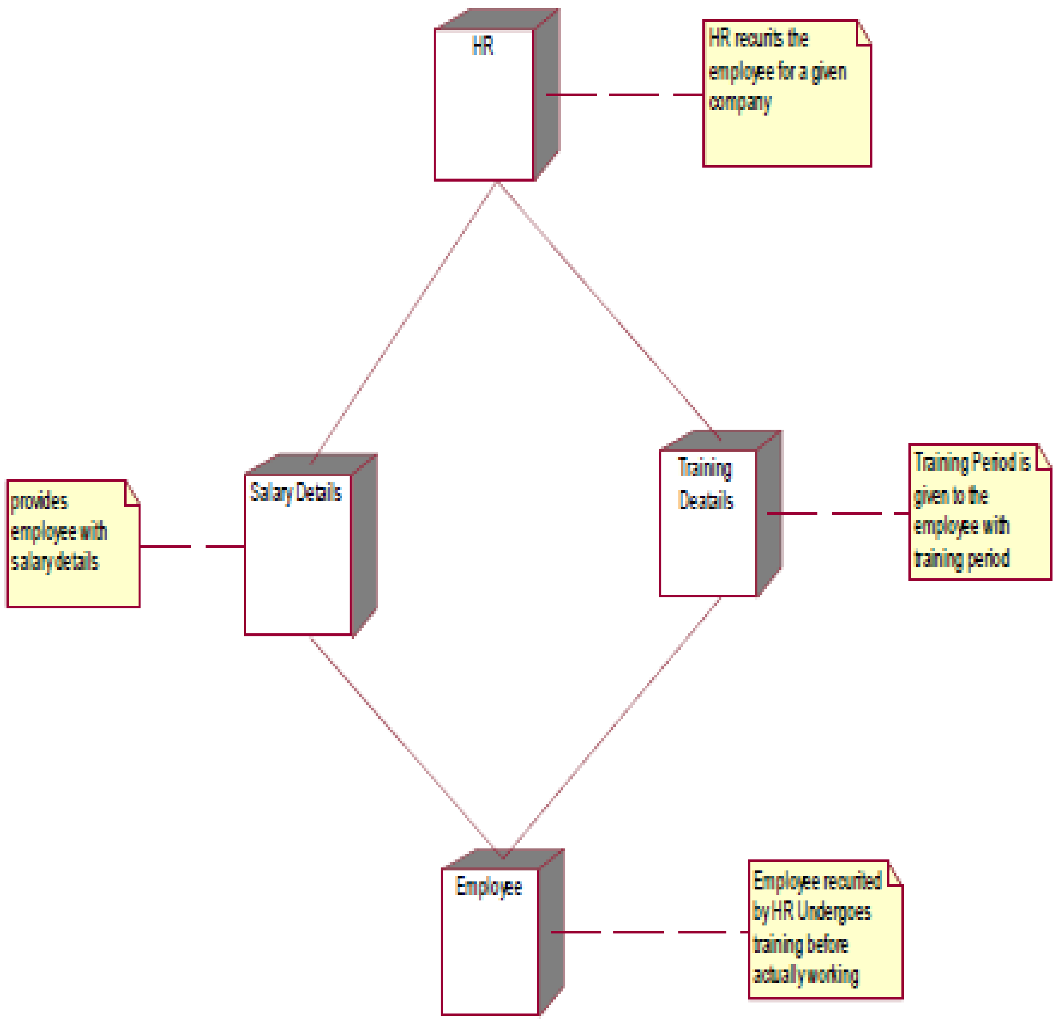
STATE TRANSITION DIAGRAM

States of object are represented as rectangle with round corner, the transaction between the different states. A transition is a relationship between two state that indicates that when an event occur the object moves from the prior state to the subsequent.



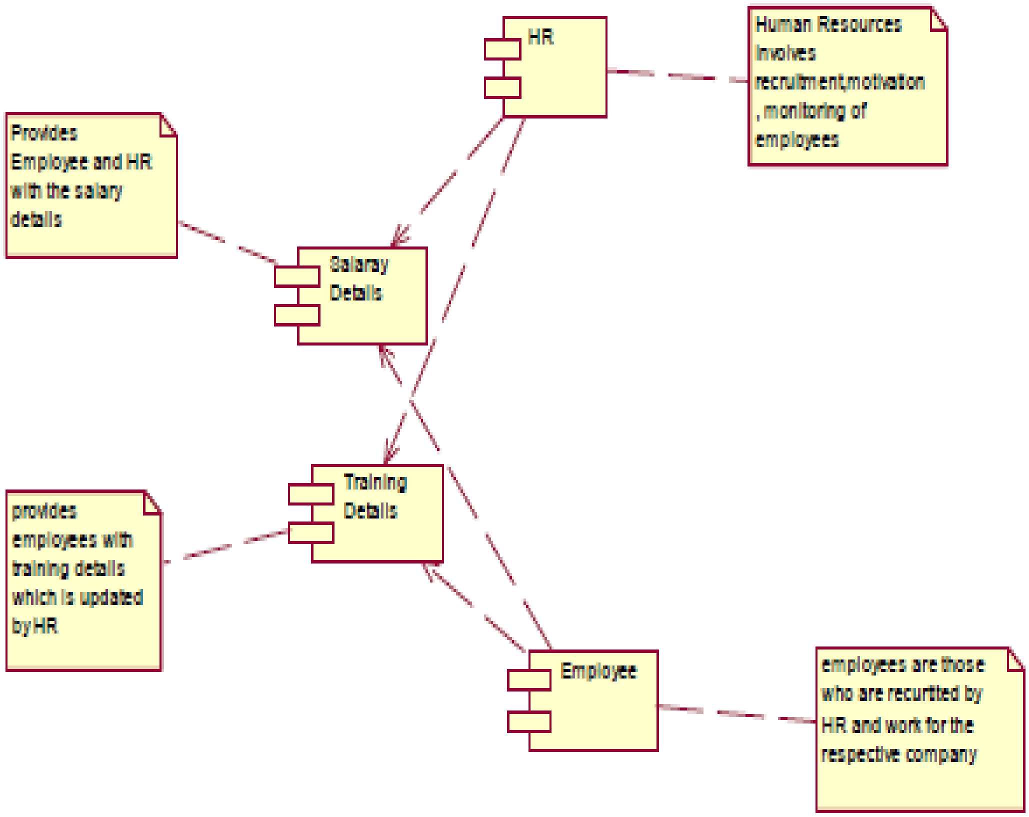
DEPLOYMENTDIAGRAMANDCOMPONENTDIAGRAM

HR recruits employee for a company employee recruited by HR goes under training before actually working. Training period is given to the employee with the training details. The salary details for the employee are provided.



COMPONENTDIAGRAM

The HR recruits, motivate and monitor the employee, HR also update the salary details and training details for reference. The employee are those who are recruited by HR and work for the company. The training details provide employees with training details which is updated by HR



RESULT

Thus the mini project for software personnel management system has been successfully executed and codes are generated.

EX.NO: 10

CREDIT CARD PROCESSING

Date:

AIM:

To create a system to perform the credit card processing

PROBLEMSTATEMENT:

Credit card processing through offline involves the merchant collecting order information(including credit card numbers),storing this in a data base on your site, and entering it using their on-site merchant credit card processing system. Takes time to manually enter credit card information for each order. This solution creates following cons:

SOFTWAREREQUIREMENTSPECIFICATION:

PRODUCT PERSPECTIVE

This solution involves signing up for a free Business Account. Once this is done and the e-commerce site is properly configured, you can accept payments from Visa, MasterCard, Amex, and Discover cards payments.

SOFTWARE INTERFACE

- **Front End Client**-The applicant and Administrator online interface is built using JSP and HTML. The Administrators 's local interface is built using Java.
- **Web Server**-Glass fish application server(SQL Corporation).
- **Back End**-SQL database.

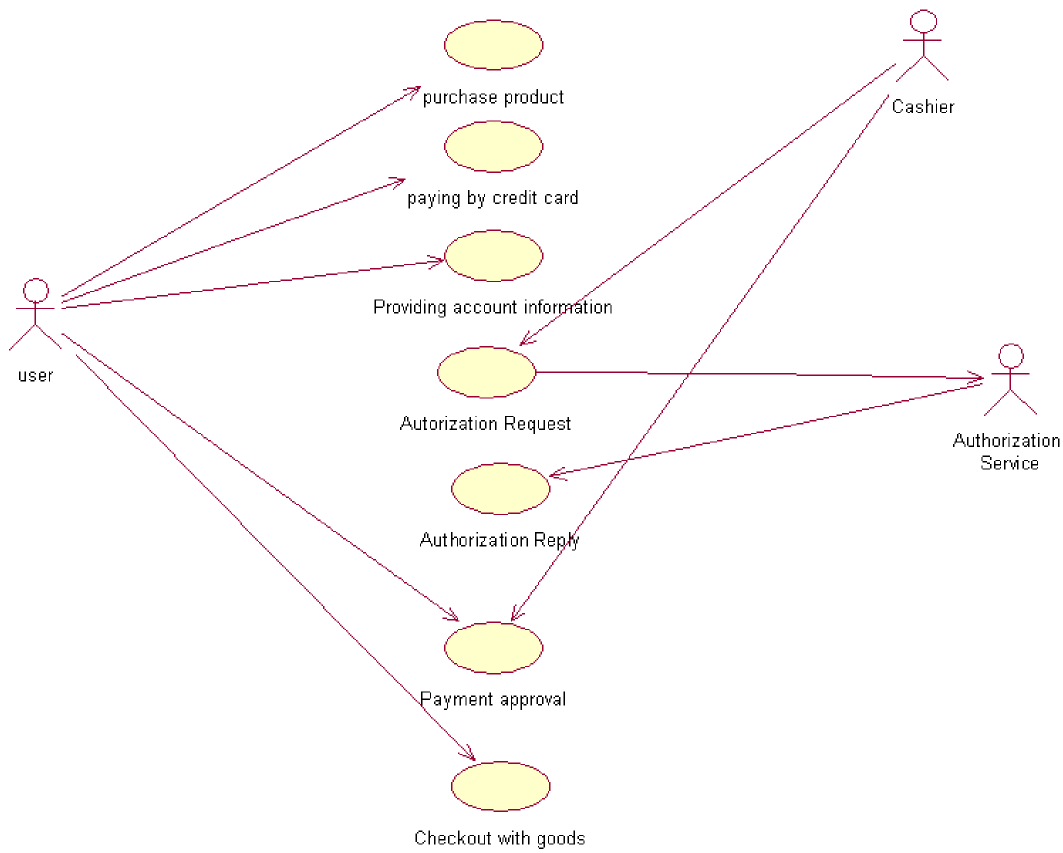
HARDWARE INTERFACE

The server is directly connected to the client systems .The client systems have access to the database in the server.

USECASE DIAGRAM:

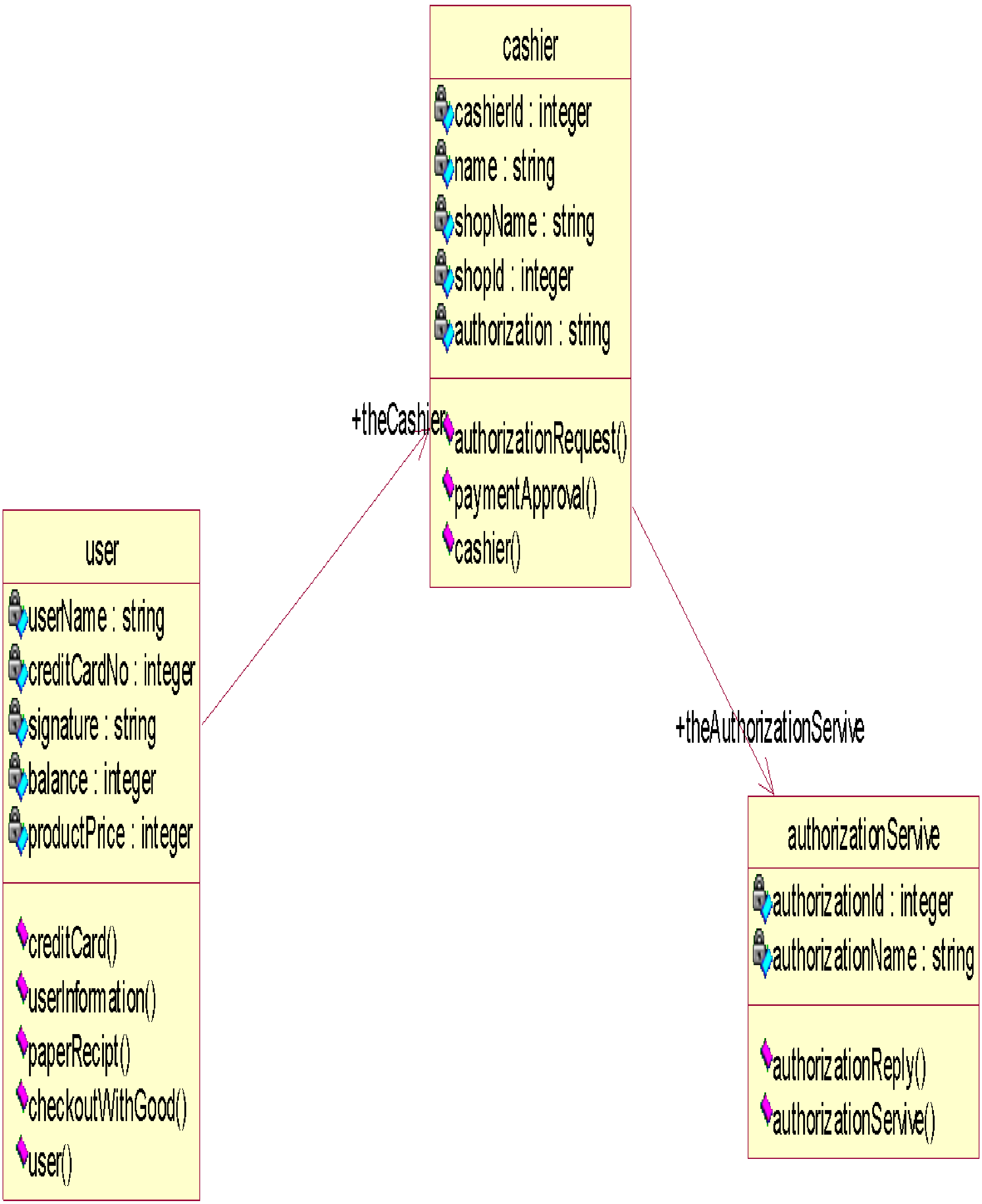
USE-CASE NAME:PAYMENT APPROVAL

The transaction details are recorded by the credit card processor and results are securely relayed to the merchant. Merchant's site receives transaction result and does appropriate actions(e.g. saves the order & shows message).

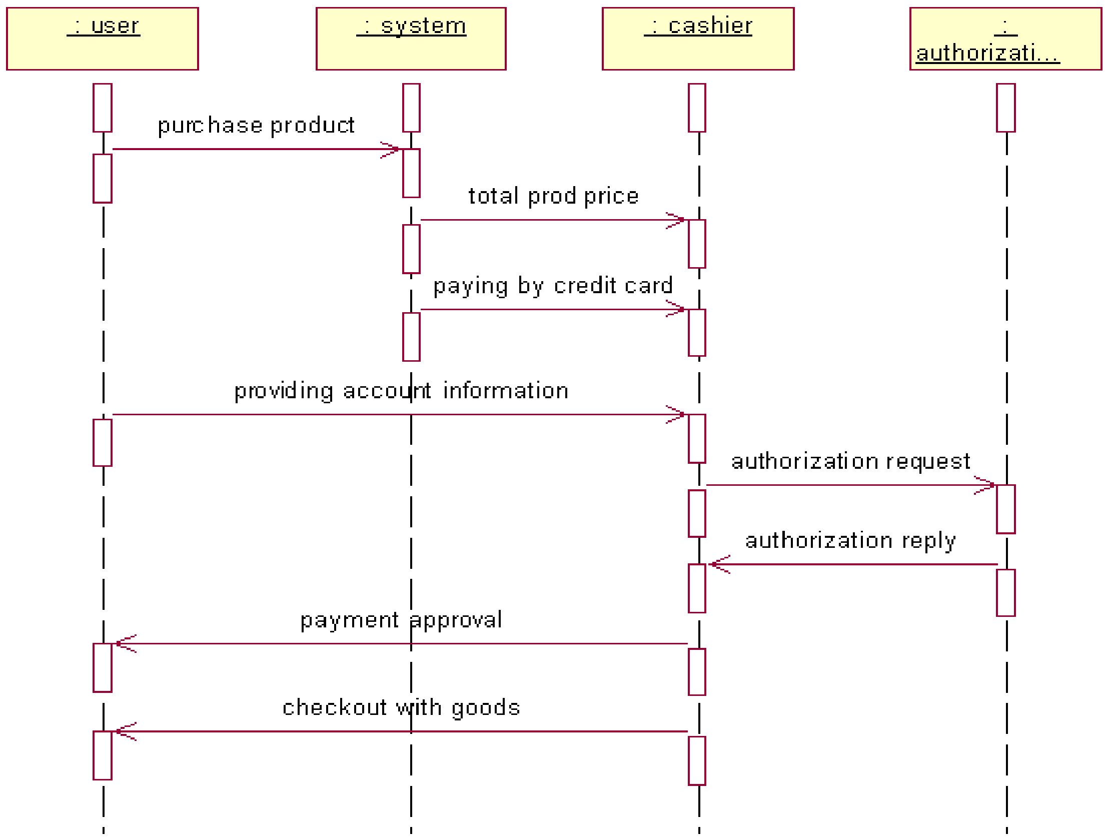


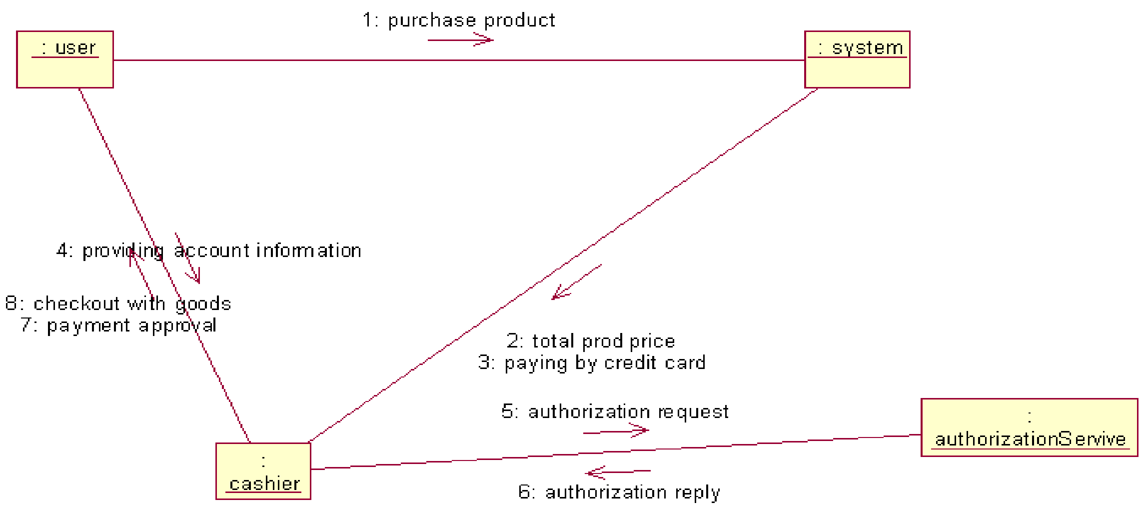
CLASSDIAGRAM:

The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.The Credit Card Processing system class diagram consists of three classes.

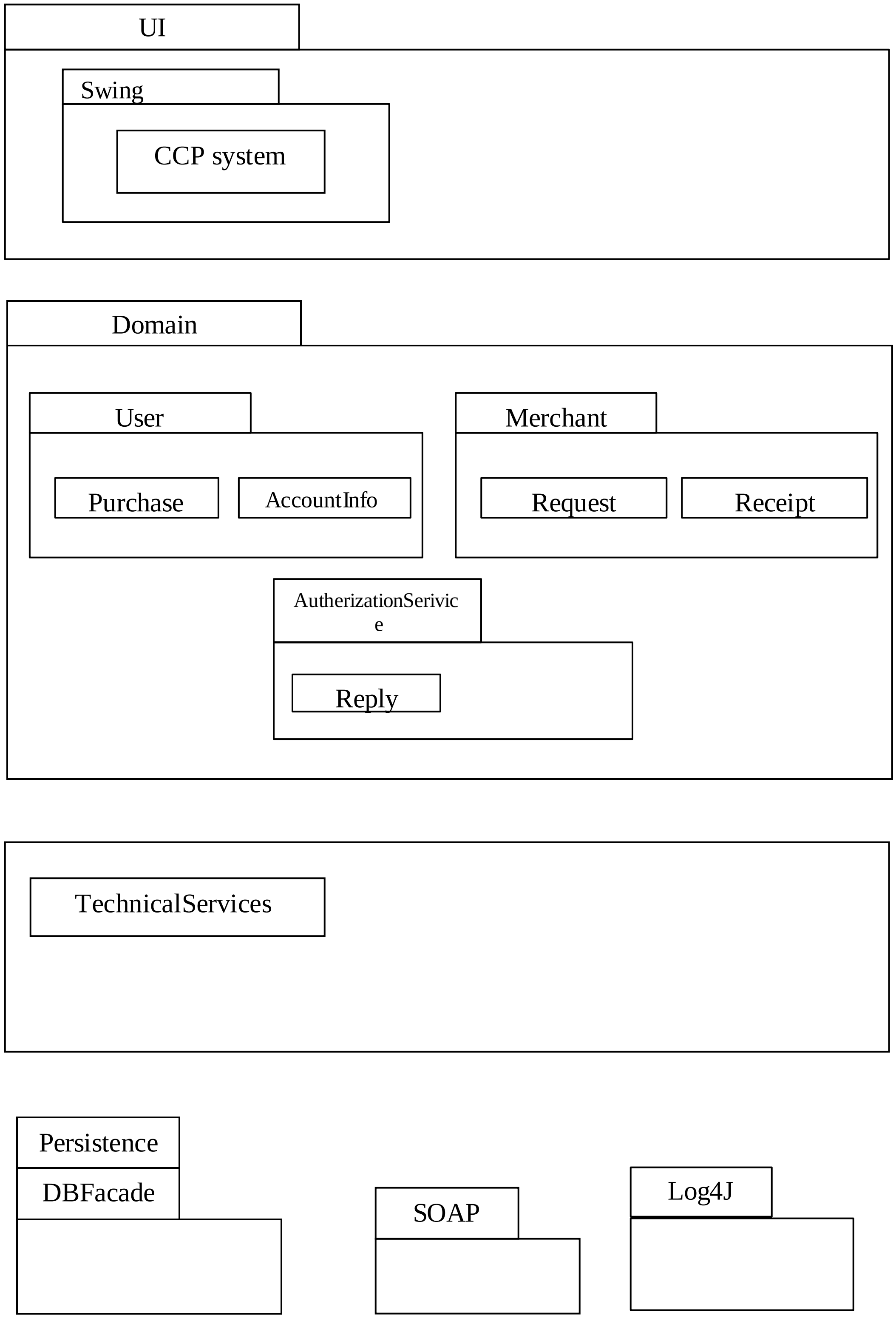

cashier
cashierId : integer
name : string
shopName : string
shopId : integer
authorization : string
authorizationRequest()
paymentApproval()
cashier()
+theCashier
user
userName : string
creditCardNo : integer
signature : string
balance : integer
productPrice : integer
creditCard()
userInformation()
paperRecipt()
checkoutWithGood()
user()
+theAuthorizationServive
authorizationServive
authorizationId : integer
authorizationName : string
authorizationReply()
authorizationServive()

INTERACTION DIAGRAM:



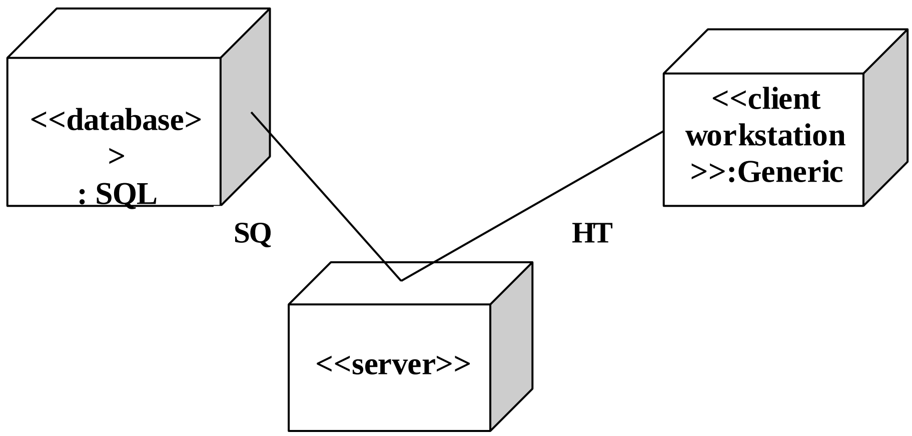


PARTIALLAYERD LOGICAL ARCHITECTURE DIAGRAM



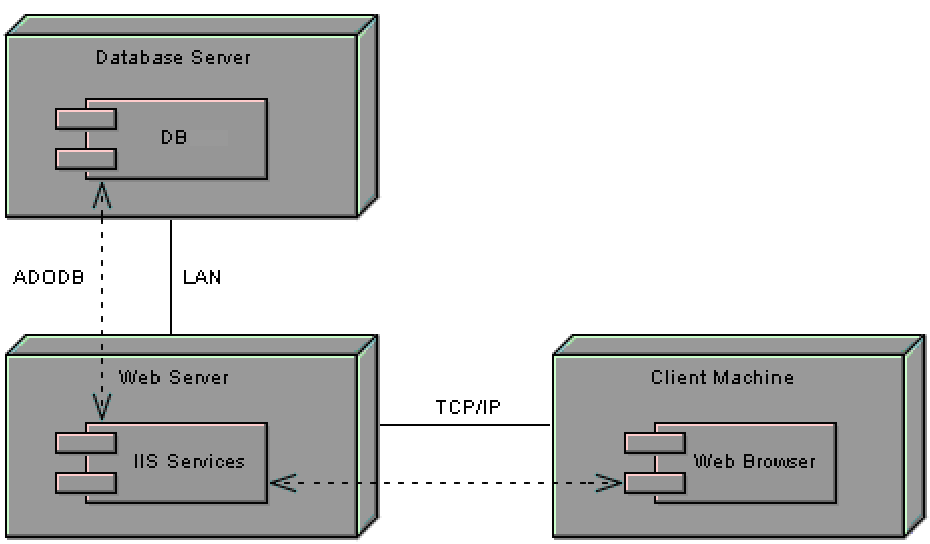
DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



COMPONENT DIAGRAM

Component diagrams are used to visualize the organization and relationships among components



RESULT:

Thus the mini project for credit card processing system has been successfully executed and codes are generated.

EX.NO: 11

E-BOOK MANAGEMENT SYSTEM

Date:

AIM:

To create a system to perform E-book Management System.

PROBLEMSTATEMENT:

An E- Book lends books and magazines to member, who is registered in the system. Also it handles the purchase of new titles for the Book Bank. Popular titles are brought into multiple copies. Old books and magazines are removed when they are out or date or poor in condition. A member can reserve a book or magazine that is not currently available in the book bank, so that when it is returned or purchased by the book bank, that person is notified. The book bank can easily create, replace and delete information about the tiles, members, loans and reservations from the system.

SOFTWARERESOURCESPECIFICATION:

OVERALL DESCRIPTION

It will describe major role of the system components and inter- connections.

PRODUCT PERSPECTIVE

The ORS acts as an interface between the user and the 'e-book manager'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the books or magazines.

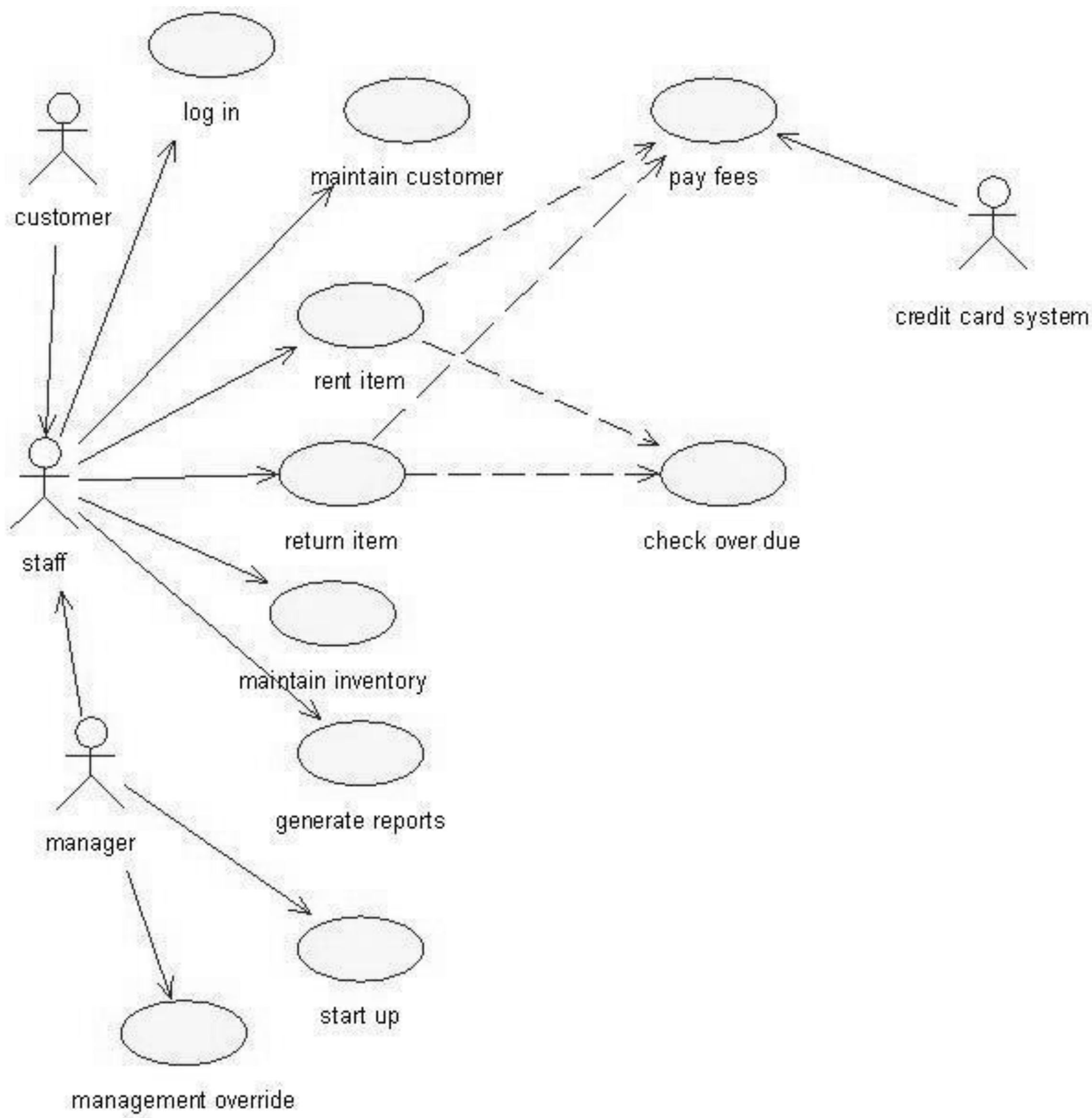
SOFTWAREINTERFACE

- **Front End Client-** The Student and Librarian online interface is built using JSP and HTML. The Librarians local interface is built using Java.
- **Web Server-**Glass fish application server (Oracle Corporation).
- **Back End-**Oracle database

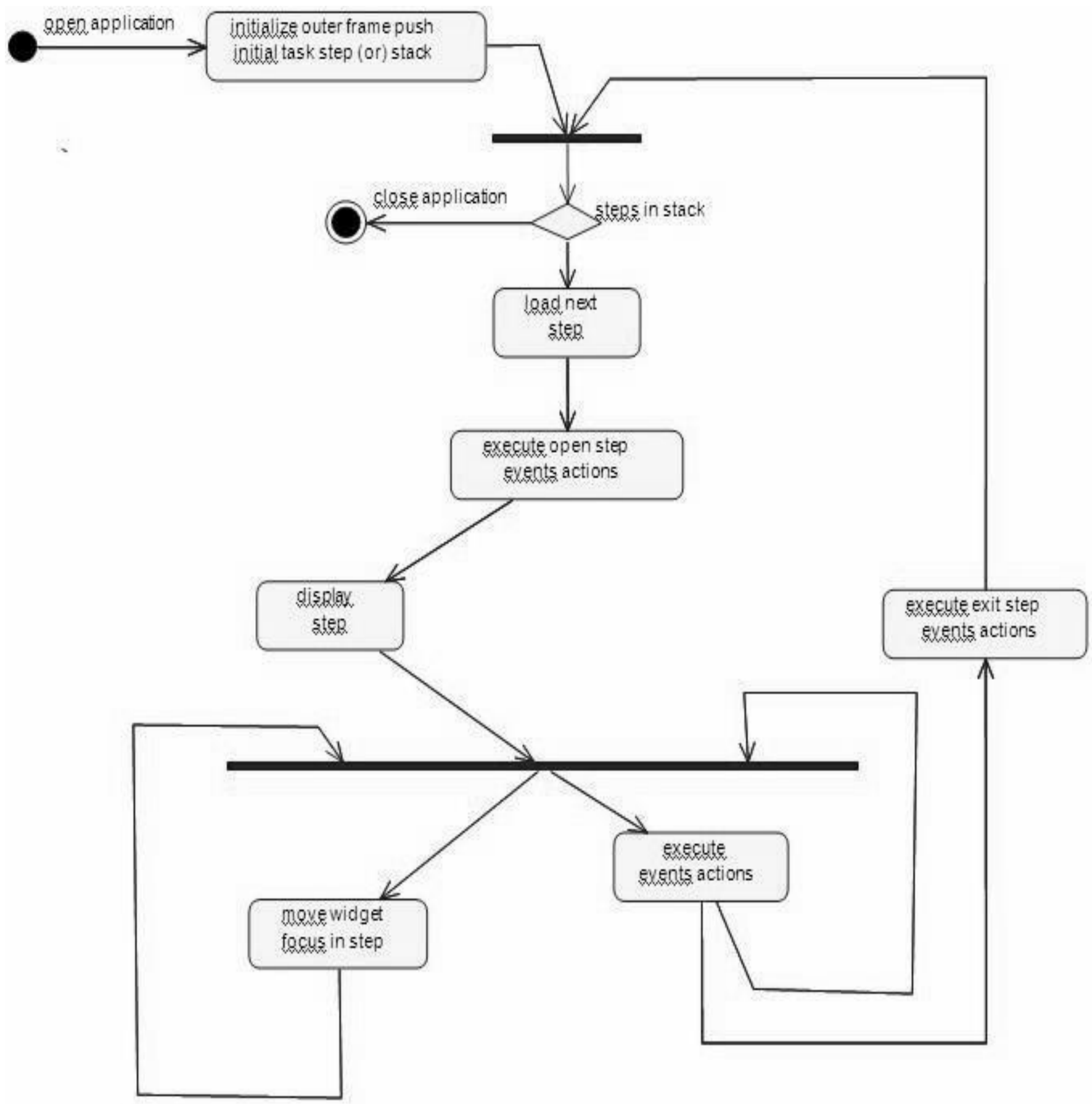
HARDWARE INTERFACE

The server is directly connect to the client systems. The client systems have access to the database in the server.

USE-CASEDIAGRAM:



ACTIVITYDIAGRAM:



CLASSDIAGRAM

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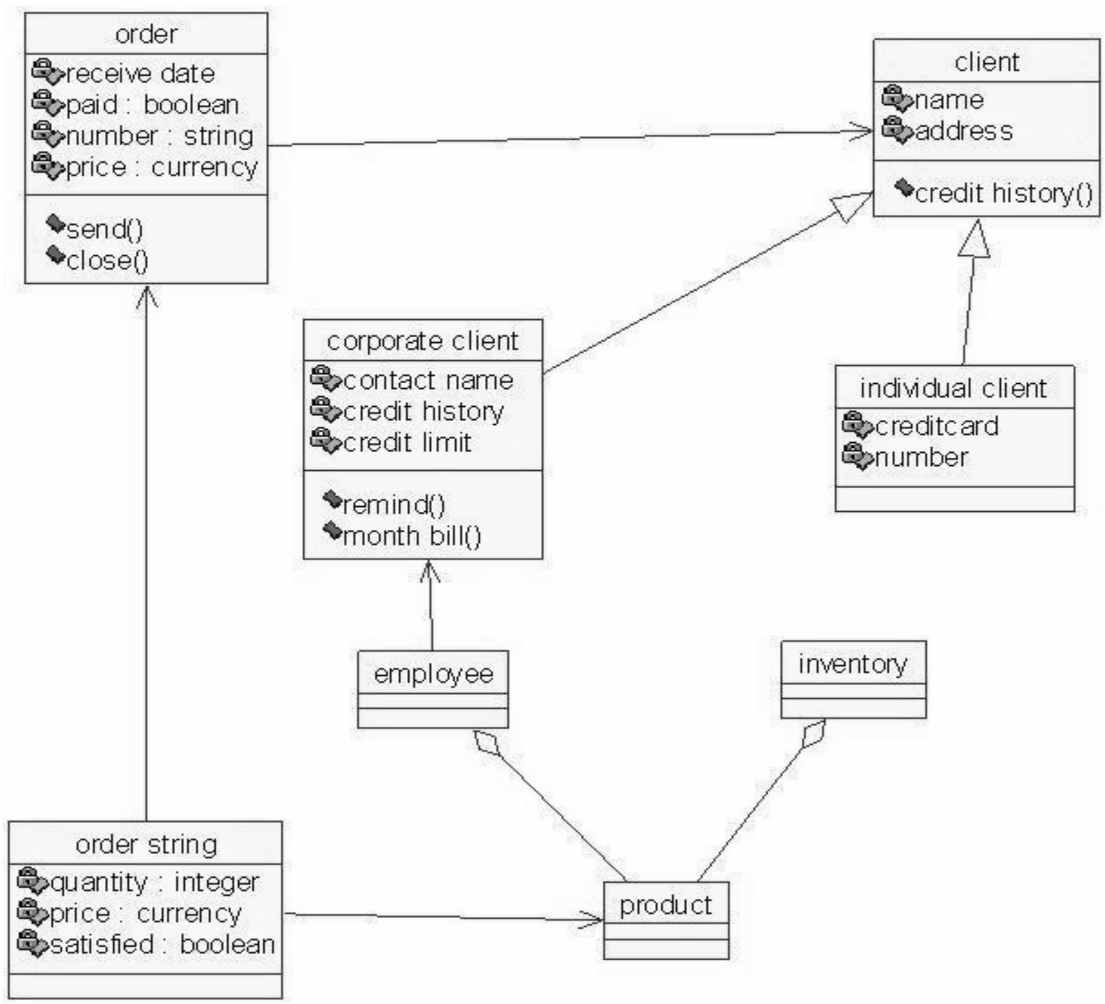


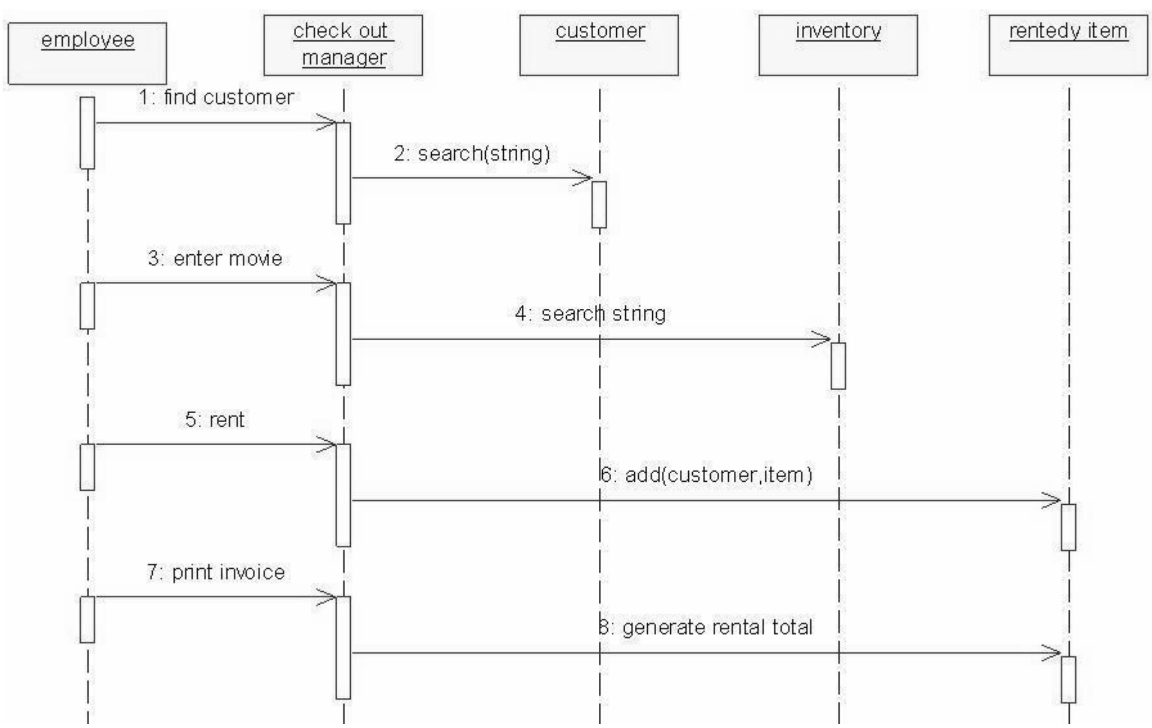
INTERACTIONDIAGRAM:

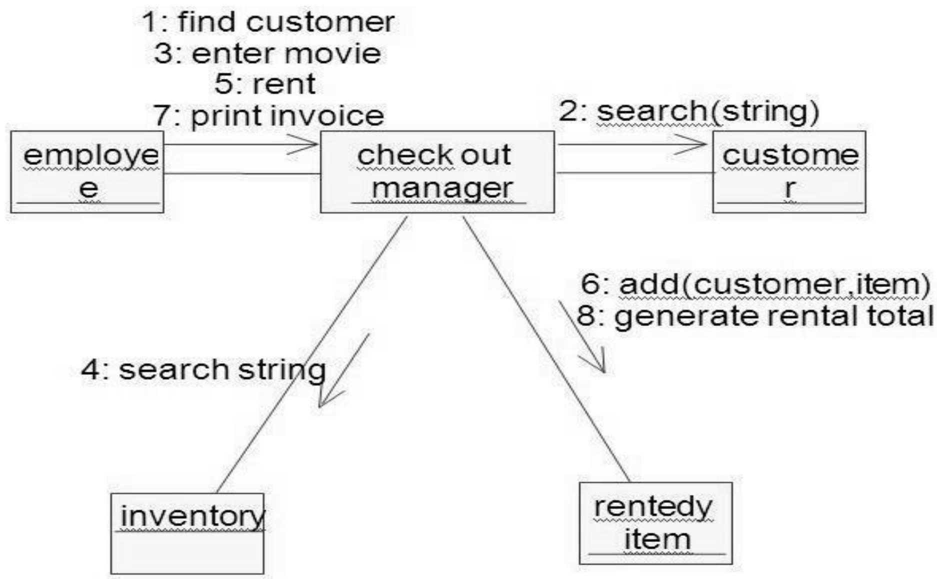
A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.

An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another, in which the “form” object is requesting an operation be performed by the “to” object. The “to” object performs the operation using a method that the class contains.

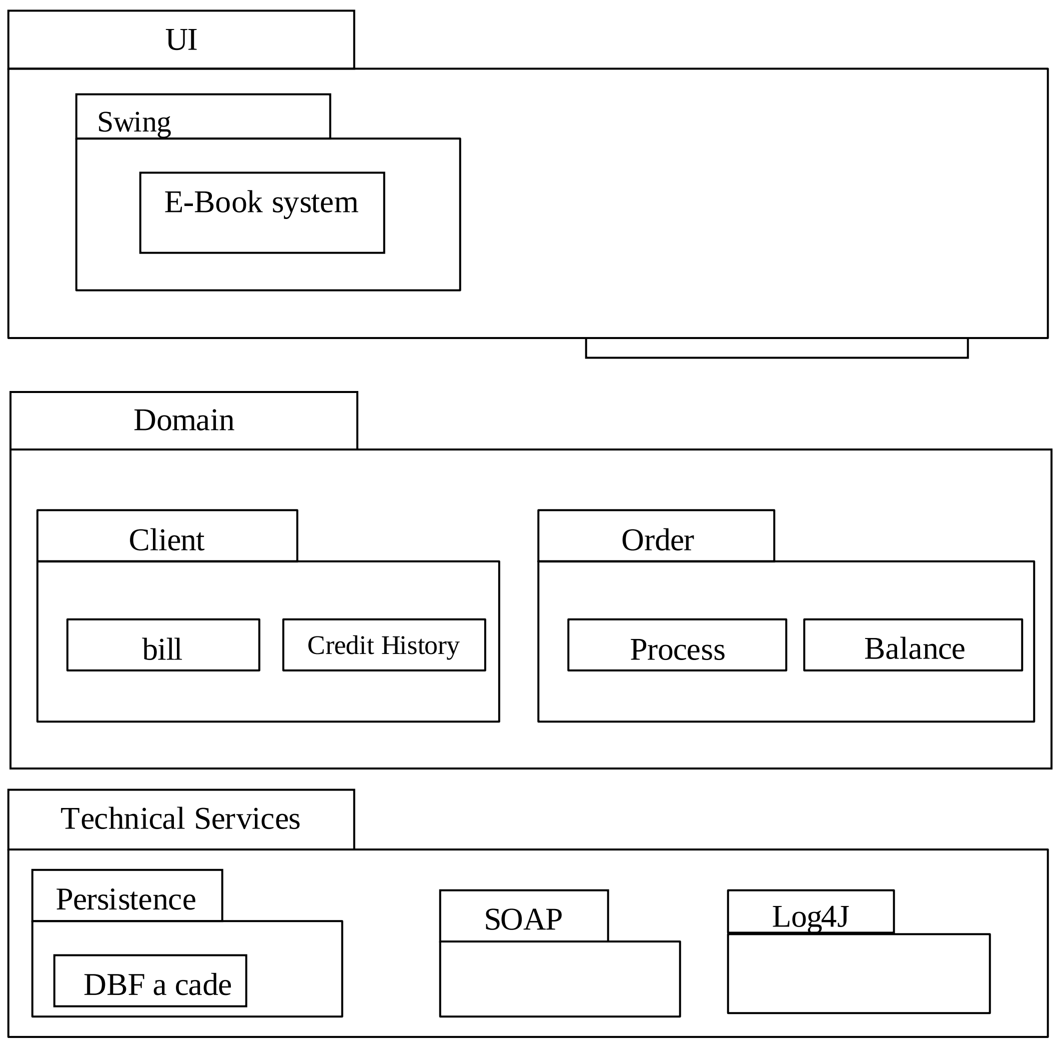
It is also represented by the order in which things occur and how the objects in the system send message to one another.

These quince diagram and collaboration diagram are given below.



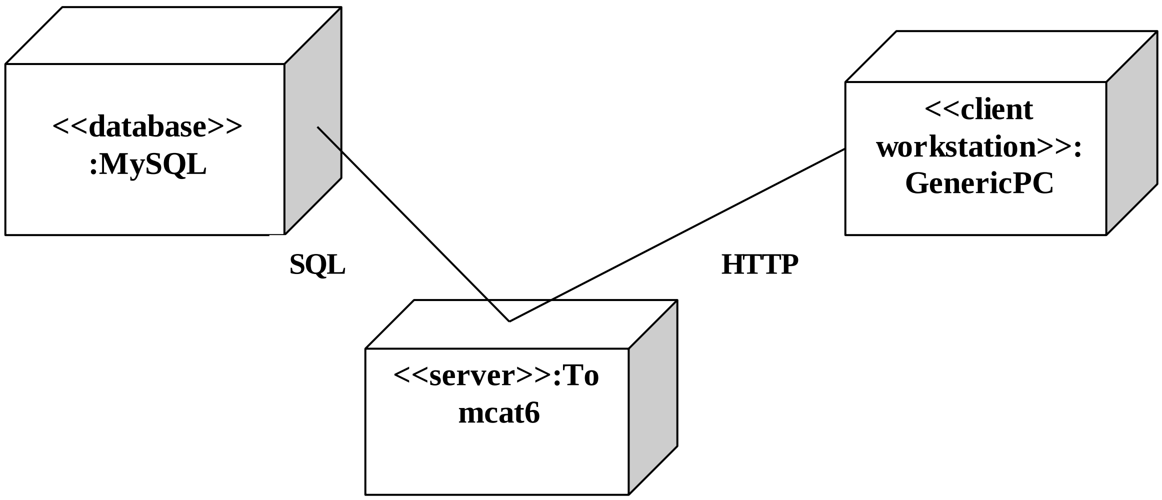


PARTIALLAYERDLOGICALARCHITECTUREDIAGRAM



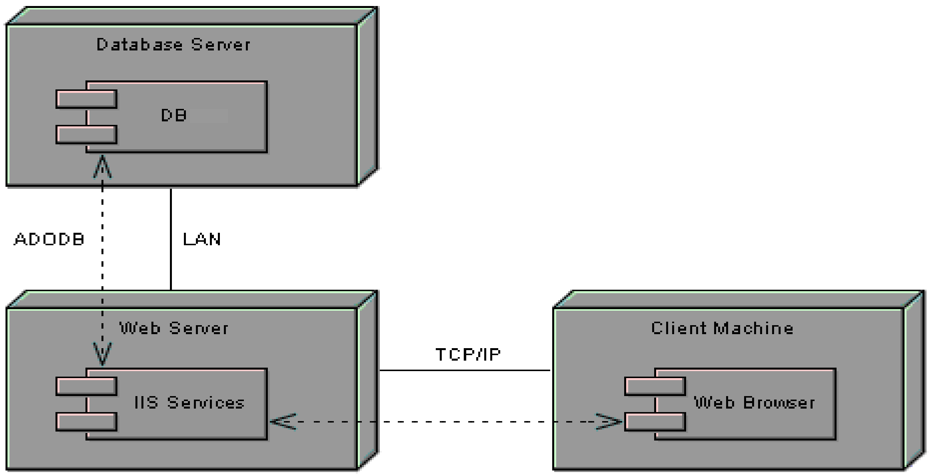
DEPLOYMENTDIAGRAMANDCOMPONENTDIAGRAM

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.



COMPONENT DIAGRAM

Component diagrams are used to visualize the organization and relationships among components in a system.



RESULT:

Thus the mini project for E-Book System has been successfully executed and codes are generated.

EX.NO: 12

LIBRARY MANAGEMENT SYSTEM

Date:

AIM

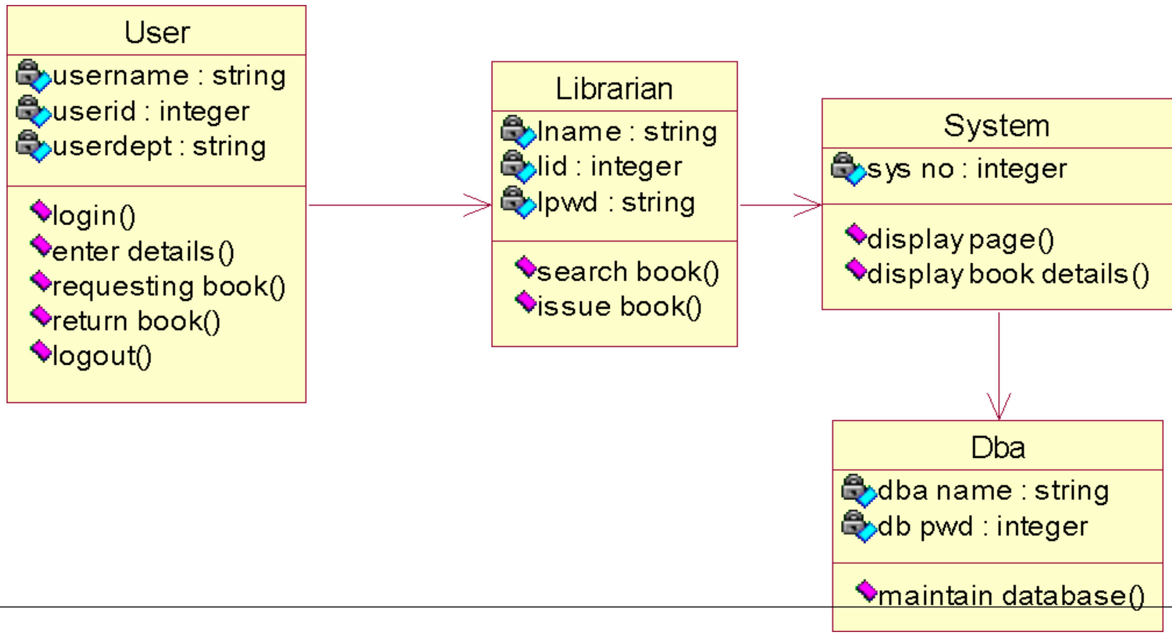
To design an object oriented model for Library Management System using Rational Rose software and to implement it using Java.

PROBLEM STATEMENT

The library management system is a software system that issues books and magazines to registered students only. The student has to login after getting registered to the system. The borrower of the book can perform various functions such as searching for desired book, get the issued book and return the book.

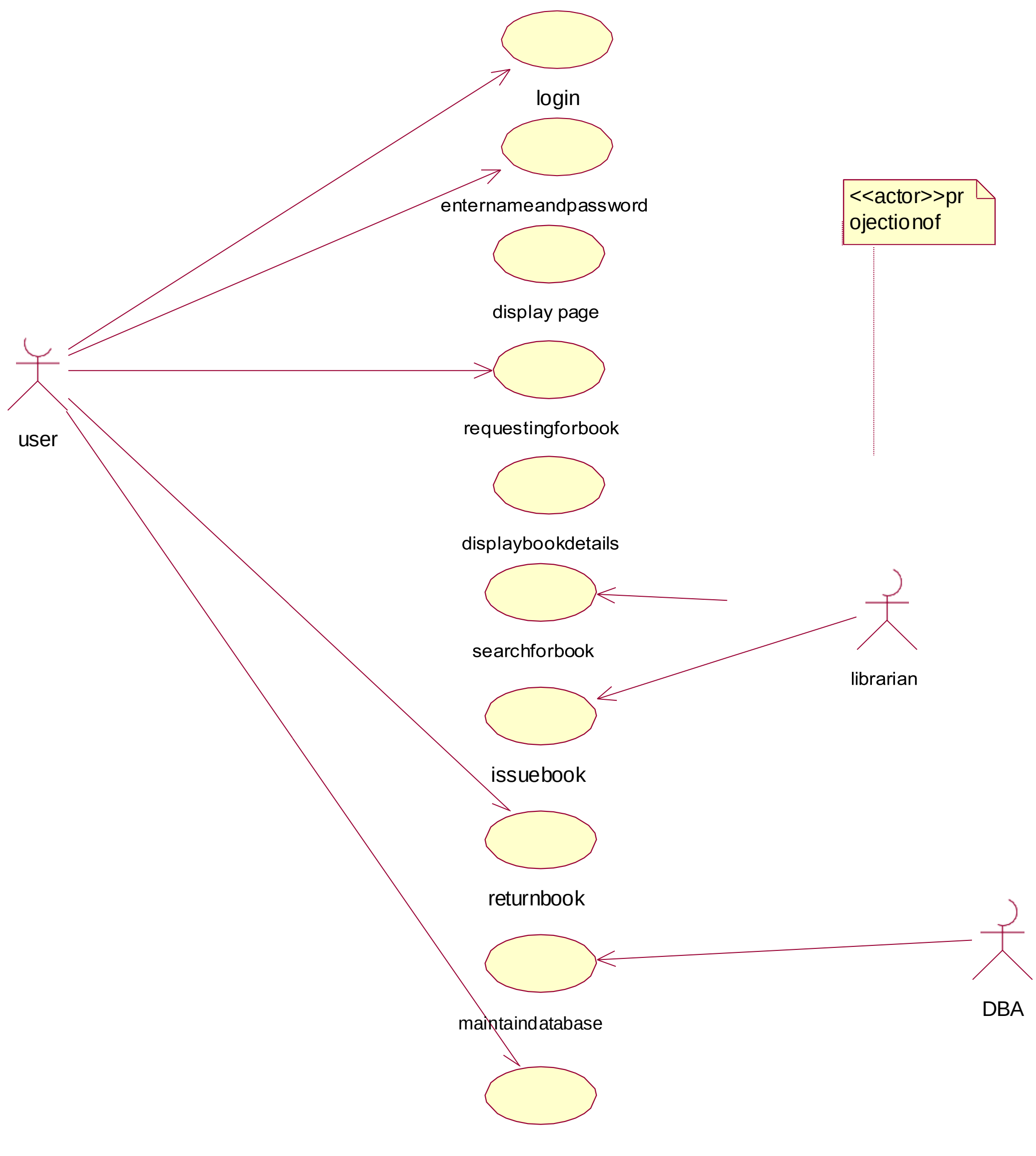
CLASSDIAGRAM

A class diagram in the unified modeling language is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations and the relationships among objects. The library management system makes use of the following classes user, librarian, system and DBA.



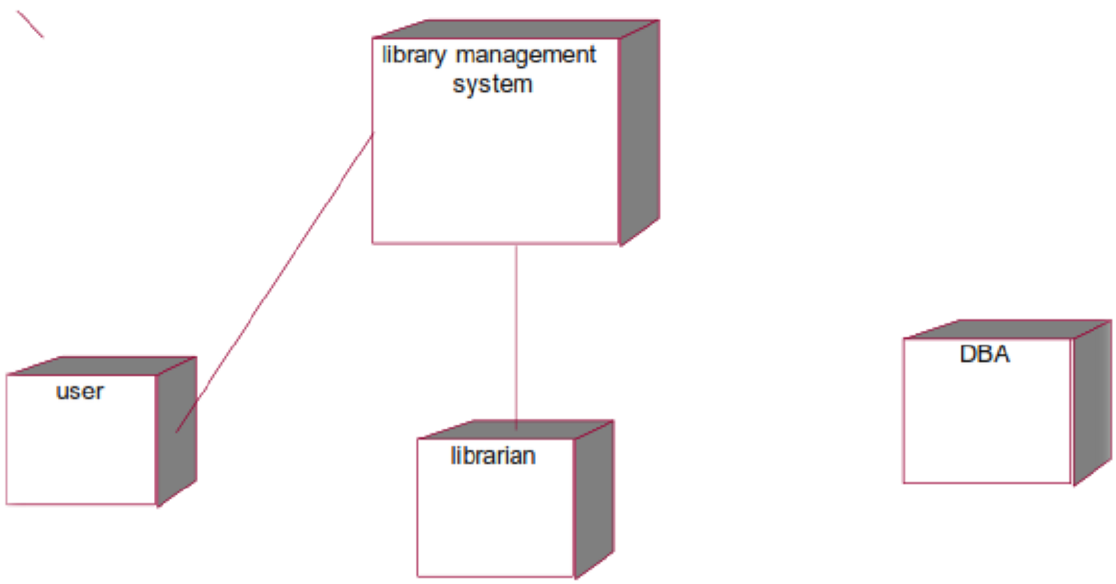
USECASEDIAGRAM

Use case is a list of actions or events. Steps typically defining the interactions between a role and a system to achieve a goal. The use case diagram consists of various functionality performed by actors like user, librarian, system and DBA.



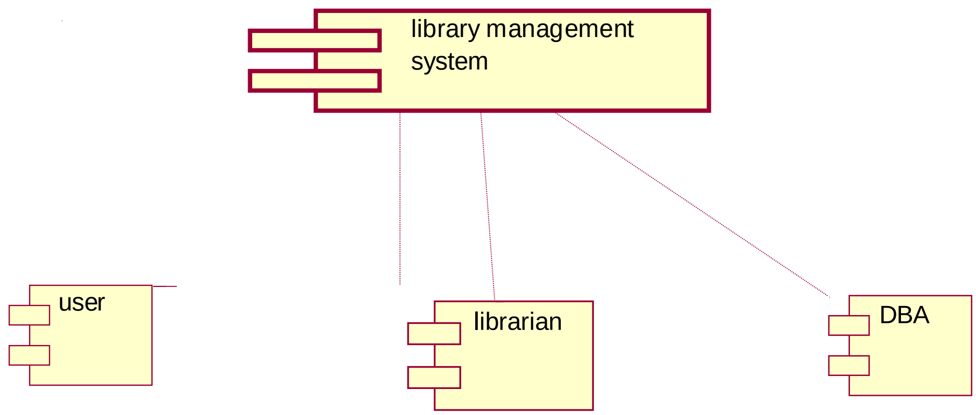
DEPLOYMENTDIAGRAM

Deployment diagram is a structure diagram which shows architecture of the system as deployment of software artifacts to deployment target. It is the graph of nodes connected by communication association. It is represented by three dimensional box. The device node is library management system and execution environment nodes are user, librarian, system and DBA.



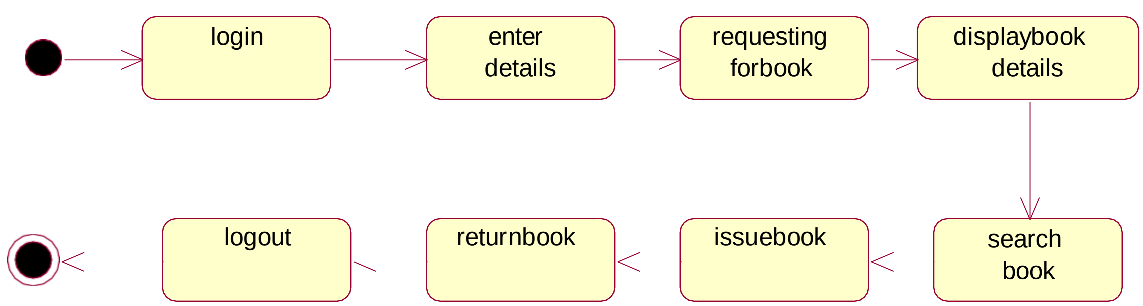
COMPONENTDIAGRAM

Component diagram shows the dependencies and interactions between software components. Component diagram carries the most important living actors of the system i.e, user, librarian and DBA.



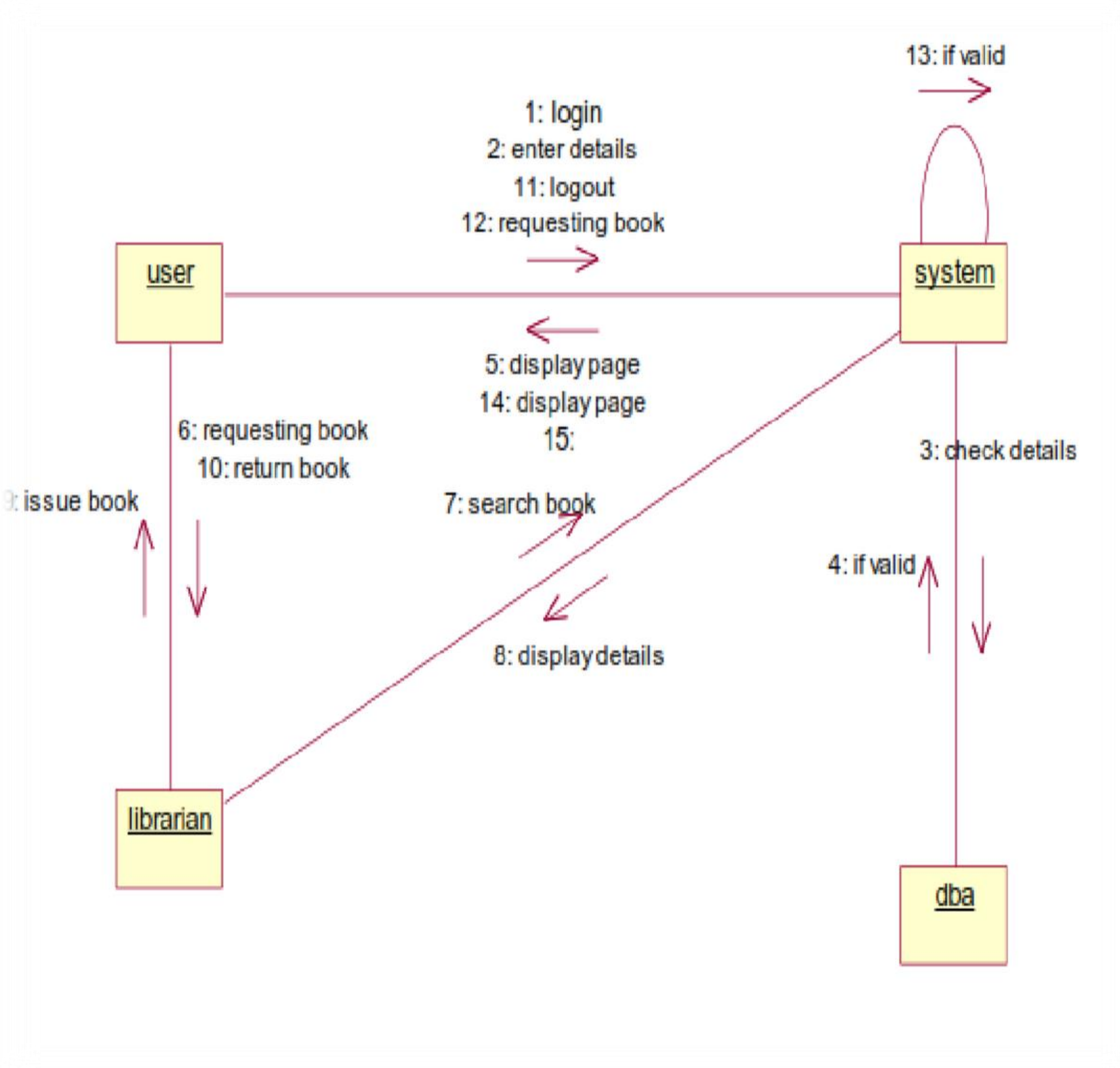
STATE CHART DIAGRAM

State chart diagram is also called as state machine diagram. The state chart diagram contains the states in the rectangular boxes and the states are indicated by the dot enclosed. The state chart diagram describes the behavior of the system. The state chart diagram involves eight stages such as login, enter details, requesting for book, display book details, search book, issue book, return book and logout.



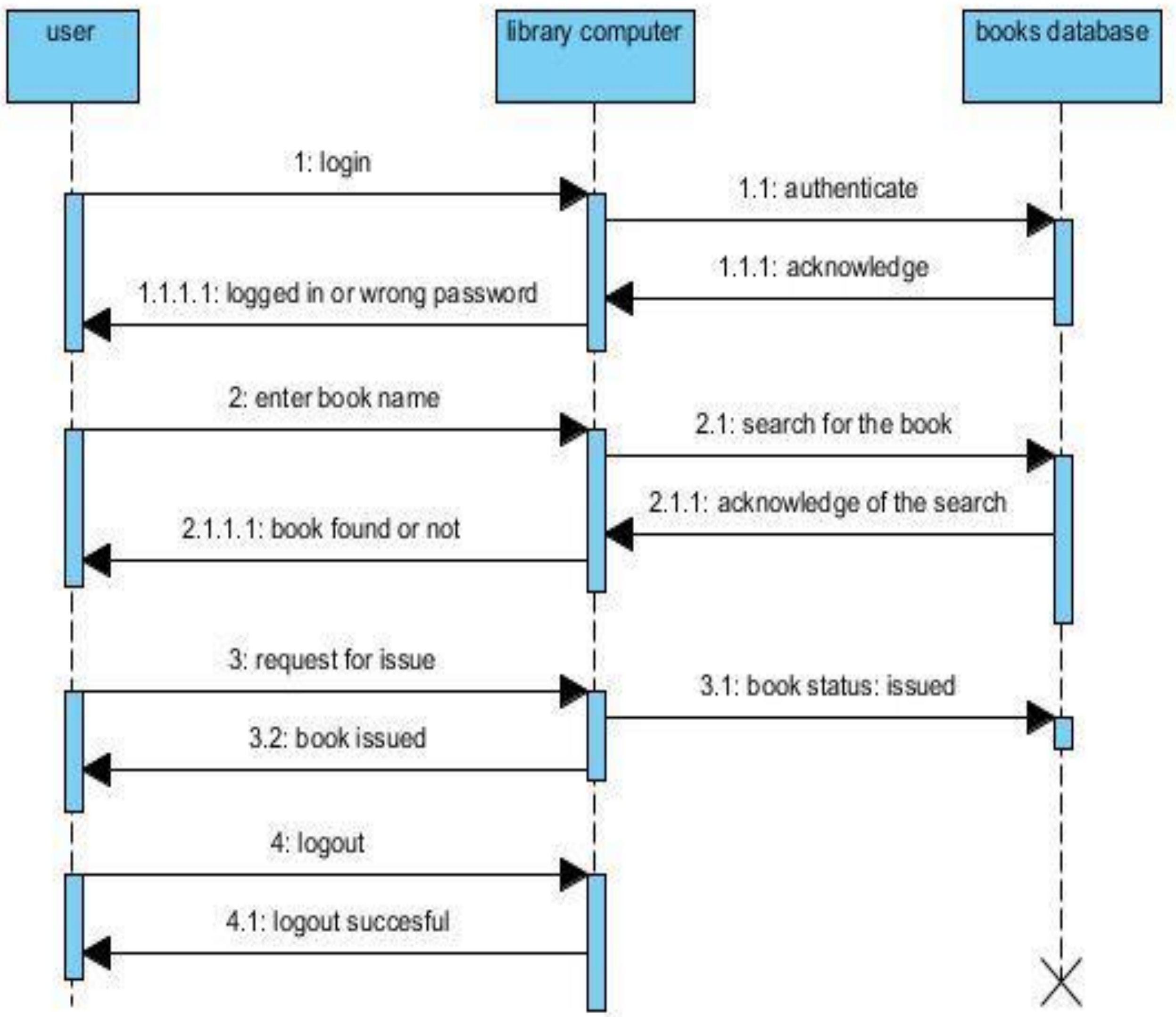
COLLABORATION DIAGRAM

Like sequence diagram collaboration diagrams are also called as interaction diagram. Collaboration diagram convey the same informations as sequence diagram but focus on the object roles instead of the times that messages are sent. Here the actions between various classes are represented by number format for the case of identification.



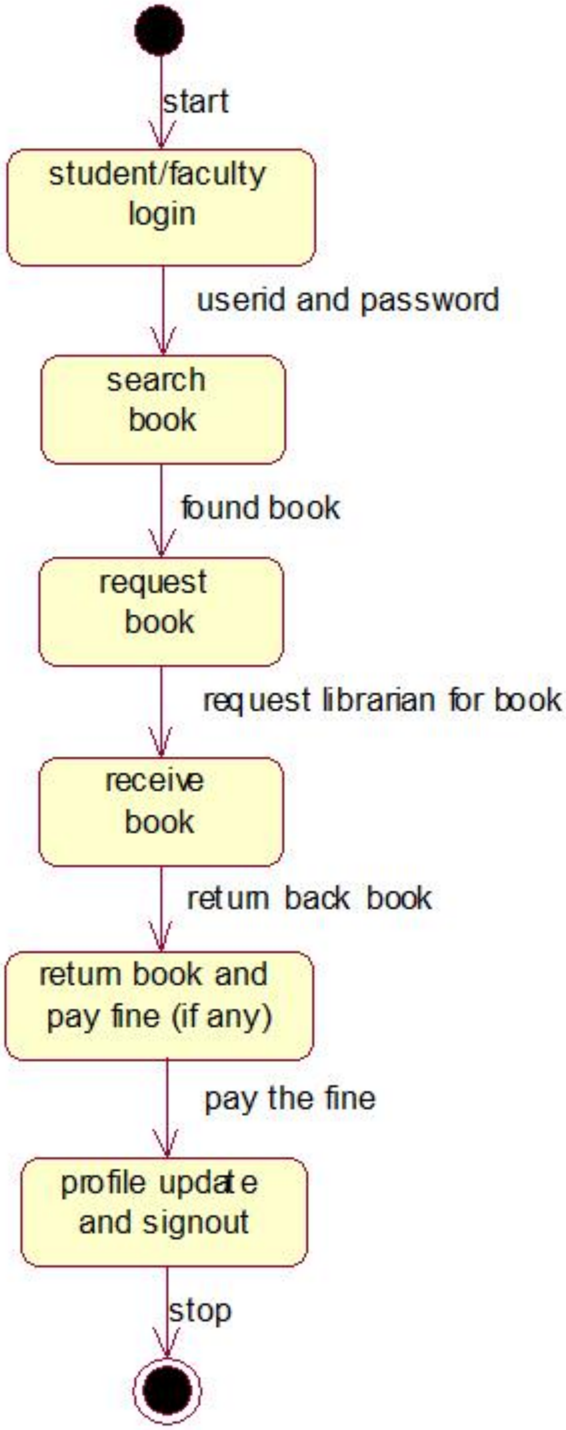
SEQUENCE DIAGRAM

A sequence diagram represent the sequence and interactions of a given use case or scenario. Sequence diagram capture most of the information about the system. It is also represent in order by which they occur and have the object in the system send message to one another. Here the sequence starts with interaction between user and the system followed by database .Once the book have been selected the next half of sequence starts between librarian and user followed by database.



ACTIVITY DIAGRAM

Activity diagram are graphical representation of workflows of stepwise activities and actions with support for choice, iteration and concurrency. Here in the activity diagram the user login to the system and perform some main activity which is the main key element to the system.



RESULT

Thus the various UML diagrams for library management system was drawn and the code was generated successfully.